

Artificial Intelligence for Early Damp and Mould Prediction and Management in Buildings: A Systematic Review and Future Directions Toward Physics-Informed Digital Twins

Mohammad H. Mehraban ^{1,2*}, Frank Ato Ghansah ¹, Hua Zhong ¹, Zainab Dangana ^{1,3}, and Chika Udejaja¹

¹ School of Construction, Property and Surveying, College of Technology and Environment, London South Bank University, London SE1 0AA, United Kingdom

² School of Management, Swansea University, Swansea, United Kingdom

³ Wates Group, Wates Sustainable Technology Services, Surrey, United Kingdom

Corresponding Author: Mohammad H. Mehraban, mehraban.smh@gmail.com

Abstract

Despite improved building standards and energy regulations, damp and mould remain pervasive due to complex interactions between hygrothermal conditions, building envelopes, heating, ventilation, occupant behavior, and environmental factors, posing significant risks to occupant health and indoor air quality. Artificial Intelligence (AI) has been increasingly applied to mould prediction; however, existing approaches remain fragmented, reactive, and weakly connected to the physical mechanisms governing fungal growth, and have not been systematically synthesized. This paper aims to critically review and consolidate AI-based methods for damp and mould prediction in buildings and to propose an integrated direction for future research. A PRISMA-guided systematic review was conducted across Scopus and Web of Science, with 79 studies meeting the inclusion criteria after database and citation searching. The reviewed studies were synthesized across six methodological categories: AI-based imaging covering surface computer vision and sub-surface electrical impedance tomography, sensor- and simulation-driven hygrothermal prediction, airborne chemical and physical indicators, housing-scale risk assessment from inspection and metadata records, material susceptibility and bioreceptivity, and integrated multimodal decision support. The review reveals limited empirical validation of mould index outputs in real buildings, fragmented modeling of inherently coupled physical and biological processes, absent Digital Twin integration, and weak linkage between indoor air indicators and surface-level mould risk. To address these gaps, the paper proposes a physics-informed, AI-integrated Digital Twin framework coupling BIM-derived building information, IoT (Internet of Things) sensing, and mould growth formulations through physics-informed loss functions and an occupant-supported, inspection-anchored validation strategy, complemented by an occupant-facing application that delivers real-time predictions, LLM-based explanation and recommendations, and photographic feedback for continuous model calibration. The framework provides a pathway toward real-time, surface-specific risk assessment and a novel direction for proactive monitoring, early intervention, and scalable mould management in occupied housing.

Keywords:

Mould prediction; hygrothermal modeling; VTT mould growth model; indoor air quality; environmental sensing; internet of things; physics informed machine learning; digital twin

1. Introduction

In developed countries, individuals spend up to 90% of their time within indoor environments, with approximately 69% of this time occurring inside residential buildings [1, 2]. Consequently, Indoor Air Quality (IAQ) is a critical determinant of human health, comfort, and quality of life [3-6].

At the same time, buildings are responsible for more than one third of global energy consumption and nearly 40% of energy and process related Carbon Dioxide (CO₂) emissions [7]. In response to these pressures, the UK government has committed to achieving net-zero carbon emissions by 2050 [8]. Improving energy efficiency within the building sector is therefore a central strategy for meeting sustainability and carbon-reduction objectives. However, energy efficiency interventions can inadvertently disrupt a building's hygrothermal performance. Measures such as enhanced insulation and increased airtightness, although effective in limiting heat loss, often reduce natural ventilation and moisture transfer. These changes can cause overheating in warmer periods and promote condensation and dampness in colder seasons, degrading IAQ and increasing mould risk [9, 10].

Recent evidence confirms that dampness and mould remain widespread in UK housing, affecting up to 27% of homes in England (approximately 6.5 million households) [11, 12], and a comparable share of dwellings in Scotland [13]. Internationally, studies report comparable prevalence, including 48% of residential dwellings in China [14] and around 50% of households in Australia [15]. In the United States, although national statistics are limited, a population weighted synthesis of studies estimates that dampness or mould affects approximately 47% of residential buildings [16].

The World Health Organization has identified damp indoor environments and mould growth as significant risk factors for a range of adverse health outcomes, including asthma, rhinitis, bronchitis, and respiratory tract infections [17]. Indoor mould exposure is associated with a 30% to 50% increase in asthma-related symptoms [18]. This issue is especially critical in the UK, which reports one of the highest global prevalence rates of adult asthma, affecting approximately 10% of the population [19].

In this context, public and political attention to damp and mould in UK housing intensified following the death of 2-year-old Awaab Ishak in 2020 as a result of a severe respiratory condition due to prolonged exposure to untreated mould in his family's flat [20, 21]. The case exposed systemic failures in identifying and remediating hazardous indoor conditions in social housing. In response, the UK Government introduced Awaab's Law in 2023, which came into force in 2025 and requires social landlords to investigate and address damp and mould hazards within legally defined timeframes [22].

Despite growing awareness of the health risks, current mould and dampness detection and management in residential buildings remain largely reactive. In practice, mould is typically addressed only after visible growth appears or occupants complain, by which point prolonged exposure may already have caused adverse health effects. This reliance on visual inspection is inherently limited, as mould growth represents a late-stage manifestation of underlying hygrothermal imbalance. Moreover, epidemiological studies show that elevated indoor dampness alone, even without visible mould, increases the risk of respiratory and allergic symptoms [23]. These challenges are particularly pronounced in the UK, which has one of Europe's oldest housing stocks; many pre-Second World War homes lack adequate insulation and are therefore more susceptible to heat loss, condensation, and consequent damp and mould problems [20, 24]. Since existing buildings are projected to comprise approximately 80% of UK housing stock by 2050 [25], reactive inspection-based strategies cannot scale to the problem, underscoring the need for early detection and real-time risk assessment.

Recent advances in AI, sensor technologies, data availability, and Digital Twin platforms offer new opportunities to transition damp and mould management from reactive visual inspection toward early, continuous, and predictive risk assessment. However, the emerging methods span visual, environmental, chemical, and contextual data domains, and remain fragmented and weakly connected to the physical mechanisms governing fungal growth. They vary widely in data requirements, modeling strategies, and real-world validation, and have not yet been systematically synthesized or critically assessed for their feasibility, scalability, and suitability for early mould prediction in occupied buildings. A consolidated evaluation of their capabilities, limitations, and integration potential is therefore needed to inform both future research and practical deployment.

In response, this study aims to systematically review and critically evaluate existing methods for damp and mould prediction in buildings, with particular emphasis on the application of AI techniques. Although AI applications in buildings have grown rapidly, prior review articles have focused on adjacent topics such as AI for smart building operation and energy management [26], hygrothermal modeling [27], occupancy behavior [28], or indoor air quality [29], rather than damp and mould prediction specifically. Existing reviews directly addressing mould have evaluated non-AI prediction models [30], summarized built-environment correlates of mould exposure [6], or surveyed dampness causes and remediation [31]. To the best of the authors' knowledge, no prior systematic review has consolidated AI-driven damp and mould methods across visual, sub-surface, hygrothermal, airborne, housing-data, and material modalities, nor articulated a physics-informed, Digital Twin-based pathway for occupied housing, a gap made increasingly urgent by Awaab's Law and the dominance of existing dwellings in the UK housing stock. Specifically, this review seeks to (1) categorize current AI-driven approaches according to data type, modeling strategy, and prediction objective; (2) assess their strengths, limitations, and suitability for early and real-time application; (3) identify key methodological and practical gaps that hinder their deployment as scalable, affordable, and proactive solutions; and (4) propose a framework for a physics-informed, AI-integrated Digital Twin that unifies building information, multi-parameter sensing, data-driven modeling, and occupant-facing feedback, providing a pathway toward early, spatially resolved, and continuous mould risk prediction in support of timely intervention and informed decision-making.

2. Background

Effective application of AI to damp and mould prediction requires a clear understanding of the physical, material, and biological processes that govern fungal growth in buildings. This section establishes the theoretical foundations underpinning the systematic review, organized around six interconnected sub-areas: the environmental drivers that initiate growth, the material properties that govern susceptibility, the moisture sources and dampness mechanisms that supply water, the airborne indicators released during active growth, the formal quantification of mould development, and the resulting implications for early and real-time detection. Figure 1 summarizes these sub-areas and the causal relationships connecting them, providing a conceptual map for the discussion that follows.

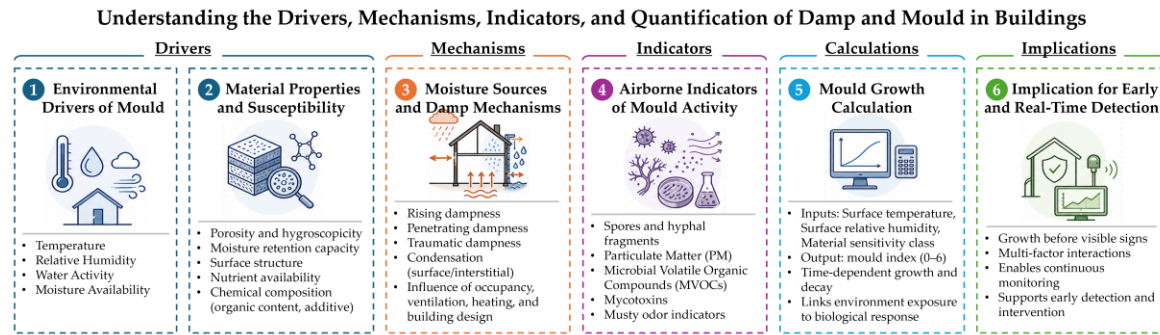


Figure 1. Overview of the damp and mould processes discussed in the background section.

2.1. Environmental Drivers of Mould Growth

Increased humidity and moisture availability are the primary factors governing mould growth on building surfaces [32]. Mould relies on unbound surface water; water locked within the material structure is not directly available for fungal use [33, 34]. In the built environment, moisture conditions are expressed in terms of Air Relative Humidity (ARH), defined as the ratio of actual water vapor in the air to the maximum amount the air can hold at a given temperature [31]. Since ARH reflects the moisture in the air, the concept of Water Activity (WA) is used to describe the amount of water available on material surfaces for microbial activity [35].

Based on their moisture requirements, mould can be classified into four groups: hydrophilic (which requires $WA > 0.9$), Mesophilic with ($0.9 > WA > 0.8$), Xerotolerant ($0.8 > WA > 0.75$), and Xerophilic ($WA < 0.75$). Most mould species thrive within a WA range of 0.90 to 0.99 [36, 37], although the critical threshold for fungal proliferation in buildings is generally taken as $WA = 0.75$ [38].

The optimal temperature range for most mould species is between 20 and 25 °C, which coincides with indoor thermal comfort levels for humans [39]. Although mould can survive across a wide temperature range, metabolic activity slows below 5 °C and may cease in dormancy, while temperatures above 46 °C are lethal for most species [40]. Dust, dirt, and the organic content of building materials supply nutrients for mould growth, but these are unavoidable and consistently present on most indoor surfaces [6, 41]. Consequently, because temperature and nutrient availability are rarely limiting under normal indoor conditions, moisture remains the primary controlling factor governing mould growth in buildings.

2.2. Material Properties and Susceptibility

Building materials vary in their susceptibility to fungal growth, governed largely by their physical and chemical characteristics [42]. Physical properties such as porosity, hygroscopicity, and water permeability govern how much moisture remains available for fungal use [43]. Porous and hygroscopic materials, including plywood, oriented strand board, and gypsum board, absorb and retain moisture from humid air, which increases their susceptibility to growth [41, 44, 45]. By contrast, hydrophobic materials such as glass, ceramics, and polymers resist water absorption and are less prone to mould formation [44, 46, 47].

The chemical properties of building materials define the amount and type of nutrients accessible for fungi. When moisture and temperature are favorable, fungi release enzymes that degrade organic polymers within the material into smaller compounds that can be absorbed as nutrients [41]. Organic-rich materials such as wood, plywood, paper-faced gypsum board, and ceiling tiles contain cellulose, hemicellulose, starch, pectin, lignin, and sugars, which provide excellent nutrient sources [40, 43, 44, 48, 49]. Conversely, inorganic or paper-free materials like lime-based plasters or uncoated gypsum are less suitable for mould

growth [44, 48, 50]. Additives also influence susceptibility: starch-based adhesives may promote growth, while phenol-formaldehyde resins, waxes, and oily coatings tend to suppress it [50, 51].

2.3. Moisture Sources and Dampness Mechanisms

As discussed previously, dampness is the primary driver of mould growth in buildings. Battersby and Ezratty [31] classify dampness into four categories. The first three (rising, penetrating, and traumatic dampness) are typically associated with building defects. The fourth, condensation dampness, is the most widespread form of moisture accumulation in buildings and arises from imbalances between indoor temperature, humidity, and ventilation. Condensation can occur either on visible surfaces or within building envelopes. Surface condensation develops when warm, moisture-laden indoor air comes into contact with cold surfaces such as window glazing or uninsulated external walls, where local Relative Humidity (RH) may reach saturation even when average room conditions appear acceptable. Interstitial condensation occurs when moisture vapor migrates through porous materials and condenses within cooler, non-porous layers of the building envelope, where it remains hidden and causes long-term material degradation.

Condensation dampness is strongly influenced by occupant moisture production behaviors from everyday activities such as cooking, bathing, and clothes drying, as illustrated in Figure 2, as well as inadequate heating and ventilation, thermal bridging, poor insulation, and increased airtightness, alongside factors related to property age, geographic location, architectural design, and construction characteristics [31, 52]. Fuel poverty further exacerbates this risk in affordable housing, particularly within UK social housing, as underheated homes are more than twice as likely to exhibit visible mould due to persistently cold and humid indoor conditions driven by financial constraints [52, 53].



Figure 2. Typical moisture generation of household activities (BS5250:2021 Management of moisture in buildings [54]).

2.4. Airborne Indicators of Mould Activity

During active growth, mould contaminates indoor air through two emission pathways: physical emissions (airborne biological particles such as spores and hyphal fragments) and chemical emissions (mycotoxins and MVOCs, a subgroup of Volatile Organic Compounds (VOCs)) [55]. The physical particles include fungal spores and mycelial fragments, which are classified as PM [56]. These particles vary in size, and their aerodynamic properties determine how long they remain suspended in air and how deeply they penetrate the respiratory system when inhaled, posing health risks. MVOCs are gaseous secondary metabolites produced during microbial metabolism and fermentation [57]. Common MVOCs span a wide range of chemical classes, including alcohols, ketones, terpenes, furans, sulphur-containing compounds, and hydrocarbons [58]. Representative compounds frequently reported in the literature include 1-octen-3-ol, 3-octanone, 2-heptanone, 2-methyl-1-butanol, geosmin, 2-methylisoborneol, limonene, pinene, styrene,

and ethylbenzene. These compounds are widely recognized as indicators of fungal metabolic activity and are largely responsible for the characteristic musty odour of mould-affected indoor environments.

To date, approximately 400 MVOCs compounds have been identified, produced by an estimated 350 fungal species [59]. Among the most hazardous indoor mould are species of the genus *Aspergillus*. It includes several highly toxic species, notably *A. parasiticus*, *A. flavus*, *A. fumigatus*, and *A. niger*, which produce aflatoxins, among the most potent known mycotoxins. In addition, *Stachybotrys chartarum* is another mould species commonly detected in indoor environments.

2.5. Mould Growth Calculation

Several prediction models are available to assess mould growth in buildings. Among these, the VTT mould index model, developed by the Technical Research Centre of Finland, is one of the most widely used. It quantifies mould growth potential as a function of Surface Temperature (ST) and Surface Relative Humidity (SRH). Originally formulated based on experimental observations of mould growth on wooden surfaces [60], the updated version extends this framework to a wider range of construction materials through the introduction of material sensitivity classes, enabling different growth and decline behaviors to be represented based on material properties [61]. The model outputs a mould index, denoted by $M(t)$, which evolves continuously over time and represents the cumulative effect of both favorable and unfavorable hygrothermal exposure.

$M(t)$ ranges from 0 to 6 (Table 1). Although treated as a continuous variable in the model, it maps onto discrete reference levels for interpretation. Because the model explicitly links environmental exposure to biological response, it has become a widely adopted indicator for assessing mould risk in hygrothermal studies. The detailed mathematical formulation of the VTT model is provided in Appendix A.

Table 1. Definition of the VTT mould index and corresponding fungal growth stages [61].

Mould Index	Description of the growth rate
0	No growth
1	Small amounts of mould on surface (microscope), initial stages of local growth
2	Several local mould growth colonies on surface (microscope)
3	Visually detected mould on surface (coverage < 10%)
4	Visually detected mould on surface (10% < coverage < 50%)
5	Visually detected mould on surface (coverage > 50%)
6	Visually detected mould on surface (coverage $\approx$ 100%)

2.6. Implications for Early and Real-Time Mould Detection

The preceding sections show that mould development arises from coupled interactions among hygrothermal conditions, material properties, moisture generation, and biological response as demonstrated in Figure 3. These processes unfold across different stages of mould development, ranging from early microbial activity governed by ST and SRH to progressive material colonization and the release of airborne biological and chemical emissions. As captured by the VTT model, biologically significant growth can occur well before visual symptoms appear, underscoring the need to monitor the underlying drivers rather than relying on visual inspection alone.

This multifaceted nature of mould growth implies that effective mould risk prediction cannot be based on a single indicator or observational scale. Instead, it requires the integration of complementary information describing environmental exposure, material susceptibility, occupancy-related moisture dynamics, and biological response over time. Recent advances in sensing technologies and data availability now enable continuous monitoring of these interacting processes, producing large volumes of

heterogeneous data. This complexity provides a strong foundation for AI methods, which are well suited to capturing non-linear relationships across diverse data streams and temporal scales. The subsequent sections therefore review AI-driven approaches to mould detection and prediction in buildings.

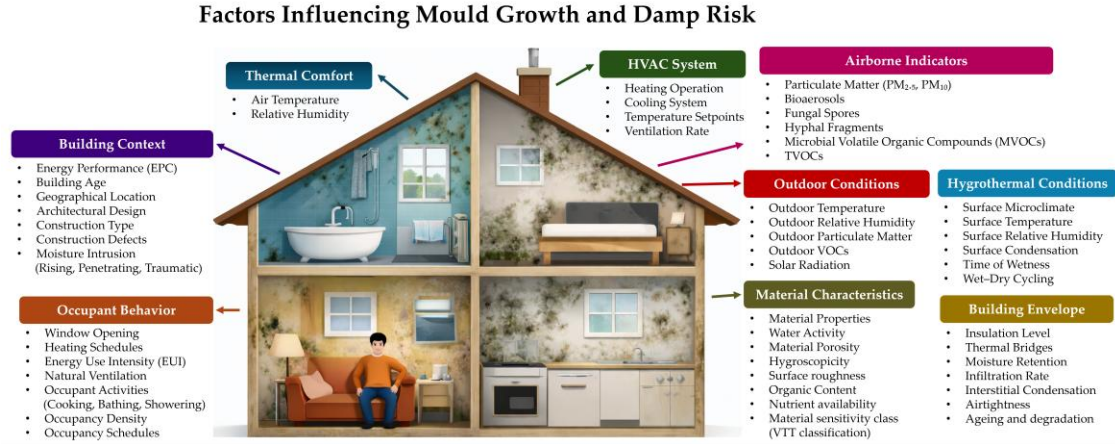


Figure 3. Key interacting factors governing dampness and mould development in buildings.

3. Methodology

The methodology comprises three sequential and interdependent stages, as illustrated in Figure 4: (1) a PRISMA-guided literature review [62] that systematically identifies, screens, and selects AI-based damp and mould studies in buildings; (2) a synthesis of the included studies through bibliometric, technical, and cross-study analyses to characterize the field and surface its outstanding research gaps; and (3) the formulation of a physics-informed, AI-integrated Digital Twin framework structured around the identified gaps. The remainder of this section details the PRISMA-guided review procedure (Stage 1); the synthesis and framework formulation stages are developed in Sections 4 and 5.

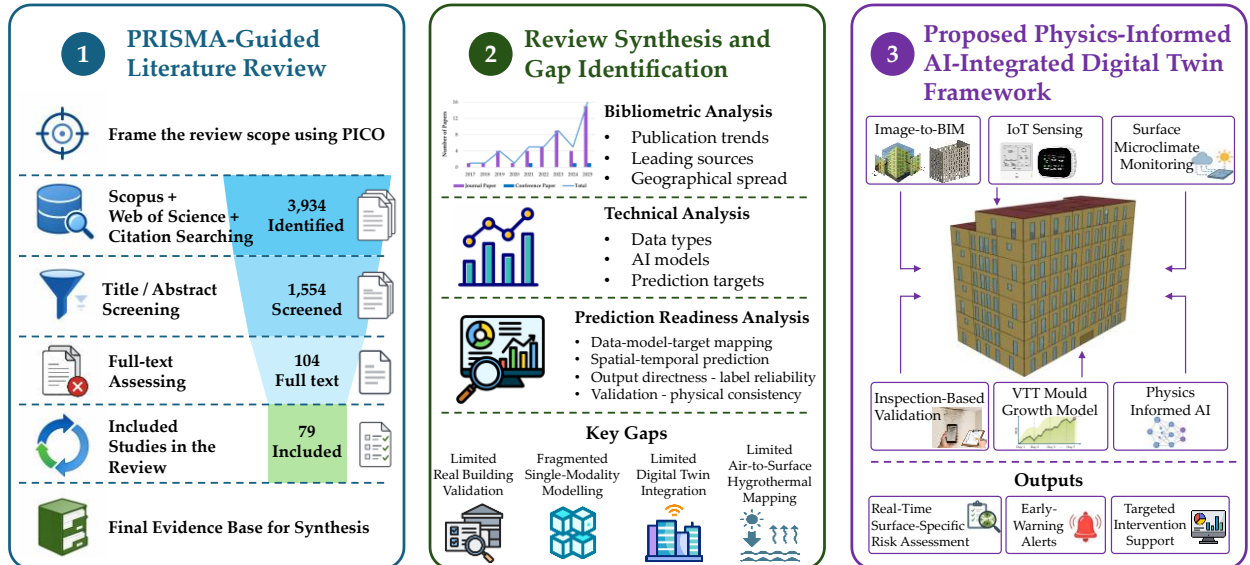


Figure 4. Methodological workflow of the study.

3.1. Review Question Framing Using the PICO Framework

The research question was conceptually framed using an adapted Population-Intervention-Comparison-Outcome (PICO) structure to define the review scope. In this review, the Population (P)

comprises residential and building environments affected by dampness, moisture-related deterioration, or mould growth, including indoor spaces, building envelopes, and occupied dwellings. The Intervention (I) refers to the application of AI-based methods, encompassing Machine Learning (ML) and Deep Learning (DL) models used for damp and mould detection, prediction, classification, early warning, or risk assessment based on environmental, visual, hygrothermal, airborne, or building-related data. A formal Comparison (C) is not specified, as the objective of this review is not to evaluate AI methods against a single baseline or conventional technique, but rather to comparatively examine and categorize diverse AI-driven approaches across different data types and modeling strategies. The Outcomes (O) include the detection or prediction of mould presence, damp risk, fungal contamination indicators, mould growth indices, or related measures relevant to early and real-time assessment in buildings.

3.2. Database Selection

The literature search was conducted using Scopus and Web of Science (WoS), which provide comprehensive indexing of peer-reviewed research across the disciplinary intersections most relevant to this review: building physics, environmental science, indoor microbiology, and computer science. Scopus was prioritized for its extensive coverage of recent publications, rapid indexing, and demonstrated superior precision and recall over alternative platforms in engineering-domain reviews [63-65], and it has been adopted as the sole database in several recent built-environment reviews [66, 67]. WoS was retained as a complementary source. Combining the two is consistent with established systematic-review practice in building and engineering science [68, 69] and reduces platform-specific indexing bias.

3.3. Search Query Development

To identify studies aligned with the scope of this review, a structured search expression was developed and adapted to the syntax requirements of each database. The search targeted titles, abstracts, and author keywords, with the database-specific search strings presented in Table 2.

Table 2. Selected databases and corresponding search queries

Source	Search Queries
Scopus	TITLE-ABS-KEY (("artificial intelligence" OR "machine learning" OR "deep learning" OR "data-driven" OR "predictive model" OR "neural network" OR "AI" OR "ML") AND ("mould" OR "mold" OR "mold growth" OR "damp" or "fungi" OR "fungus") AND ("detection" OR "prediction" OR "predictive" OR "detecting") AND ("building" OR "construction" OR "housing" OR "home"))
Web of Science	TS=("artificial intelligence" OR "machine learning" OR "deep learning" OR "data driven" OR "predictive model" OR "neural network" OR "AI" OR "ML") AND TS=("mould" OR "mold" OR "mold growth" OR "damp" OR "fungi" OR "fungus") AND TS=("detection" OR "prediction" OR "predictive" OR "detecting") AND TS=("building" OR "construction" OR "housing" OR "home")

Note: Document types are limited to journal articles (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (SRCTYPE , "j")) in Scopus; DT=("Article" OR "Proceedings Paper") in Web of Science. Also the search was restricted to English language (LIMIT-TO (LANGUAGE,"English")) in Scopus; LA=("English") in Web of Science.

3.4. Technical Inclusion Criteria

Eligibility was assessed based on the following criteria.

Subject-matter relevance. The study had to address mould growth, dampness, moisture-related deterioration, fungal contamination, or related indoor microbial risk within building environments. "Building environments" was interpreted broadly to include indoor air, interior surfaces, in-wall and inter-

layer hygrothermal conditions, and exterior building envelope surfaces, since moisture and mould phenomena traverse all of these zones.

Methodological relevance. The study had to employ an AI- or ML-based approach that learns patterns from data and reports predictive, detection, or classification performance using a defined validation procedure, such as a train/test split, cross-validation, hold-out testing, or external validation. Eligible studies therefore had to demonstrate both data-driven model development and evaluation on data not used for model training. Studies were excluded when they relied solely on deterministic hygrothermal calculations, fixed empirical formulations, descriptive statistical analysis, laboratory assays without predictive modeling, or unvalidated curve-fitting approaches.

3.5. Quality and Reporting Assessment

To support the reliability of evidence synthesis, each included study underwent a structured quality and reporting assessment across six dimensions: (1) data adequacy, whether the dataset was of sufficient size and representativeness and whether data sources were clearly described; (2) outcome definition, whether the predicted outcome (e.g., visible mould, mould index, fungal concentration, MVOCs profile, surface moisture) was clearly defined and whether it represented a direct measurement or a correlated proxy signal; (3) model development, whether training, hyperparameter tuning, and feature selection were appropriately conducted and reported; (4) validation strategy, whether internal and, where applicable, external validation were performed; (5) performance reporting, whether quantitative metrics suitable to the task (e.g., RMSE, MAE, R^2 , accuracy, precision, recall, F1, AUC) were reported with appropriate context; and (6) reporting transparency, whether sufficient methodological detail was provided for the study to be conceptually reproducible. This assessment was used to systematically characterize data adequacy, outcome definition, model development, validation, performance reporting, and transparency across studies, and informed the qualitative interpretation of findings; however, it was not a formal risk-of-bias tool, and no study-level risk ratings or summary scores were assigned.

3.6. PRISMA Selection Process

The study-selection process followed the PRISMA framework [70] and is summarized in Figure 5. It comprised identification through database and citation searching, screening, eligibility assessment, and inclusion. The database search, conducted on 20 August 2026, identified 324 records, including 191 from Scopus and 133 from WoS. After removing 104 duplicates, 220 records remained for title screening. Of these, 145 were excluded as outside the review scope, including civil engineering, materials, and infrastructure unrelated to damp or mould ($n = 45$); agriculture, food, and forestry ($n = 37$); health, biomedical, and clinical applications ($n = 27$); manufacturing, sensors, and general engineering ($n = 16$); and other out-of-scope topics ($n = 20$).

The remaining 75 records underwent abstract screening, of which 30 were excluded because they did not address damp or mould ($n = 13$), included no prediction, detection, or assessment task ($n = 15$), or were unrelated to the built environment ($n = 2$). Full texts were retrieved for all 45 remaining records and assessed against the eligibility criteria. Sixteen were excluded because the outcome was not related to damp or mould ($n = 3$), no predictive modeling was performed ($n = 2$), predictive modeling did not meet the AI/ML criterion defined in Section 3.4 ($n = 9$), the publication was not in English ($n = 1$), or methodological reporting was insufficient ($n = 1$). This resulted in 29 studies included through database searching.

To improve coverage, the database search was supplemented by backward and forward citation searching [71]. This was particularly important because relevant studies may use algorithm-specific,

building-physics, or computer-science terminology instead of broad AI terms. Accordingly, the search strategy incorporated terms such as “predictive model,” “neural network,” “data-driven,” “AI,” and “ML.” Citation searching was conducted on 20 August 2026 using Scopus and WoS and, consistent with TARCis guidance [72], was limited to one generation of backward and forward searching.

Citation searching identified 3,610 records, comprising 1,250 Scopus backward citations, 721 Scopus forward citations, 1,089 WoS backward citations, and 550 WoS forward citations. Before screening, 2,276 records were removed as duplicates ($n = 1,738$), non-English publications ($n = 14$), non-journal publications ($n = 498$), or overlaps with the database search ($n = 26$). The remaining 1,334 records underwent title and abstract screening, resulting in 1,274 exclusions because they were unrelated to damp or mould ($n = 901$), addressed damp or moisture outside the built environment ($n = 94$), contained no prediction or detection task ($n = 72$), or used predictive modeling that did not meet the AI/ML criterion ($n = 207$).

Full texts were sought for 60 records, of which one could not be retrieved. The remaining 59 were assessed for eligibility, and 18 were excluded because the outcome was not related to damp or mould ($n = 16$), methodological reporting was insufficient ($n = 1$), or the publication was not in English ($n = 1$). This resulted in 41 additional studies. A further nine relevant studies known to the authors were assessed against the same eligibility criteria and included. Overall, 79 studies were included in the review, comprising 29 identified through database searching, 41 through citation searching, and 9 through author knowledge.

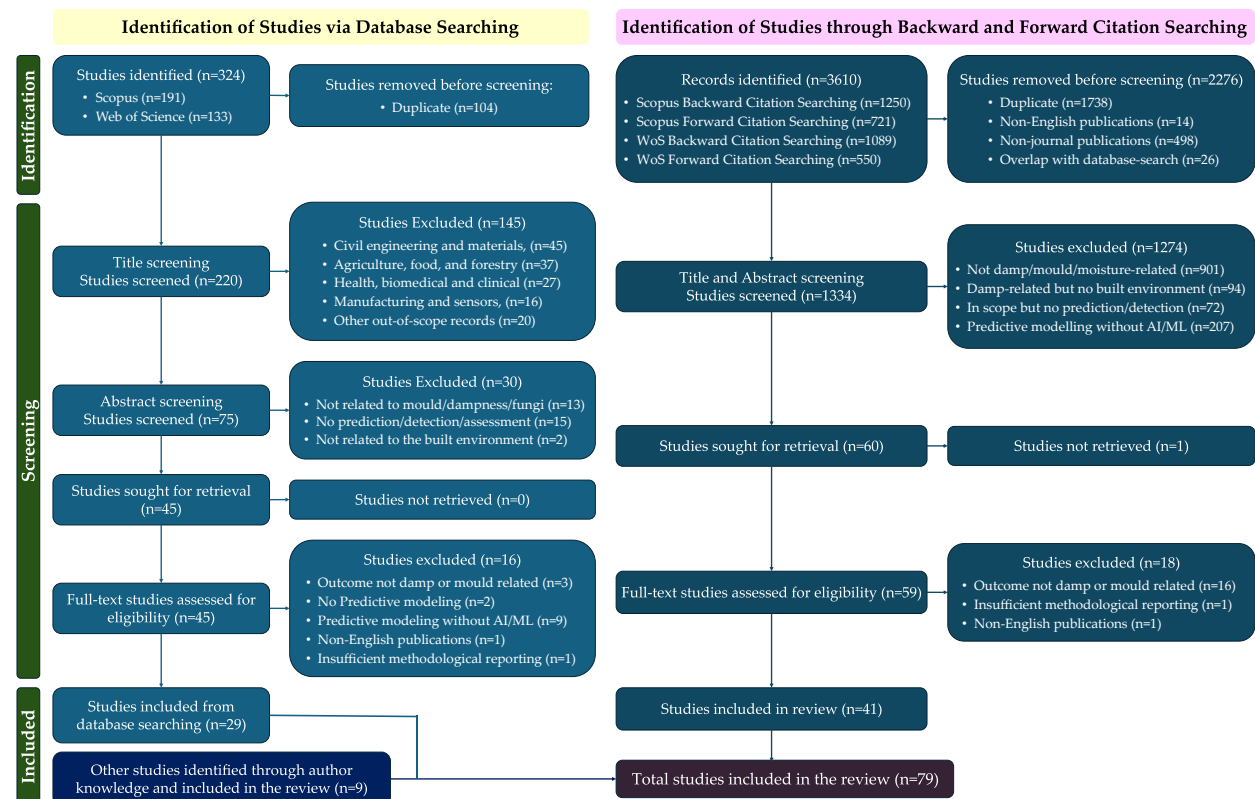


Figure 5. PRISMA selection process summarizing the flow of records through identification, screening, full text assessment, and final inclusion stages for studies on AI-based mould and damp detection.

4. Results

This section presents the findings of the systematic review based on the final set of 79 studies. The results are organized into two complementary categories. The first, bibliographical characteristics, provides

an overview of publication patterns across time, outlets, geographical origins, and keyword co-occurrence relationships within the literature. The second, technical characteristics, examines the methodological and analytical features of the included studies and culminates in a cross-study synthesis interrogating the evidence base against the central question of this review.

4.1. Bibliometric Analysis

The included studies are distributed across 29 countries and span the period 2017 to 2026, as shown in Figure 6. The inset chart indicates a clear growth in research activity on AI-based damp and mould prediction, rising from two publications in 2017 to a peak of 20 in 2025, with the upward trend continuing into 2026. The intensity of the purple shading in map reflects the number of publications originating from each country. Contributions come from a broad range of countries, led by China (13) and Poland (11), followed by the United Kingdom, United States, and Canada (7 each) and South Korea (6). European contributions are the most geographically dispersed, spanning 14 countries, while East Asian output is concentrated in fewer but more productive centers. Counts exceed the number of studies because multi-country collaborations are attributed to each contributing country.

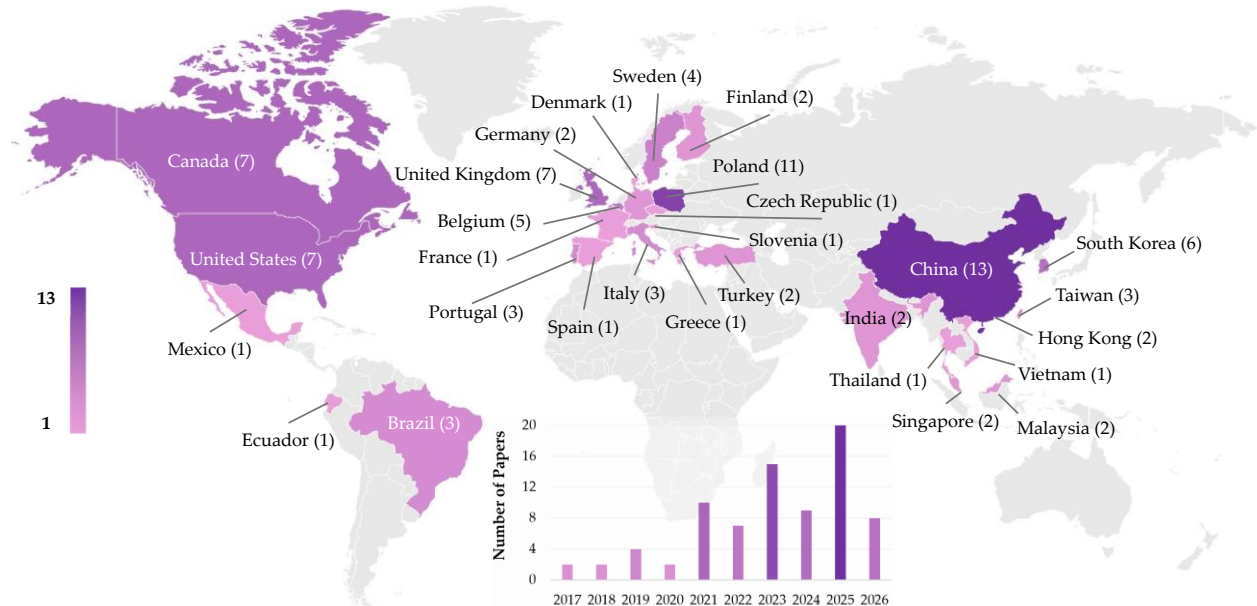


Figure 6. Geographical and temporal distribution of the included studies. Shading intensity indicates the number of publications originating from each country; the inset shows the number of publications per year from 2017 to 2026.

The distribution of publication outlets provides insight into the main sources disseminating research on AI-based mould and damp analysis. As illustrated in the treemap in Figure 7, Elsevier accounts for the largest share of publications, followed by MDPI, while Springer, Nature, IEEE, and several other publishers also contributed to the field. The figure also highlights the individual journals within each publisher, where the relative size of each block corresponds to the number of published studies. Overall, the results demonstrate that research on this topic is distributed across a diverse range of journals and publishers spanning building science, environmental research, and AI-related disciplines.

tasks, and reported performance metrics, enabling systematic cross-study comparison of model effectiveness across different data modalities and application contexts.

4.2.2 Types of Data

The reviewed studies draw on a diverse set of data sources reflecting the multidisciplinary nature of mould and damp prediction. Figure 9 summarizes the frequency of each data type used across the included papers. As several studies relied on more than one source of information, the total count exceeds the number of reviewed papers.

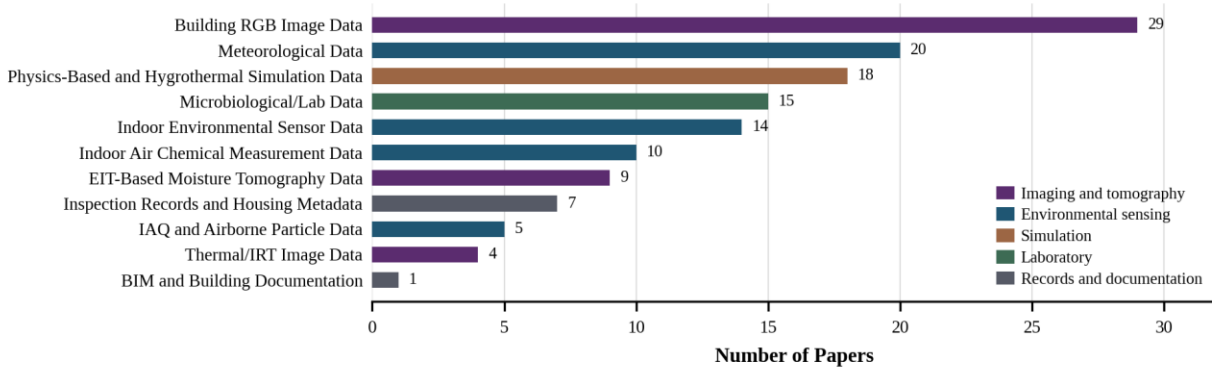


Figure 9. Distribution of data types used across the included studies.

Building RGB image data were the dominant source, used in 29 studies and spanning photographs, smartphone and UAV imagery, video frames, orthophotos, and web images for detecting visible mould, biological growth, or damp and moisture related surface deterioration. Meteorological data (20 studies) included historical, reanalysis, typical-year, and projected climate data, primarily temperature, relative humidity, wind-driven rain, solar radiation, wind, and precipitation, used either directly as model inputs or as boundary conditions for physics-based analyses. Physics-based and hygrothermal simulation data (18 studies) were generated mainly using DELPHIN, WUFI, and heat-air-moisture models, with CFD and flood simulations, to represent wall temperature and moisture responses, mould risk, and indoor moisture or airflow conditions. Microbiological/laboratory data (15 studies) included culture-based fungal counts, qPCR and fluorescence-based bioaerosol measurements, fungal culture and headspace experiments, and laboratory growth or material-durability tests, often providing biological reference or target data. Indoor environmental sensor data (14 studies) primarily captured temperature and relative humidity, alongside variables such as CO₂, pressure, and airflow at room, wall, or chamber scale. Indoor air chemical measurements (10 studies) used electronic noses, gas-sensor arrays, and GC-MS-based techniques to characterize VOC and MVOC signatures associated with fungal contamination. EIT-based moisture tomography (9 studies) used inter-electrode voltage measurements, frequently combined with FEM-generated training data, to reconstruct conductivity-based moisture distributions within walls. Inspection records and housing metadata (7 studies) incorporated building characteristics, EPC and facility-management information, HVAC attributes, and housing or spatial descriptors. IAQ and airborne particle data (5 studies) captured variables such as particulate matter, particle-size distributions, and fluorescence-based bioaerosol measurements for assessing airborne fungal contamination. Thermal/IRT image data (4 studies) were used to identify surface temperature patterns and moisture-related thermal anomalies in building elements. BIM and building documentation data (1 study) provided spatial and contextual information, including building geometry, room zoning, sensor locations, construction details, and diagnostic records.

4.2.3 AI Algorithms Employed in the Reviewed Studies

In this review, artificial intelligence refers to computational approaches designed to emulate human reasoning in tasks such as pattern recognition, prediction, and decision-making in building performance and condition assessment applications [73]. ML represents a subset of AI focused on learning patterns from data, while DL is a further specialization of ML that utilizes multi-layer neural network architectures [74]. As shown in Figure 10, the identified algorithms are organized into three broad paradigms: classical machine learning, neural networks, and foundation models, with individual algorithms grouped into model families and bubble size indicating their frequency of use.

Within neural networks, feed-forward MLP/ANN models were the most frequently reported individual architecture, appearing in 17 studies across classification, regression, hygrothermal modeling, and EIT moisture reconstruction. CNN-based methods formed the broadest family, spanning image classification, object detection, segmentation, and time-series modeling, with YOLO variants widely used for mould, moisture, efflorescence, biological growth, and other surface defects. Recurrent models, particularly LSTM and BiLSTM, were applied to temporal prediction and EIT reconstruction, while hybrid architectures such as CNN-LSTM, CNN-Transformer, GNN-Transformer, and YOLO-Swin Transformer were used less frequently.

Within classical ML, tree-based and ensemble methods were the most prominent family. Random Forest was the dominant individual algorithm, appearing in 16 studies, followed by SVM in 7 studies and Decision Tree and XGBoost in 6 studies each. These methods supported damp and mould classification, fungal and MVOC detection, building-level risk assessment, and hygrothermal surrogate modeling. Regularized and linear regression methods, including Linear Regression, Elastic Net, LASSO, and PLS, were also common, while instance-based, discriminant, probabilistic, unsupervised, and composite approaches appeared less frequently.

Foundation models showed very limited adoption. One study used CLIP as a vision-language model for zero-shot and few-shot building-defect classification and detection, while another employed the Gemma 3 large language model within an Agentic RAG framework for damp and mould assessment. Although several studies combined multiple data sources or neural architectures, integrated multimodal and foundation-model approaches remain rare. This highlights an emerging opportunity to combine visual, hygrothermal, chemical, sensing, and building-context information within unified AI frameworks, directly supporting the physics-informed, AI-integrated Digital Twin framework proposed in Section 5.6.

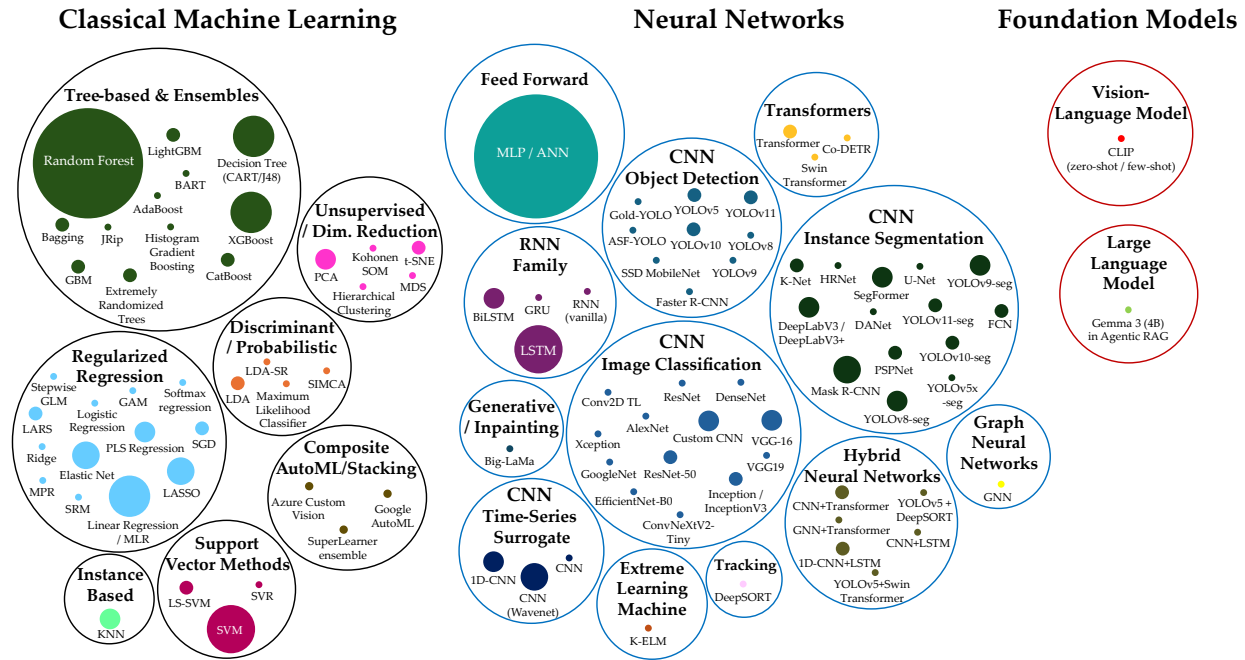


Figure 10. Distribution of AI models used across the included studies.

5. Synthesis and Discussion

This section synthesizes the findings of the systematic review by critically examining how existing AI approaches have been applied to damp and mould prediction and by evaluating their suitability for early and real-time application. Sections 5.1 to 5.6 systematically review and compare the six principal categories of AI-based methods, AI-based imaging, hygrothermal performance assessment, airborne indicators, housing and inspection records, material susceptibility, and integrated multimodal decision support, and analyze their underlying methodologies, data requirements, and reported performance. For each category, key limitations are identified, particularly in relation to physical interpretability, validation against real building conditions, spatial resolution, and predictive capability. Section 5.7 then draws these findings into a cross-study synthesis, comparing all 79 studies by spatial resolution, prediction lead time, deployment readiness, and the directness of their predicted targets. Building on this comparative analysis, Section 5.8 identifies a set of outstanding research gaps that extend across individual methodological domains. In response to these gaps, Section 5.9 proposes a physics-informed, AI-integrated Digital Twin framework that combines building information, IoT sensing, physically grounded mould modeling, data-driven learning, and an occupant-facing application for real-time prediction, explanation, and feedback, with the aim of supporting early, spatially resolved, and real-time mould risk prediction in occupied residential buildings.

5.1. AI-Based Imaging for Damp and Mould Detection

Imaging-based approaches represent the largest methodological category in the reviewed literature, accounting for 38 of the 79 included studies. These approaches use AI to transform measurements of the building fabric into spatial representations of damp- and mould-related conditions. Two main branches are identified: surface imaging using computer vision and sub-surface moisture imaging using EIT. Surface imaging detects visible or near-surface manifestations from optical, thermal, or spectral imagery, whereas EIT reconstructs moisture concealed within the wall from electrical measurements [75-79].

5.1.1. Surface Imaging by Computer Vision

Computer vision combines image acquisition and AI to automatically detect, classify, and segment visual features in building inspection imagery, offering a faster and more consistent alternative to manual surveys [80]. Across the 29 studies in this group, data acquisition extends beyond conventional RGB photography to smartphone imagery, UAV surveys, infrared thermography, visible-thermal fusion, and hyperspectral imaging [76, 77, 81-84]. Accordingly, the detected conditions include visible mould and mildew, damp and moisture stains, seepage, rising damp, efflorescence, and broader biological colonization [76, 77, 83, 85-89].

The reviewed studies show a progression toward increasingly precise spatial characterization. Image- and patch-level classification assigns a single defect label to an image or region [81, 90-94]. Object detection introduces localization through bounding boxes, particularly using YOLO-based architectures, for mould, efflorescence, and other building defects [85, 86, 95, 96]. Semantic and instance segmentation further provide pixel-level defect maps for applications including rising damp, moisture stains, seepage, efflorescence, and biological growth [77, 83, 88, 89, 97-101]. Other studies explore few-shot learning, contrastive learning, and vision-language models to reduce dependence on large labeled datasets [102, 103].

A common workflow underpins these approaches, as illustrated in Figure 11. Images are acquired using handheld, smartphone, UAV, thermal, or spectral sensors; annotated at image, bounding-box, or pixel level; and used to train and evaluate AI models. Several studies extend this workflow into practical inspection systems, including mobile applications, drone-based inspection, AR/BIM integration, automated thermographic analysis, severity-ranking platforms, and photogrammetric 3D defect mapping [81, 83, 84, 87, 89, 95, 104-106].

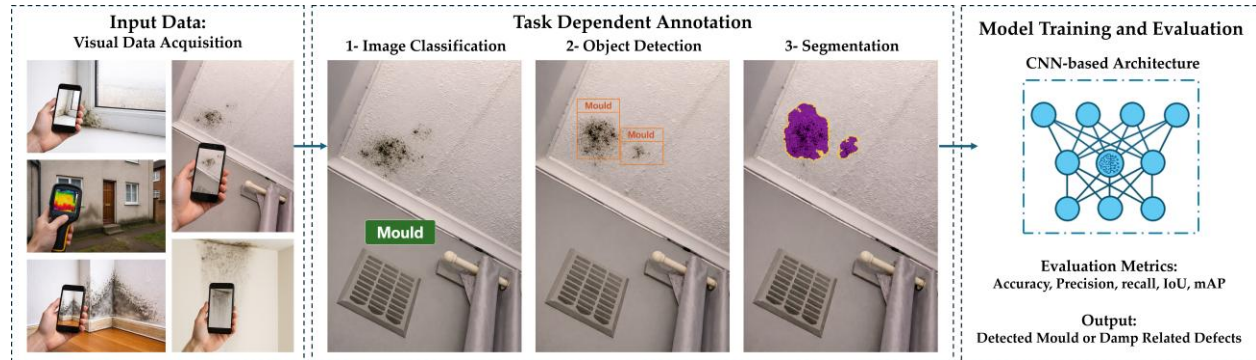


Figure 11. Overview of the computer vision workflow for damp and mould detection in buildings.

The central limitation is consistent across studies. Computer vision methods detect existing visible defects rather than predict mould risk or progression, rely on single-image inference, and are disconnected from the hygrothermal drivers that govern moisture accumulation and biological growth. They are therefore effective for automated detection of visible symptoms but insufficient as a standalone basis for proactive or predictive mould management in buildings.

5.1.2. Sub-Surface Moisture Tomography

EIT provides a complementary approach by reconstructing moisture concealed within walls before it becomes directly visible at the surface. Electrode arrays apply low-amplitude electrical currents and measure boundary voltages, from which the internal conductivity distribution is reconstructed as a two- or three-dimensional moisture map [78, 79]. Because EIT reconstruction is an ill-posed inverse problem, all nine studies employ ML or DL models to reconstruct the internal conductivity field [75, 78, 79, 107-112],

with several studies also benchmarking these approaches against conventional numerical reconstruction methods [78, 79, 111].

Applications focus primarily on historic masonry and rising damp, with field evaluation using moisture meters, thermography, or limited physical measurements [75, 78, 107-109, 111, 112]. However, conductivity is influenced not only by moisture but also by salts and material heterogeneity [107], and independent validation remains limited. Like surface imaging, EIT currently provides current-state information and does not forecast future deterioration. Its particular value lies in detecting hidden moisture that can precede visible damp and mould, suggesting potential for future integration with hygrothermal and mould-growth prediction models.

5.2. Hygrothermal Performance Assessment and Mould Index Evaluation

Hygrothermal performance assessment directly characterizes the environmental conditions that govern fungal growth within building envelopes, focusing on the evolution of temperature and RH at material surfaces and within wall assemblies, where condensation, moisture retention, and biological activity are initiated. The 20 studies in this category integrate mould index models, most commonly the VTT model, with AI-based predictive frameworks, either by forecasting temperature and RH at critical locations or by directly estimating mould index values, while a few predict the underlying hygrothermal states without computing an index [113-116]. Based on the origin of the hygrothermal data, the literature falls into two branches of sensor-driven models and simulation-driven metamodels.

5.2.1. Sensor-Driven AI Models for Hygrothermal Performance Assessment

Sensor-driven studies use in-situ monitoring of temperature and RH to train AI models that predict hygrothermal states at mould-critical locations, and they span three levels of temporal ambition. At the simplest level, an ANN estimated same-time temperature and RH inside a green-facade concrete wall on a laboratory weathering rig, as a basis for non-invasive moisture monitoring [115]. At the forecasting level, a CNN-Transformer predicted indoor temperature and RH one to seven days ahead in a hospital office and embedded the VTT mould index, alongside energy use and thermal comfort, in an NSGA-III optimization of air supply settings, reporting an 88% improvement in mould control [117], while a companion GNN-Transformer study on the same monitored venue extended the forecasts spatially across six ceiling positions, targeting the uneven conditions where mould risk concentrates [113]. At the longest horizon, a 1D-CNN-LSTM trained on two years of in-wall monitoring of a timber-framed office building then predicted in-wall temperature and RH for three years after monitoring ended, using updated outdoor climate as input, and derived orientation-specific VTT indices as a cost-effective substitute for continued sensor operation [118].

The principal limitation is the absence of biological or visual validation. Although mould index values are computed from measured or predicted hygrothermal conditions, none of the reviewed studies verify whether predicted index levels correspond to observed mould growth on monitored surfaces. Given that VTT index thresholds are explicitly defined in terms of microscopic or visible surface coverage, the lack of validation through surface inspection or microbiological analysis represents a critical gap.

5.2.2. Simulation-Driven Hygrothermal Metamodels

Simulation-driven studies use numerical heat and moisture transfer models, most often Delphin and WUFI, to generate large datasets describing temperature, RH, and moisture evolution within envelope assemblies under varied material configurations, boundary conditions, and climatic scenarios. AI

surrogates trained on these datasets follow two formulations: direct prediction of aggregate indicators such as the maximum mould index from scenario descriptors [119-126], and reproduction of the hourly hygrothermal time series inside the wall, from which the mould index is post-computed [116, 127-132]. The algorithms range from PLS regression [121, 124, 126], support vector regression [129, 132], and random forests [119, 120, 122, 125] to AutoML [123] and convolutional and recurrent time-series networks [116, 127, 128, 130, 131], with one study predicting material moisture content from surface thermal gradients to support quantitative infrared diagnosis [114].

AI surrogates trained on simulation data have been shown to cut prediction time from days to seconds while maintaining high accuracy across diverse climates and envelopes [133]. Because AI evaluates a scenario in a fraction of a second, these metamodels enable analyses that direct simulation cannot: probabilistic reliability, sensitivity, and decision analyses over ten million sampled scenarios [119, 122, 125], design guidance derived from over 135,000 metamodel evaluations spanning 96 wall compositions and 54 European climates [128, 130], moisture severity ranking and reference year screening under historical and future climate ensembles [121, 124, 126, 129], regional mould risk mapping [127], and quantification of workmanship defects such as extruded mortar joints [119, 120].

The predictive validity of these metamodels is constrained by their dependence on numerically generated data. Predicted mould risk inherits the assumptions of the underlying hygrothermal models, including prescribed boundary conditions, idealized material properties, and simplified moisture loads and air leakage. Fifteen of the sixteen studies never test their predictions against physical measurements; the single exception, a model trained on simulations and validated against one year of field monitoring, saw the R2 for in-wall RH fall from 0.98 on simulated data to 0.82 on monitored data [127]. It therefore remains unclear whether predicted mould index levels correspond to actual mould development under operational conditions influenced by occupant behavior, ventilation variability, and material aging.

5.3. Airborne Indicators of Fungal Contamination

Airborne indicators capture signals released into indoor air before or alongside visible mould growth on building surfaces, offering the potential for continuous, non-invasive monitoring under occupied conditions. The 12 studies in this category exploit two complementary classes of indoor air features: chemical signatures associated with microbial emissions, and physical particulate indicators linked to bioaerosol dynamics.

5.3.1. Chemical-Based Detection Using MVOCs and Electronic Nose Systems

MVOCs are low molecular weight gases released as by-products of microbial metabolism, particularly from fungi colonizing damp building materials and dust. The reviewed literature follows two methodological paths. The first relies on laboratory-based analytical chemistry, particularly Gas Chromatography coupled with Mass Spectrometry (GC-MS), which provides detailed identification and quantification of fungal VOCs and captures the broader chemical fingerprint of microbial metabolism across species, substrates, and growth conditions [134, 135]. Despite their analytical precision, these techniques depend on specialized equipment, controlled sampling, trained personnel, and off-line processing, and are therefore unsuitable for continuous, real-time mould monitoring in occupied buildings. The second path involves electronic noses, which capture the overall VOC pattern in the air using arrays of metal oxide semiconductor sensors with additional temperature and humidity sensing [59, 136] and more recently UV-activated SnO₂ nanowire chemiresistors [137] and multi-modal metal-organic framework and carbon nanotube arrays combining mass, resistance, and impedance channels [138]. AI models trained on

these sensor responses infer quantitative MVOC concentrations [59, 138] categorical mould-affected versus clean classifications [135, 136], or mould species identity [135, 137], enabling lower-cost, rapid, and potentially real-time deployment, though performance is sensitive to environmental variability, sensor drift, and non-fungal VOC sources.

Across the reviewed literature, AI plays a central role in interpreting MVOC patterns, which are inherently complex and non-linear, and reported performance is consistently high. SVM regression on MOS sensor responses approximated GC-MS reference concentrations of eight fungal markers with an R^2 near 0.98, showing that low-cost sensors can approach laboratory measurements [59]. Random Forest separated fungal MVOCs from non-microbial VOCs with 100% accuracy in GC-MS data, and a simplified model using five key compounds retained 96% [134]. CatBoost distinguished fungal culture headspace from control gas at 96% on a multi-modal sensor array [138], and LDA ensembles identified mould species from nanowire e-nose signals with an F1 above 98% [137]. At the room scale, Random Forest classified mould-threatened and clean indoor spaces with about 100% accuracy [136], and in whole-building indoor air analysis, SVM separated problem from reference houses with accuracy above 91% while a SIMCA screening model recognized contaminated homes with overall accuracy above 84% [135].

The key limitation concerns the predictive horizon. MVOC-based methods remain limited to state recognition, identifying contamination at the time of sampling instead of anticipating its emergence or progression. MVOC emissions arise from microbial activity once favorable growth conditions are already established and do not represent the physical drivers that initiate mould growth. Without concurrent information on moisture availability, surface temperature, ventilation, and airborne particle dynamics, MVOC signals cannot be temporally contextualized to distinguish early metabolic onset from persistent contamination or transient chemical interference, limiting their ability to support proactive, pre-visible intervention. Integration with hygrothermal and physical airborne parameters would allow MVOCs to function as early biochemical indicators within a predictive risk assessment framework, not as isolated classifiers of indoor air status.

5.3.2. Physical-Based Prediction Using IAQ Parameters, Particulate Matter, and Bioaerosols

PM and bioaerosols represent the physical component of IAQ and provide a complementary pathway for detecting mould activity beyond chemical emissions. Indoor particle concentrations are shaped by ventilation, occupant activity, resuspension of settled dust, and the presence of dampness and mould [139, 140]. Because fungal spores and fragments fall within the size ranges of PM_{2.5} ($\leq 2.5 \mu\text{m}$) and PM₁₀ ($\leq 10 \mu\text{m}$), and frequently become attached to, embedded within, or physically mixed with airborne PM, these fractions serve as central indicators of mould-related aerosols [141].

The reviewed studies quantify airborne fungi through three ground-truth routes. The first, illustrated in Figure 12, relies on impaction and culture-based sampling [142-145]: airborne particles are collected onto agar media and incubated in the laboratory before colony-forming units are counted and expressed as CFU/m³. In the workflow shown, a six-stage Andersen impactor separates particles by aerodynamic diameter, followed by seven days of incubation at 25 °C [144, 145]. This provides reliable ground truth limited to viable culturable organisms, but is slow, labor-intensive, dependent on specialized equipment, and constrained to single-point measurements that cannot capture rapid fluctuations in airborne fungal activity. The second route replaces culturing with a gene-based qPCR assay, a culture-independent alternative that captures both viable and non-viable fungi [146]. The third, illustrated in Figure 13, involves optical and fluorescence-based bioaerosol detection [147], which classifies particles as bacteria-like, fungi-

like, or pollen-like based on light-scattering and ultraviolet-excited fluorescence responses, without culturing or chemical analysis. This enables minute-by-minute tracking of bioaerosol dynamics, but the instruments cannot resolve species or distinguish indoor from outdoor sources, and their cost (around USD 200,000) restricts deployment to research settings.

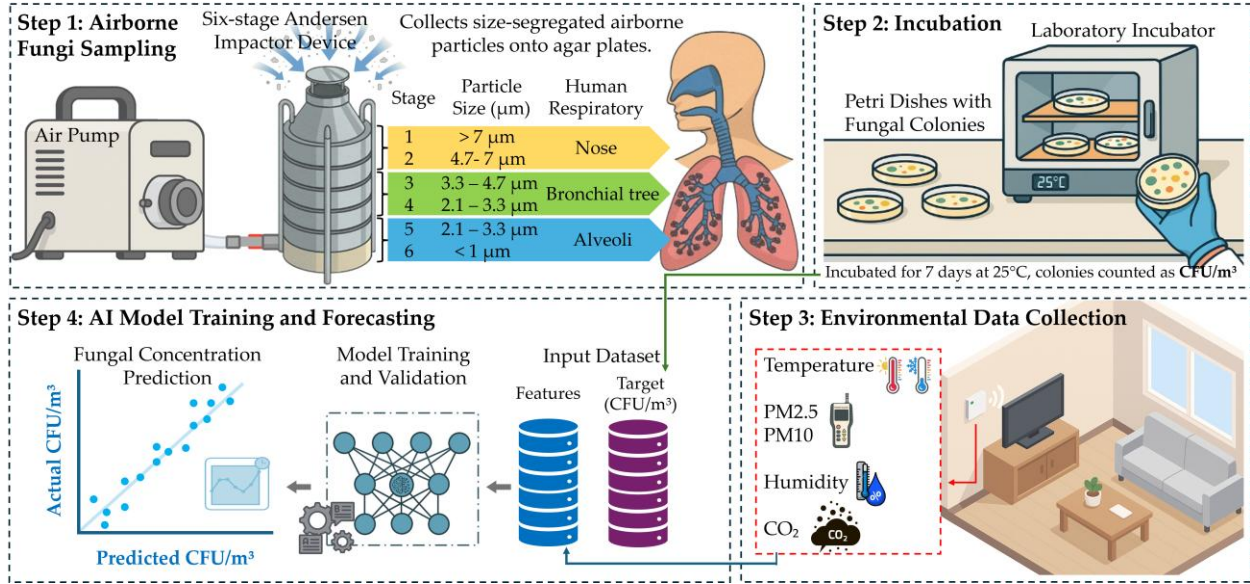


Figure 12. Overview of the culture based workflow used for predicting airborne fungal concentration.

AI models use these physically measurable indicators to estimate fungal concentrations and bioaerosol dynamics. An ANN trained on PM2.5, PM10, temperature, humidity, and CO₂ predicted culturable fungi with 83% accuracy, defined within a $\pm 30\%$ tolerance band [145], and a kernel-based extreme learning machine reached an R² of 0.76 with indoor PM10 and RH as the preferred minimal input pair [144]. Gradient boosting predicted gene-based fungal concentrations across seven public facilities with an R² of 0.78, with SHAP identifying facility type, temperature, humidity, coarse PM, and antecedent precipitation as the main drivers [146], while Random Forest related airborne mould to temperature and humidity alone across 4,048 samples from ten facility types, reaching a balanced accuracy of 78% for exceedance of the Korean caution threshold [143]. A low-cost multisensor prototype combining PM, MOX gas, and thermal channels classified chamber contamination levels against impaction-based CFU counts with near-perfect accuracy [142], and LSTM models trained on high-frequency fluorescence data predicted fungi-, bacteria-, and pollen-like particle concentrations with 60 to 80% accuracy and PM dynamics with about 90%, forecasting up to 60 minutes ahead [147]. Across models, PM fractions, humidity, and temperature recur as the dominant predictors in varying combinations [143-146].

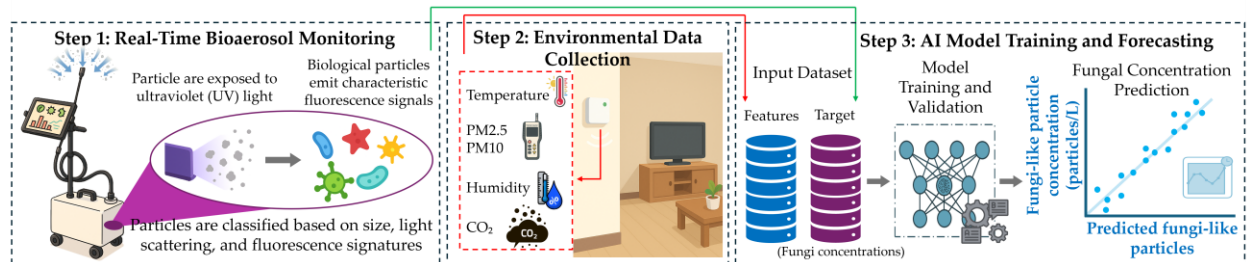


Figure 13. Overview of fluorescence based workflow for fungal concentration prediction.

The central question is whether PM and bioaerosol based methods can reliably detect mould growth in real buildings. The evidence indicates that although these physical indicators are highly sensitive to changes in airflow, outdoor infiltration, resuspension, and occupancy patterns, they cannot independently confirm the presence of active indoor mould. One study found indoor PM levels closely tracking outdoor PM (R^2 up to 0.97), suggesting that a substantial portion of airborne fungal particles originates outdoors, not from indoor microbial amplification [145], and elevated PM₁₀ or CFU measurements may simply reflect human movement, cleaning, or other mechanical disturbances that resuspend settled spores. The predictive horizon is also constrained, as eleven of the twelve airborne studies operate at the moment of sampling, and the single forecasting study reaches only 60 minutes ahead [147]. Consequently, PM and bioaerosol based systems are most effective as screening tools, signaling periods of elevated exposure or unusual particle dynamics, and cannot yet provide a definitive diagnosis of indoor mould growth.

5.4. Data driven housing and inspection risk models

Data-driven housing and inspection risk models approach damp and mould assessment at a macro scale by leveraging existing administrative, inspection, and contextual datasets instead of direct environmental sensing or hygrothermal modeling. The four studies in this category integrate heterogeneous inputs such as building age and typology, construction characteristics, energy performance indicators, inspection frequency and outcomes, HVAC attributes, flood exposure, geospatial and neighborhood features, and health or occupancy-related records [148-151]. ML models then learn patterns linking these features to reported damp presence [148, 151], mould and moisture severity ratings [150], or, in one post-flood application, laboratory-measured indoor spore concentrations [149], inferring risk indirectly through correlations with historical records without simulating fungal growth mechanisms. This supports portfolio-wide screening of housing stock and prioritization of inspections and interventions where direct monitoring or simulation would be impractical.

Performance across the four studies spans the widest range in this review, and the spread carries the lesson. A severity model trained on a 2,494-building facility-management portfolio predicted its mould and moisture index well with R^2 of 0.88 [150]. A damp classifier for English housing association homes from inspection and Energy Performance Certificate records, with heating cost, energy consumption, and wall energy efficiency as the strongest predictors, achieved 0.79 accuracy but a recall of only 0.24, too low to support inspection triage [148]. A post-flood model explained 38% of the variation in spore counts across 60 hurricane-affected homes, with airtightness as the dominant feature [149], and a stacked ensemble model built on health records and geospatial data predicted in-home mould no better than guessing with AUC of 0.54 [151]. The pattern is that models work when the label comes from the same records they learn from, such as an inspection rating predicted from inspection data [150], and they fail when the label is a sparse self-report far removed from those records [151].

The principal limitation concerns temporal sensitivity and label reliability. Because these models rely primarily on static or slowly changing attributes, they are insensitive to the short-term variations in temperature, humidity, ventilation, and moisture conditions that drive active mould growth; all four studies are current-state assessments, better suited to strategic risk prioritization than to operational real-time detection. In addition, with the exception of the post-flood study, whose spore counts were laboratory-measured [149], model outputs are trained on reported damp or inspection-derived labels that may reflect reporting bias, inspection frequency, or subjective assessment instead of actual fungal presence or severity [148, 150, 151], and the absence of direct environmental or biological validation further limits the

interpretability of predicted risk scores in terms of true mould activity. Housing-level risk models therefore provide valuable decision-support tools for large-scale asset management, but their effectiveness for early detection of active damp and mould conditions depends on integration with complementary sensing, hygrothermal, or inspection-based evidence.

5.5. Material Susceptibility and Bioreceptivity

In the previous categories, the question is whether a given building, room, or wall currently has, or will develop, damp and mould. The studies in this category ask a different question: how easily a given material supports biological growth. AI models learn the relationship between intrinsic material properties, such as chemical composition, density, porosity, roughness, surface pH, and pigment or binder content, and biological outcomes measured directly on laboratory specimens. The four studies in this category cover fungal and termite mass loss in wood-fiber insulation panels [152], microalgae growth on porous brick facades [153], algal colonization time on mortar [154], and mould presence across the surface of a biodeteriorated tempera painting [155]. The scope accordingly extends beyond mould to algae and other biodeterioration organisms, the same colonizers that the surface imaging category detects once they appear on facades, and the practical aims are design-stage material selection, maintenance planning, and conservation prioritization.

Studies in this category first measure a biological outcome on material specimens, then train an AI model to predict that outcome from material properties and identify which properties matter most. One study predicted how much mass insulation panels lose to fungal and termite attack (R^2 up to 0.99), with the ratio of holocellulose to lignin in the wood emerging as the main driver of decay [152]. Another predicted how many days algae need to reach a given coverage on mortar, accurate to about 4.6 days, from roughness, porosity, pH, permeability, and cement type [154]. A third modeled algae growth on brick facades hour by hour (R^2 0.82) and, when combined with climate data, showed that wind-driven rain roughly doubles the facade area colonized in a year [153]. The fourth searched 535 samples of a historic painted ceiling for factors linked to mould, pointing to position in the room, the coating layer, and certain binders and pigments, although its 62% accuracy was barely above the 51% baseline, so its value lies in identifying those factors, not in prediction [155].

The main limitation is that this evidence comes from the laboratory, not from buildings. Three of the four studies test small numbers of specimens under controlled conditions [152-154], and the one study on a real object barely beat guessing [155]. The models also produce a one-off susceptibility score for a material; they do not track growth over time, and the only model that tries advances one hour per step and misses the slow and dormant phases of growth [153]. The strength of this category is the opposite of its weakness: it is the only one that measures the biology directly, as mass loss, surface coverage, or confirmed growth, whereas the hygrothermal category models real walls but never checks whether mould actually grew. So far the two sides have not been joined: no bioreceptivity result has been used to update the material sensitivity classes in mould growth models such as VTT, and none is connected to monitoring in real buildings. For now, these models help with choosing materials and scheduling cleaning; their future value is providing the material inputs that an integrated predictive framework would need.

5.6. Integrated Multimodal Decision Support

This final category contains a single study, the only one in the review that combines several data sources into one working decision-support system [156]. Tested on a real residential building, it merges

five months of temperature and RH monitoring from four indoor locations, a BIM model linking each sensor to its room, and written expert diagnostic reports, and uses a large language model (LLM) to answer questions about damp risk, its root causes, and remediation. Over 50 diagnostic questions, the system reached 95.2% completeness and 94.6% accuracy [156].

Two points matter for the review. First, this is the only operational pilot outside the surface imaging category, and it shows the integration layer works: sensor, spatial, and expert evidence can be fused and queried in plain language. Second, it integrates data, not prediction: its answers rest on descriptive statistics, threshold checks, and rules over past measurements, no model projects conditions forward, and none of the predictive methods from the previous categories sits inside it. Its evaluation also scores answer quality, on one building monitored outside the heating season, not detection. The shell of an integrated system now exists, and the predictive engines of the previous five categories are what remains to be placed inside it, which is the starting point for the framework proposed at the end of this paper.

5.7. Cross-Study Synthesis: Evidence for Early and Real-Time Mould Prediction

Building on Sections 5.1 to 5.6, this section evaluates the reviewed evidence against the central question of this review: the extent to which existing studies support early, real-time, and spatially resolved mould prediction in buildings. Figure 14 and 15 provide the basis for synthesis across five dimensions: relationships between data, model family, and prediction target; spatial resolution; prediction lead time and temporal scale; the directness of predicted outputs and reliability of training labels; and the source of validation evidence and degree of physical consistency within each modeling approach.

Figure 14 maps the relationships between 11 data modalities, 12 AI model families, and 10 prediction targets across the 79 studies. RGB image data are predominantly linked to CNN-based vision models and target visible mould, biological growth, and damp or moisture-related surface defects. In contrast, meteorological, simulation, and indoor sensor data connect to a broader range of models, including feed-forward networks, recurrent and CNN time-series models, hybrid architectures, tree-based ensembles, and regression methods, with outputs concentrated on hygrothermal conditions and VTT-based mould risk. Chemical, microbiological, and IAQ data are mainly associated with classical ML and feed-forward networks for MVOC signatures and airborne fungal indicators, while inspection and housing records primarily support building- or room-level damp and mould risk assessment. Overall, the flows reveal a fragmented modeling landscape in which particular data modalities remain strongly associated with specific model families and targets, with limited integration across visual, hygrothermal, airborne, and building-context information.

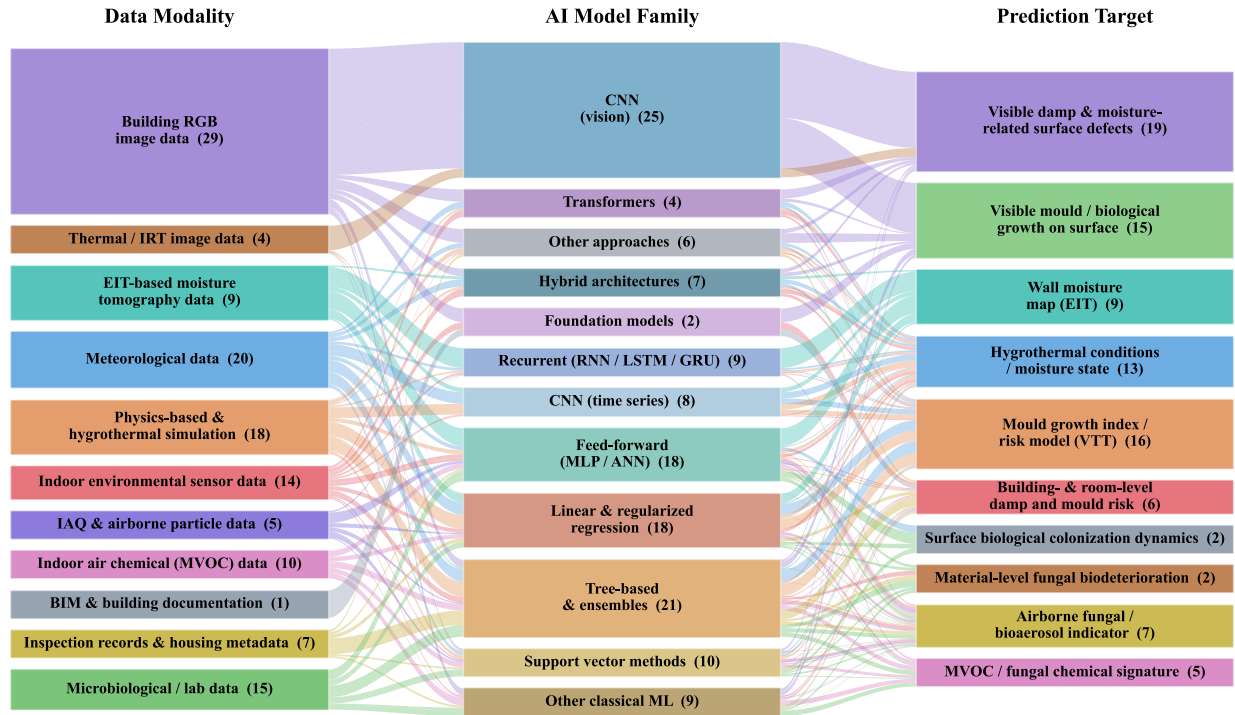


Figure 14. Alluvial diagram linking data modalities, AI model families, and prediction targets across the 79 reviewed studies. Each study contributes a total weight of one, divided equally across all data–model–target combinations associated with that study; ribbon widths therefore represent aggregated fractional study weights, while values within nodes indicate the number of studies associated with each category. Small nodes are displayed at a minimum height for label readability.

Figure 15 extends the synthesis by relating the predicted target to the spatial prediction unit (left panel) and prediction lead time (right panel), with bubble size representing deployment readiness and bubble color indicating manuscript category in the spatial panel and primary data type in the lead-time panel.

The spatial panel reveals clear specialization according to data modality and modeling approach. Surface-imaging studies occupy the three finest scales, from pixel-level segmentation to bounding-box detection and image- or patch-level classification, and target visible mould, biological growth, or damp-related surface defects. EIT studies operate exclusively at the sub-surface volumetric scale, while VTT mould-index and hygrothermal studies concentrate at the envelope/wall-layer scale, with only the two sensor-driven studies of Jiang et al. [113, 117] extending to within-room multi-point prediction. Airborne fungal and MVOC studies are predominantly associated with air-sample volumes, except the room-scale e-nose study of Lagod et al. [136]; the building-scale post-flood model of Pakdehi et al. [149] also targets airborne fungal concentration but belongs to the records category. Record-based approaches operate at the building or dwelling level, while the integrated decision-support system of Arslan et al. [156] operates at room level. Overall, spatial resolution is strongly determined by the underlying data-acquisition or modeling framework. No reviewed study integrates surface-level evidence of mould with room- or building-level contextual information within a unified predictive model, revealing a key structural fragmentation in the field.

The temporal distribution is considerably more restrictive and represents a central finding of the comparison. Of the 79 studies, 62 perform current-state or contemporaneous prediction, with two additional non-temporal laboratory regressions. Eight studies predict outcomes for predefined hygrothermal simulation scenarios; these support design-stage assessment, not forecasting, because the

climate trajectory is supplied as an input. Only seven studies look beyond the present state: five forecast with explicit lead times, and two evaluate long-term extrapolation beyond the model-development period. Among the five forecasting studies, lead times extend to 60 min for indoor bioaerosol concentrations [147], one hourly step for algal surface growth [153], 6 h for a simulation-derived mould-risk index [132], and up to 7 days for indoor temperature and RH in the two studies by Jiang et al. [113, 117]. The two long-term extrapolations extend much further: Meng et al. [118] predict three years beyond a two-year monitoring period using updated outdoor climate data, while Bansal et al. [129] train on the first five years of simulated climate-response data and extrapolate over the remaining 26 years using corresponding climate variables.

More importantly, none of these seven studies forecasts surface mould that is subsequently observed. Their outputs comprise predicted temperature and RH, with VTT mould indices subsequently derived in three of them, a simulation-derived mould-risk index, indoor bioaerosol concentrations, or laboratory-derived algal surface growth. Meanwhile, all visible-mould, damp-related surface-defect studies remain current-state assessments. The evidence therefore reveals a clear spatial-temporal trade-off: fine-resolution methods provide localization without predictive lead time, while forecast-ahead approaches operate at wall-layer, within-room, air-sample, or material scales but do not directly predict and subsequently validate future surface mould development.

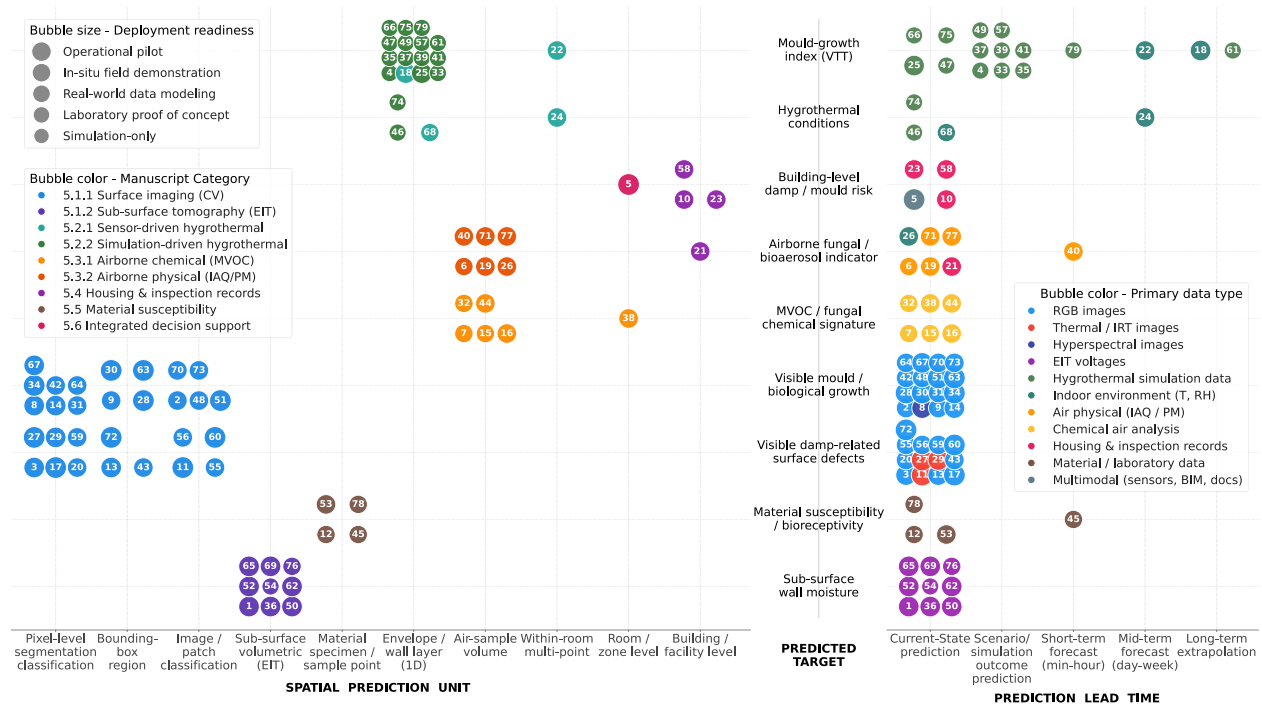


Figure 15. Cross-study comparison of predicted targets against spatial prediction unit (left) and prediction lead time (right) across the 79 reviewed studies. Each bubble represents one study; bubble size indicates deployment readiness, while bubble color represents manuscript category in the left panel and primary data type in the right panel

Deployment readiness adds a further distinction between the methodological categories. Although 11 studies reached operational pilot status, this maturity is highly concentrated: ten are computer-vision inspection systems, while Arslan et al. [156] represents the only non-vision pilot. Across the wider corpus, 15 studies are classified as in-situ field demonstrations, 28 as real-world data-modeling studies, and 9 as laboratory proofs of concept, while 16 remain simulation-only. VTT-index modeling is particularly concentrated at the simulation stage: of the 16 studies whose predicted target is the VTT mould index, 13

are simulation-only. This distribution indicates that translation into practice has progressed furthest for image-based inspection, whereas hygrothermal and mould-risk modeling remains largely at the simulation, modeling, or field-demonstration stages. Deployment readiness is therefore highly uneven across the field and strongly dependent on the underlying methodological category.

A related distinction concerns how directly the predicted target represents mould and how physical knowledge enters the models. The corpus spans directly observed evidence, such as annotated surface mould and measured airborne fungal indicators, and indirect or model-derived indicators, including MVOC signatures, hygrothermal states, VTT indices, and record-derived risk scores. Physical knowledge is incorporated mainly upstream or downstream of the learning step: hygrothermal metamodels are trained on physics-based simulation outputs, while sensor-driven models derive VTT mould indices from predicted temperature and RH during post-processing. These approaches provide physical grounding, but the underlying physical relationships remain external to model training. Across the reviewed literature, no study embeds mould-growth or moisture-transport dynamics directly within the learning objective, for example through a physics-informed loss function, while simultaneously anchoring predictions to empirical mould observations. This highlights a distinct methodological gap between physically grounded modeling, data-driven learning, and empirical validation.

Across all dimensions, the literature demonstrates strong capability within each specialized task but not the combination required for early, real-time, and spatially resolved mould prediction. No reviewed study unites direct surface-level mould prediction, an explicit future lead time, and operational deployment. Fine-resolution approaches document conditions that already exist, while future-oriented models depend on hygrothermal, airborne, or material indicators at coarser scales. Closing this gap requires forecast targets anchored to observed mould outcomes, fusion of visual, environmental, and building-context data, and physical knowledge embedded in the learning process itself.

5.8. Outstanding Research Gaps in Light of Existing Literature

In addition to the method-specific limitations discussed in Sections 5.1 to 5.6, the reviewed studies reveal broader gaps that hinder the application of AI for early and real-time mould detection in buildings. The cross-study synthesis in Section 5.7 showed that no study unites direct surface-level mould prediction, an explicit future lead time, and operational deployment; the four gaps identified below are the reasons this combination remains out of reach.

First, mould prediction models in general have been developed based either on controlled laboratory experiments or on theoretical and empirical assumptions regarding the relationships between temperature, RH, material susceptibility, and fungal growth mechanisms [30]. While several studies have attempted to compare mould models' predictions with observations from real buildings, they frequently report discrepancies between predicted and observed mould occurrence, indicating that further investigation under real operational conditions is still required [157, 158]. For example, the VTT mould index defines an index value of $M = 4$ as visually detectable mould coverage exceeding 10%; however, systematic verification and data-driven recalibration of model thresholds and response behavior using monitored conditions in occupied dwellings remain limited. The reviewed studies follow the same pattern: none of the hygrothermal studies verified predicted mould index values against observed growth on the monitored surfaces (Section 5.2), and the only category whose ground truth is a directly measured biological outcome remains confined to laboratory specimens (Section 5.5), whose results have never been used to update the material sensitivity classes of models such as VTT. This gap matters important because real buildings

operate under non-ideal conditions influenced by occupant behavior, ventilation variability, material ageing, and spatially heterogeneous moisture distribution. Recent advances in sensing technologies and data availability create a clear opportunity to validate, recalibrate, or adapt the mould models for application in real residential buildings. However, relying solely on purely data-driven ML models to address this issue introduces new challenges, as such models may produce predictions that lack physical or biological meaning and may violate known mould growth constraints. Physics-informed Machine Learning (PIML) offers a principled alternative by combining the strengths of physics-based and data-driven approaches within a unified modeling framework [159]. PIML integrates fundamental physical laws and domain knowledge, such as conservation principles, causal relationships, and governing equations, directly into the learning process. This integration can be achieved by embedding physical relationships into model architectures, loss functions, model parameters, or training algorithms, allowing ML models to learn from data while remaining physically consistent [159]. As a result, PIML has been shown to improve generalization to unseen conditions, reduce data requirements, and enhance interpretability compared to purely data-driven models when applied across a range of building and environmental domains, including building energy modeling [160], building performance simulation [159], heating and cooling systems [161], and weather and climate modeling [162]. Despite this progress, no study in the reviewed corpus applies PIML to mould prediction, consistent with the finding in Section 5.7 that no study embeds mould-growth or moisture-transport dynamics within the learning objective. This represents a significant opportunity, as mould growth is inherently governed by known physical and biological processes yet exhibits strong sensitivity to real building operation. In this context, a mould model such as VTT can serve as a physically grounded baseline, while physics-informed learning can be used to refine its parameters, thresholds, or response functions using real-world monitoring data. Such an approach preserves physical interpretability while enabling data-driven adaptation, making it particularly well suited for improving early and real-time mould risk prediction in occupied buildings.

Second, As demonstrated in Figure 3, mould growth is a multifaceted process governed by the combined interaction of indoor environmental conditions, occupant behavior, hygrothermal exposure, heating and ventilation operation, surface microclimate, material properties, moisture generation from daily activities, and airborne chemical and physical indicators, alongside visual manifestations on building surfaces. Each of these factors also becomes measurable at a different stage of mould development, so no single sensor class captures the full temporal arc from favorable hygrothermal conditions through microbial emissions to visible growth. However, the reviewed literature predominantly addresses these factors in isolation, with individual studies focusing on a single domain such as temperature and RH sensors [117, 118, 127], surface images [85], MVOCs [59, 134, 135], or inspection records [148, 151]. The one reviewed study that combines several data sources shows the same limitation. Arslan et al. [156] merge temperature and RH monitoring, a BIM model, and written expert reports into a single system, but that system only describes past measurements: it contains no model that predicts mould, and no surface images or airborne indicators feed into it (Section 5.6). This fragmented treatment limits the ability of AI models to represent the coupled mechanisms that drive mould initiation, progression, and potential dissemination within real buildings. In practice, hygrothermal conditions, IAQ, occupant behavior, and material response through MVOC and PM emissions evolve simultaneously and continuously influence one another. As a result, reliance on a single indicator or data source can lead to incomplete, biased, or misleading predictions of mould risk. The absence of integrated frameworks that jointly incorporate environmental sensing, surface conditions, air-based indicators, and contextual building information therefore remains a

fundamental gap, particularly for achieving robust early and real-time mould prediction in occupied buildings.

Third, recent research in the built environment has increasingly focused on integrating BIM, AI, and IoT technologies to monitor indoor conditions and optimize building performance through the Digital Twin paradigm [163]. In this context, AI-BIM studies demonstrate their potential to support envelope-sensitive energy prediction and multi-objective optimization of energy use and thermal comfort [164, 165]. Digital Twins have therefore been successfully applied in domains such as building energy prediction [166], real-time IAQ management [167], HVAC energy use reduction [168], and indoor thermal comfort [169], where they enable continuous data exchange between physical assets and their virtual representations. Systematic reviews in this domain have consolidated the enablers and applications of Digital Twins for smart facility management [170, 171], while more recent work has begun extending the paradigm through generative AI to enable adaptive, self-evolving building operation [172]. Despite these advances, the application of Digital Twin frameworks to damp and mould prediction remains notably underexplored. Most existing mould-related studies operate as standalone analytical models, disconnected from building geometry, material properties, spatial context, and dynamic operational states. The closest step in the reviewed studies is Arslan et al. [156], whose BIM layer links each sensor to its room; it shows that spatial anchoring works, but it carries no forward prediction and no envelope or surface representation. The absence of Digital Twin integration limits the ability to localize mould risk spatially, track its temporal evolution, and evaluate intervention strategies in real-time. Incorporating AI-driven mould prediction within a Digital Twin environment would allow sensor data, hygrothermal conditions, and building information to be continuously synchronized, enabling context-aware, room-level, and surface-specific mould risk assessment. The lack of such integrated Digital Twin-based approaches therefore represents a critical gap in advancing real-time, proactive mould management in occupied buildings.

Fourth, as demonstrated in the background section, mould growth depends on surface conditions, the ST and SRH that the VTT mould index requires at the material interface, not on bulk indoor air conditions. However, the majority of monitoring systems in buildings typically measure Air Temperature (AT) and ARH, not surface conditions. The relationship between air and surface hygrothermal conditions is non-linear and influenced by factors such as thermal bridging, insulation quality, ventilation patterns, radiant heat exchange, and transient moisture loads, however, this mapping is rarely modeled explicitly in existing studies. As a result, mould predictions based solely on air measurements may fail to capture localized surface conditions where condensation and microbial growth actually occur [158]. The reviewed studies reflect this blind spot: the only two studies with within-room multi-point resolution predict air temperature and RH at ceiling positions, not surface states [113, 117], and the in-wall predictions available for real buildings are driven by outdoor climate [118], so the surface microclimate at thermal bridges and cold corners, where growth actually initiates, remains both unmonitored and unmodeled. Data-driven and AI-based modeling approaches offer a promising pathway to learn the complex mapping between AT and ARH and corresponding surface hygrothermal states using monitored data, building characteristics, and operational context.

Collectively, these four gaps are interconnected, and they prevent AI-driven mould prediction from being deployed reliably in occupied buildings. Their resolution therefore cannot be achieved in isolation but requires an integrated framework that combines established physical theory, multi-stage sensing, spatially resolved building representation, and explicit air-to-surface modeling. Figure 16 illustrates how

each component of the framework presented in Section 5.6 maps onto and addresses the identified gaps, providing the logical bridge from this synthesis to the proposed solution.

Identified Research Gaps	Framework Response
1- Lack of Real-Building Validation Mould calculation models have been derived from controlled laboratory experiments and theoretical assumptions, with limited empirical validation and calibration using observed mould conditions in dynamic occupied buildings.	Physics-Informed ML with Inspection Anchoring Integrates VTT mould-growth dynamics with Digital Twin sensor data and inspection-based observations within a physics-informed learning framework to support validation, calibration, and physically consistent prediction.
2- Fragmented Single-Modality Data Existing studies typically rely on a single data modality, whereas mould development unfolds across distinct temporal stages, each captured by a different class of sensors.	Multi-Modal IoT Sensor Integration Combines BIM-derived data, hygrothermal, IAQ, MVOC, PM, imaging, occupancy, and ventilation sensing within a unified IoT layer to capture the full temporal arc of mould development.
3- Limited Digital Twin Integration Current approaches remain standalone predictors with limited Digital Twin integration, although mould is spatially heterogeneous, changes dynamically over time, and may concentrate on specific rooms, walls, or surfaces.	Digital Twin with BIM-Anchored Spatial Mapping Links real-time sensor streams to BIM-defined rooms, surfaces, and envelope elements, enabling spatially resolved monitoring, model deployment, and decision support.
4- Air-to-Surface Hygrothermal Mapping Room-level air temperature and humidity can appear acceptable while local surface conditions at thermal bridges or cold spots remain favorable for mould growth.	AI-Based Air-to-Surface Modeling Learns the non-linear relationship between indoor air conditions and surface microclimates using monitored data, building context, and physics-informed constraints.

Figure 16. Mapping between the four research gaps identified in Section 5.5 and the corresponding components of the proposed framework presented in Section 5.9.

5.9. Proposed Physics-Informed AI-Integrated Digital Twin Framework for Early and Real-Time Mould Prediction

Building on the identified research gaps, this section proposes an integrated framework for early and real-time mould prediction in buildings. The framework combines BIM, IoT-based environmental sensing, physically grounded mould modeling, data-driven learning, and an occupant-facing application layer within a Digital Twin platform. Its objective is to move beyond isolated, single-modality approaches by providing a pathway toward spatially resolved, time-dependent assessment of mould risk that remains consistent with known hygrothermal and biological mechanisms and is communicated to occupants in real time.

5.9.1. Building Information Model generation using Image-to-BIM

The first step is creating the target building BIM model, which serves as the geometric and semantic backbone of the Digital Twin, as illustrated in Figure 17. The BIM model can be generated either manually, when detailed drawings or surveys are available, or automatically using the Image-to-BIM method, which reconstructs BIM models from RGB images using Neural Radiance Fields (NeRF) for 3D scene reconstruction followed by semantic modeling within the Revit environment [173]. Detailed building information is then added, including construction age, envelope assemblies, and material properties (layer composition, thickness, thermal conductivity, thermal mass, and thermal resistance) for external walls, roofs, floors, windows, and doors. HVAC system characteristics are incorporated where available. The completed model is exported in gbXML format and imported into OpenStudio to define thermal zoning and spatial relationships required for subsequent hygrothermal analysis and Digital Twin integration.

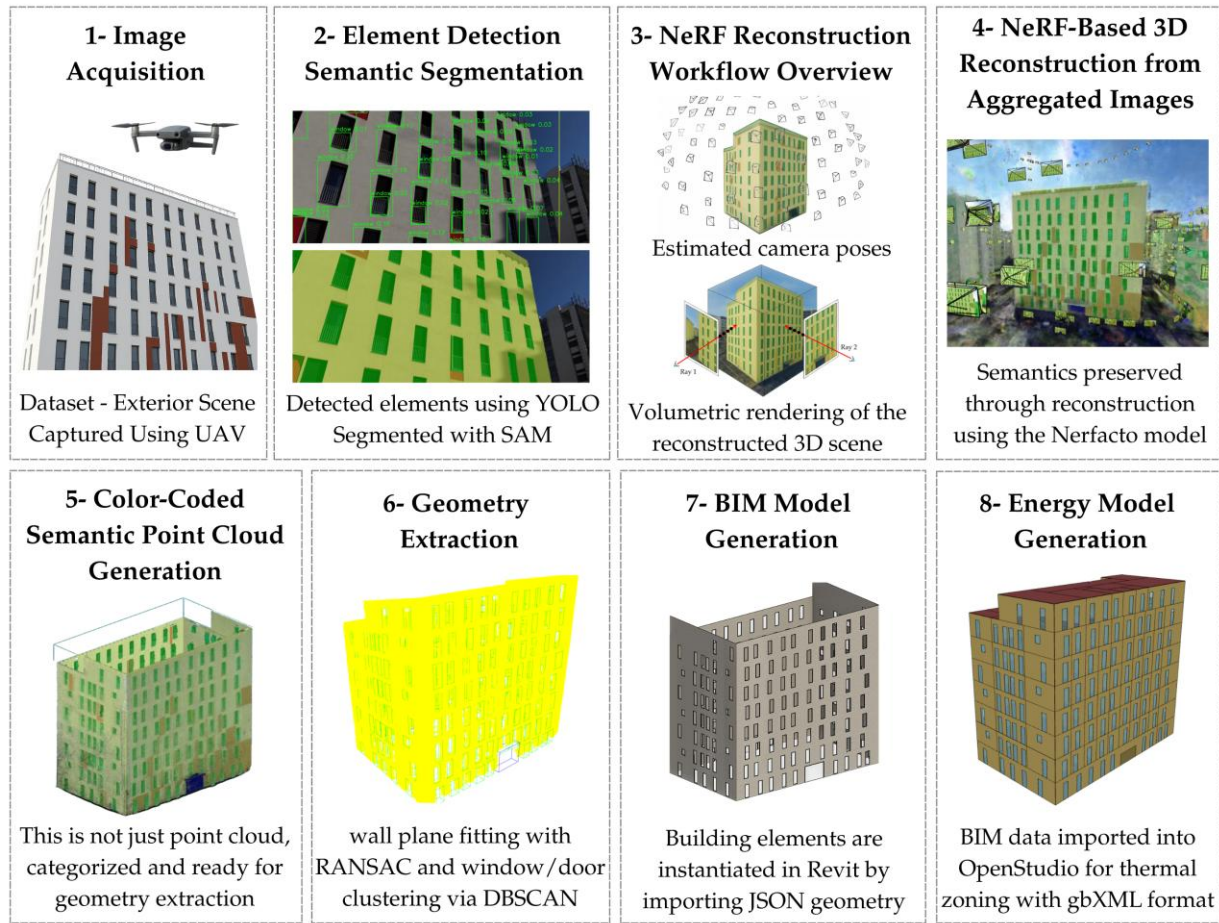


Figure 17. Image-to-BIM workflow for generating building information model used as basis of the Digital Twin [173].

5.9.2. IoT Sensor Deployment and Digital Twin Platform Development

The second step deploys an IoT sensor network and establishes a Digital Twin platform to enable real-time, spatially explicit monitoring of damp and mould-related processes. To capture the multifactorial drivers of mould growth identified in the review, the sensor system should cover eight categories: (1) energy use: whole-building and circuit-level meters identify conditions associated with fuel poverty and underheating; (2) indoor thermal comfort: sensors measuring indoor AT and ARH across thermal zones; (3) indoor air quality: CO₂ and PM_{2.5}/PM₁₀ to assess ventilation and airborne contamination; (4) VOC patterns: an electronic nose to capture TVOC signatures associated with microbial activity; (5) building envelope: ST, SRH, and heat flux at thermally vulnerable locations identified via thermal imaging, enabling modeling of the non-linear ARH-to-SRH relationship; (6) occupancy: inferred non-intrusively from motion, ambient temperature, and luminosity; (7) natural ventilation: contact sensors on windows and doors; and (8) outdoor environment: AT, ARH, CO₂, PM, and solar radiation to contextualize indoor conditions.

Once commissioned, all sensor streams are integrated into the Digital Twin platform, which links real-time time-series data to the spatial and semantic structure of the BIM model (Figure 18). Each sensor is mapped to specific rooms, surfaces, and envelope elements, enabling spatially resolved visualization of environmental conditions. The platform synchronizes heterogeneous streams in time, manages data quality, and provides a unified interface for monitoring, analysis, and model deployment, forming the operational backbone for the physics-informed AI modeling described in the following section.

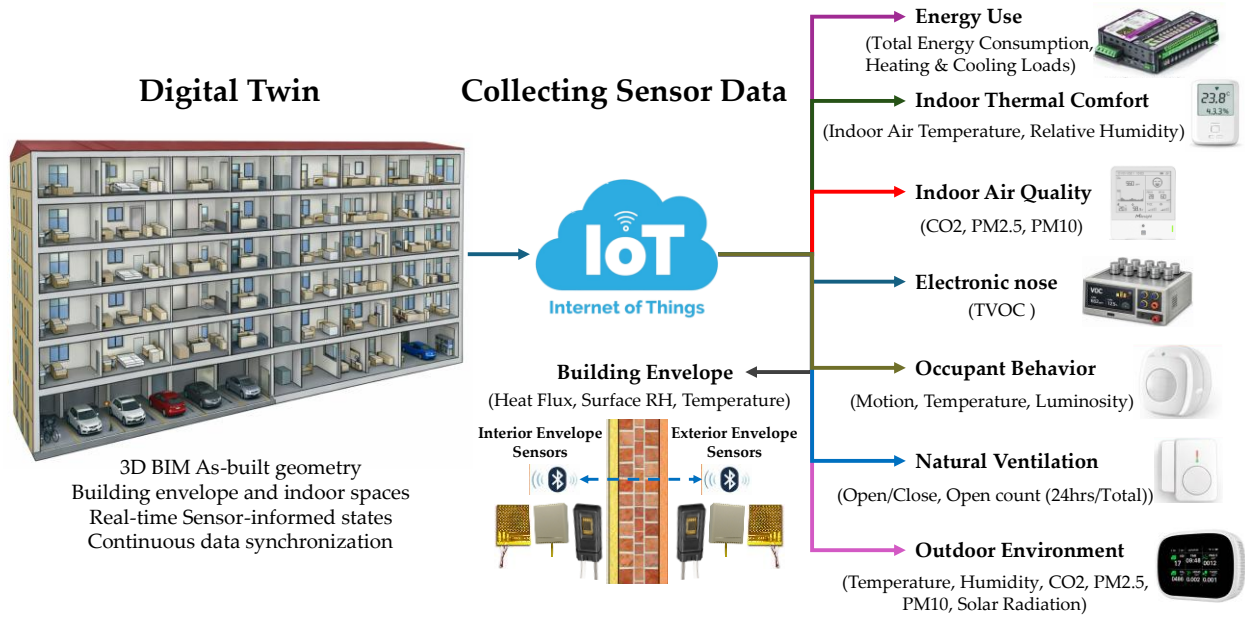


Figure 18. IoT sensor deployment and Digital Twin data integration for real-time mould risk monitoring.

5.9.3. Physics-Informed AI-Based Time-Series Mould Growth Prediction and Validation

Once the Digital Twin platform is operational, building envelope sensors provide time-series readings of $ST(t)$ and $SRH(t)$, the fundamental physical drivers in the VTT model. Mould growth is treated as a progressive, time-dependent process and evaluated over rolling weekly intervals: the mould index is updated continuously, and its end-of-week value is recorded as the representative mould level for that period.

Predicted values are interpreted against established VTT thresholds (e.g., $M = 5$ corresponds to visually detectable mould coverage exceeding 50% of the surface). At the end of each weekly interval, predictions would be validated primarily through surface photographs submitted by the occupants themselves through the application described in Section 5.9.4, supplemented by on-site visual inspection where professional confirmation is required. Because residents photograph the flagged surfaces and decide what to share, validation evidence can be collected without anyone entering the home, which protects household privacy and lowers inspection cost. For lower mould index values ($M = 1-3$), where growth is not yet visible to the naked eye, microscopic inspection is performed to assess early-stage fungal development. Air sampling is conducted in parallel, since surface colonization does not necessarily translate into measurable airborne dissemination. Holme et al. [174] reported no statistically significant relationship between visible surface mould and airborne fungal levels in a large-scale study of approximately 400 Swedish dwellings, confirming that visual surveys alone cannot reliably assess indoor fungal burden [175].

The combined outcomes of occupant-submitted images, visual inspection, microscopic analysis, and air sampling would provide empirical evidence that could be compared directly with Digital Twin predictions. To address expected discrepancies under real operational conditions and improve predictive performance, the framework incorporates a PIML model. The VTT mould growth formulation is retained as the physically grounded baseline governing equation, while additional Digital Twin data streams enhance adaptability (Figure 19).

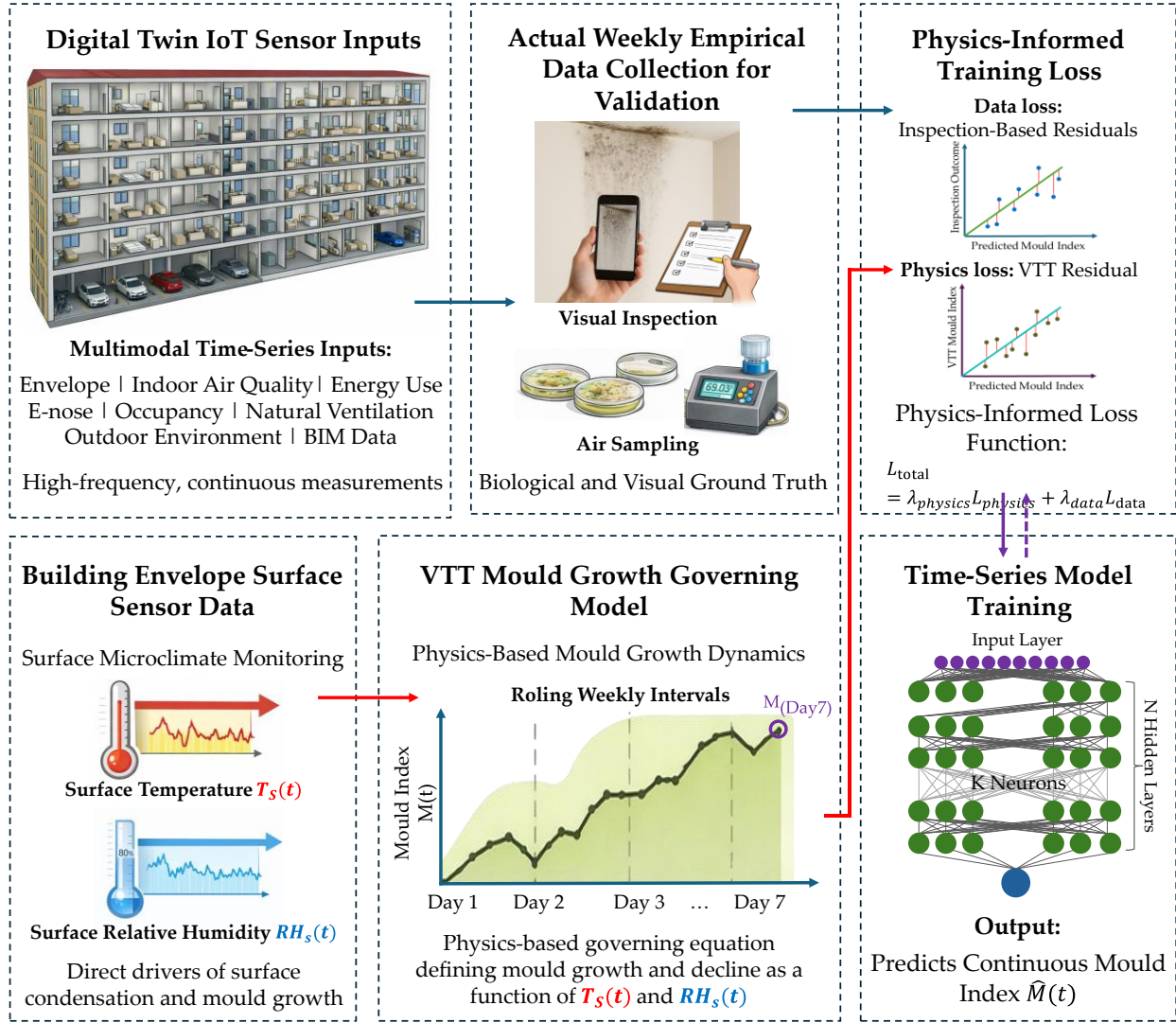


Figure 19. Physics-informed time-series learning framework for mould growth prediction.

The learning model receives dynamic sensor inputs (ST, SRH, indoor AT/ARH, IAQ indicators, VOC patterns, ventilation behavior, occupancy signals, energy use, and outdoor conditions), static BIM-derived attributes (room geometry, envelope materials, construction properties, surface location), and previous mould state, and predicts the mould index trajectory $\hat{M}(t)$ for each monitored surface on the same 0–6 scale as the VTT model.

Physical consistency is enforced by embedding the VTT governing dynamics into the learning process through a physics-informed loss function. In the VTT formulation, mould development is governed by $ST(t)$ and $SRH(t)$, and its temporal evolution can be expressed as a first-order dynamical system, as given in Equation (1), where $f_{VTT}()$ represents the VTT-defined growth or decline rate depending on whether surface hygrothermal conditions exceed the critical humidity threshold. The detailed VTT growth and decline equations defining $f_{VTT}()$ are provided in Appendix A.

$$\frac{dM}{dt} = f_{VTT}(M, ST, SRH) \quad (1)$$

Instead of solving Equation (1) explicitly, it is used to regularize training by constraining the predicted temporal evolution. Deviations from VTT-consistent evolution are penalized through an L2-based physics residual loss, defined in Equation (2), in parallel with a data-driven loss anchoring predictions to empirical

observations from occupant submissions and building inspections defined in Equation (3). The total training objective is a weighted sum of the two terms (Equation (4)), where $\lambda_{physics}$ and λ_{data} control the trade-off between physical consistency and empirical fit. Additional Digital Twin sensor streams that are not embedded in the governing equations enter through the data-driven term, while the physics-informed loss constrains predicted mould index dynamics to remain biologically and hygrothermally plausible.

$$L_{physics} = \frac{1}{N} \sum_{i=1}^N ([\hat{M}(t_i + \Delta t) - \hat{M}(t_i)] - \Delta t f_{VTT}(\hat{M}(t_i), ST(t_i), SRH(t_i)))^2 \quad (2)$$

$$L_{data} = \frac{1}{K} \sum_{n=1}^K (\hat{M}(t_n) - y(n))^2 \quad (3)$$

$$L_{total} = \lambda_{physics} L_{physics} + \lambda_{data} L_{data} \quad (4)$$

Through this integrated formulation, the framework is designed to move from retrospective mould risk assessment toward proactive, time-resolved prediction. Once deployed and validated, the resulting physics-informed, AI-enabled Digital Twin would provide spatially and temporally explicit mould risk trajectories at room and surface levels, with the potential to support early warning, targeted inspection, and informed decision-making for mould prevention and management.

5.9.4. Occupant-Facing Application for Real-Time Prediction, Explanation, and Feedback

The final component turns the framework outputs into a two-way service for the people who live in the building. Once trained and validated, the PIML model serves as the computational backend of a web-based application accessible from a phone or computer. The Digital Twin passes each new sensing interval through the model, so the application displays the current mould index and its predicted weekly trajectory for every monitored room and surface in real time, to both occupants and housing providers.

In the other direction, occupants use the same application to submit photographs and short condition reports of their own surfaces. These submissions supply the inspection-based observations required by the data-driven loss term in Equation (3), so the model continues to learn from real conditions after deployment. Because residents take the photographs themselves and control what is shared, no third party needs to enter the home, and the feedback loop operates at a fraction of the cost of scheduled professional visits.

A large language model (LLM) is integrated as the interaction layer of the application. The LLM retrieves the relevant sensor records, model predictions, and building context before answering, and lets occupants ask in plain language about the current risk level, its likely causes, and its recent trend. Whereas the LLM-based system reviewed in Section 5.6 [156] answers questions over past measurements only, here the LLM communicates forward-looking predictions produced by the physics-informed model, placing a predictive engine inside the integration shell that Sections 5.6 and 5.7 identified as empty.

To make each prediction explainable, SHapley Additive exPlanations (SHAP) [176] quantify how much every input feature, such as AT, ARH, SRH, ventilation behavior, occupancy-derived moisture generation, envelope properties and other features, contributes to the predicted mould index of a given surface. The LLM translates these attributions into plain-language insight and practical recommendations delivered through the application, for example increasing ventilation after moisture-generating activities, raising the heating setpoint in a specific room, or reporting a suspected leak for repair. The application therefore closes the loop of the framework: it communicates risk before growth becomes visible, explains

which conditions drive that risk, and collects the occupant evidence that keeps the underlying model calibrated.

5.9.5. Implementation Challenges

Translating the proposed framework to real-world deployment depends on resolving practical challenges, notably cost, data quality, and privacy, which are recurring barriers to Digital Twin implementation in facility management [177]. Five principal challenges and their corresponding solution pathways are outlined below.

First, BIM information accuracy. Predictive performance depends on accurate BIM geometry and material properties, which are often unavailable for existing residential buildings. The Image-to-BIM workflow described in Section 5.9.1 partially addresses this through automated 3D reconstruction [173], although manual refinement is typically required. A practical solution is to build the BIM model in stages: first the geometry of rooms and surfaces, which is required to anchor each sensor to a specific location, then the material and thermal properties, which are populated progressively from inspection records, archive drawings, and standard housing-typology databases. This staged approach allows the framework to begin operating with partial BIM information and to refine its predictions as additional data become available.

Second, cost of large-scale IoT deployment. Dense sensor networks across entire housing portfolios are not economically feasible. Cost can be reduced through risk-prioritized sensor placement using preliminary thermal imaging to identify thermal bridges and condensation-prone surfaces, hybrid use of low-cost sensors with periodic cross-calibration against reference instruments, and transfer learning across dwellings of similar typology, which has been shown to effectively address data-scarcity challenges in building applications, enabling models trained on data-rich reference buildings to be adapted to diverse new buildings and climates with as little as 1% of target-domain data [178].

Third, data privacy in occupancy sensing. Prior to any data collection, ethical approval should be obtained from the relevant institutional ethics committee, and informed consent must be secured from all building occupants, with the option for residents to opt out of specific sensor categories or withdraw at any time. The framework deliberately avoids cameras, microphones, and personally identifiable data. Occupancy is inferred from non-intrusive indirect indicators (motion, ambient temperature, luminosity, CO₂), and window-opening behavior is captured through binary contact sensors. Occupant-submitted images follow the same principle: photographs enter the system only when a resident chooses to take and submit them through the application, and no camera is ever installed in the dwelling. Data are processed locally or aggregated at building level before transmission, in line with privacy-by-design principles and applicable data protection regulations.

Fourth, sensor calibration and drift. Low-cost sensors exhibit measurement drift over time, particularly for RH, gas, and PM sensors. This is addressed through periodic cross-calibration against reference-grade instruments during maintenance visits, redundancy across co-located sensors for outlier detection within the Digital Twin data-quality layer, and the inherent robustness of the PIML architecture, which anchors predictions to physics-based VTT dynamics instead of raw sensor signals alone.

Fifth, inspection-based validation at scale. Universal application of visual inspection, microscopic analysis, and air sampling is not feasible across entire housing portfolios, and it is also not required by the framework. The purpose of inspection-based validation is to verify and, where necessary, refine the VTT mould index predictions through physics-informed data-driven calibration; once this calibration objective is achieved for a representative range of building typologies and surface conditions, the validation effort

does not need to be repeated at scale across the wider portfolio. In practical deployment, validation is therefore applied selectively to dwellings and surfaces flagged as high-risk by the predictive model, integrating naturally with periodic housing inspection schedules required under emerging social housing regulations such as Awaab's Law. Within this selective regime, the occupant-submitted images described in Section 5.9.4 serve as the first line of evidence, flowing in continuously between scheduled visits, with professional visual inspection triggered when submitted images or model predictions indicate elevated risk. Microscopic analysis and air sampling are reserved as secondary methods, applied where feasible to a smaller subset of longitudinal validation sites to provide ground truth for early-stage growth assessment and continuous model refinement, without imposing intensive sampling at portfolio scale.

5.10. Limitations of the Review

Although this review followed a structured PRISMA-guided process, the broad and interdisciplinary nature of damp and mould prediction means that relevant studies may be indexed using different terminology across building physics, indoor air quality, microbiology, computer vision, and AI research. As detailed in Section 3.3, this risk was mitigated by incorporating broad umbrella terms and supplementing the database search with citation searching. Nevertheless, the possibility that some relevant studies fell outside the network of the retrieved studies cannot be entirely excluded. This limitation may be particularly relevant for studies that used highly specific algorithm names, domain-specific hygrothermal terminology, or related indoor air quality terms without explicitly referring to AI, machine learning, damp, or mould. In addition, the search was limited to Scopus and WoS and to English-language journal articles, which may have excluded relevant studies indexed elsewhere. The synthesis should be interpreted with this consideration in mind and as a systematic and transparent review of the identifiable AI-based literature within the defined search boundaries, not as an exhaustive account of every potentially relevant study.

A further consideration extends beyond the review process to the proposed framework itself. The framework is presented as a conceptual architecture that synthesizes the directions emerging from the reviewed literature into a coherent structure for future research. It has not yet been implemented and its development and evaluation in occupied buildings represent the primary direction for future work.

6. Conclusion

This review has examined the state of AI driven approaches for detecting, predicting, and assessing dampness and mould in buildings, synthesizing evidence from 79 studies across six methodological categories: AI-based imaging spanning surface computer vision and sub-surface electrical impedance tomography, sensor- and simulation-driven hygrothermal modeling, airborne chemical and physical indicators, data-driven housing and inspection risk models, material susceptibility and bioreceptivity, and integrated multimodal decision support. While recent studies demonstrate that AI can achieve high predictive accuracy within individual data domains, the review reveals that most existing approaches remain fragmented, reactive, and insufficiently grounded in the physical mechanisms governing fungal growth: only five of the 79 studies forecast with an explicit lead time, none predicts surface mould that is subsequently observed, and no study unites surface-level prediction, a future horizon, and operational deployment.

A consistent limitation across the literature is the weak coupling between room-scale environmental measurements and surface-level mould risk, alongside the limited validation and calibration of mould

index predictions against real fungal observations in buildings under occupancy. Many studies rely on static thresholds, single-modality inputs, or laboratory-controlled datasets, which restrict their applicability to occupied buildings with heterogeneous usage patterns, ventilation behavior, and material assemblies. In addition, the majority of reviewed works treat AI models as standalone predictors, with no integration into building information models or operational Digital Twin environments capable of supporting continuous monitoring and decision making.

To address these gaps, this paper articulates a physics informed, AI integrated Digital Twin framework that unifies BIM-based building context, multi-parameter IoT sensing, mould growth dynamics, and an occupant-facing application layer within a single operational platform. By embedding hygrothermal constraints through physics-informed loss functions and anchoring predictions to observed mould conditions collected through occupant-submitted photographs and targeted professional inspection, the framework is designed to move beyond black-box prediction toward physically consistent, spatially and temporally resolved mould risk assessment. Through the application layer, the trained physics-informed model runs as a real-time backend, an integrated LLM allows occupants to query current and predicted mould risk in plain language, and SHAP-based feature attributions are translated into practical recommendations, so predictions reach the people who can act on them while their photographic feedback keeps the model calibrated. This integration is intended to support continuous monitoring, closed-loop model updating, and spatially resolved decision support, reframing mould prediction as an operational Digital Twin capability instead of a retrospective analytical task. The proposed approach therefore provides a pathway toward proactive detection, surface-specific risk mapping, and adaptive intervention strategies, with the longer-term aim of supporting scalable deployment in occupied residential buildings and bridging the gap between academic model development and real-world damp and mould management.

A key contribution of this work lies in explicitly addressing the persistent validation and calibration limitations identified across the reviewed literature. In many studies, mould risk is determined directly from calculated model outputs, while only limited effort is made to align these predictions with observed mould conditions in real buildings, and the underlying model parameters are seldom updated using monitored field data despite being originally derived from controlled laboratory experiments or theoretical assumptions. By positioning occupant-supported, inspection-anchored validation and data-driven calibration as integral components of the proposed framework, this review clarifies the need for a shift toward systematically verifiable and operationally credible mould prediction models suitable for deployment in real buildings.

The proposed framework outlines a future research direction by emphasizing time-resolved and surface-specific mould risk trajectories, in place of static, room-averaged, or threshold-based assessments. This temporal and spatial resolution is intended to support early detection of emerging dampness and mould conditions before visible manifestation, and to enable preventive intervention in place of post-damage identification.

From a methodological perspective, this work introduces physics-informed neural networks as a foundational element for mould risk prediction, a direction not applied to mould prediction in any of the reviewed studies. By incorporating hygrothermal theory and mould growth formulations directly into the AI learning process through physics-informed loss functions, the framework is designed to constrain AI predictions to physically plausible regimes and to enhance robustness, interpretability, and generalizability beyond purely data-driven approaches.

Future research should prioritize pilot deployment of the proposed framework in occupied residential buildings to enable quantitative validation under real operational conditions, alongside multimodal data fusion, long-term in-situ validation across diverse housing typologies, and integration with regulatory and asset management workflows, particularly in the context of emerging housing quality standards such as Awaab's Law. By aligning AI methods with building physics and Digital Twin paradigms, the field can shift from post-damage identification toward scalable, preventative mould management that protects occupant health and housing performance.

Abbreviations

AI	Artificial Intelligence	MDS	Multidimensional Scaling
ANN	Artificial Neural Network	ML	Machine Learning
ARH	Air Relative Humidity	MLP	Multilayer Perceptron
AT	Air Temperature	MVOCs	Microbial Volatile Organic Compounds
AUC	Area Under the Curve	NeRF	Neural Radiance Fields
AutoML	Automated Machine Learning	PCA	Principal Component Analysis
BiLSTM	Bidirectional Long Short-Term Memory	PICO	Population–Intervention–Comparison–Outcome
BIM	Building Information Modeling	PIML	Physics-Informed Machine Learning
CAM	Class Activation Map	PLS	Partial Least Squares
CFU	Colony Forming Unit	PM	Particulate Matter
CNN	Convolutional Neural Network	PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
CO ₂	Carbon Dioxide	ResNet	Residual Network
DL	Deep Learning	RGB	Red, Green, Blue (image data)
EIT	Electrical Impedance Tomography	RH	Relative Humidity
FCN	Fully Convolutional Network	RMSE	Root Mean Square Error
GAM	Generalized Additive Model	RMSEP	Root Mean Square Error of Prediction
GBDT	Gradient Boosted Decision Trees	RNN	Recurrent Neural Network
GBM	Gradient Boosting Machine	R ²	Coefficient of Determination
gbXML	Green Building XML	SIMCA	Soft Independent Modeling of Class Analogy
GC-MS	Gas Chromatography coupled with Mass Spectrometry	SOM	Self-Organizing Map
GNN	Graph Neural Network	SRH	Surface Relative Humidity
GRU	Gated Recurrent Unit	SSD	Single Shot Detector
GWO	Grey Wolf Optimizer	ST	Surface Temperature
HTC	Hybrid Task Cascade	SVM	Support Vector Machine
HVAC	Heating, Ventilation, and Air Conditioning	SVR	Support Vector Regression
IAQ	Indoor Air Quality	t-SNE	t-Distributed Stochastic Neighbor Embedding
IoT	Internet of Things	TVOC	Total Volatile Organic Compounds
K-ELM	Kernel Extreme Learning Machine	UAV	Unmanned Aerial Vehicle
KNN	k-Nearest Neighbor	UK	United Kingdom
LASSO	Least Absolute Shrinkage and Selection Operator	VGG	Visual Geometry Group
LDA	Linear Discriminant Analysis	VOCs	Volatile Organic Compounds
LightGBM	Light Gradient Boosting Machine	VTT	Technical Research Centre of Finland (Valtion Teknillinen Tutkimuskeskus)
LS-SVM	Least Squares Support Vector Machine	WA	Water Activity
LSTM	Long Short-Term Memory	WHO	World Health Organization
MAE	Mean Absolute Error	XGBoost	Extreme Gradient Boosting
MAPE	Mean Absolute Percentage Error	YOLO	You Only Look Once

Appendix A

Mould growth is permitted only when SRH exceeds a critical threshold that depends on ST. This threshold represents the minimum moisture level required for biological activity to initiate. The critical RH is defined in Equation (5).

$$RH_{crit}(ST) = \begin{cases} -0.00267ST^3 + 0.160ST^2 - 3.13ST + 100, & ST < 20^\circ C \\ 80, & ST \geq 20^\circ C \end{cases} \quad (5)$$

For non-wood materials, the updated model introduces an additional material-dependent minimum humidity threshold RH_{min} . Mould growth is therefore allowed only when the SRH satisfies the relevant class-based threshold. When hygrothermal conditions are favorable, the mould index increases according to a first-order differential equation that accounts for temperature, humidity, material sensitivity, and growth saturation as demonstrated in Equation (6).

$$\frac{dM}{dt} = \frac{k_1 k_2}{7e^{(-0.68 \log_e ST - 13.9 \log_e SRH + 0.14W - 0.33SQ + 66.02)}} \quad (6)$$

Where W is the reference wood species indicator, SQ represents surface quality and is set to zero for non-wood materials, k_1 is a material-dependent growth intensity factor, k_2 moderates growth as the mould index approaches its upper limit. To prevent unrealistically unlimited growth, the model introduces a maximum attainable mould index M_{max} , shown in Equation (7), which depends on how far SRH exceeds the critical threshold.

$$M_{max} = A + B \left(\frac{RH_{crit} - SRH}{RH_{crit} - 100} \right) - C \left(\frac{RH_{crit} - SRH}{RH_{crit} - 100} \right)^2 \quad (7)$$

The constants A , B , and C depend on the material sensitivity class. The saturation effect is enforced through the moderating term demonstrated in Equation (8), which progressively reduces the growth rate as M approaches M_{max} .

$$k_2 = \max(1 - e^{[2.3(M - M_{max})]}, 0) \quad (8)$$

As outlined in Table A1, in order to extend the original wood-based formulation to general construction materials, the updated VTT model groups materials into four sensitivity classes.

Table A1. Material sensitivity classes and corresponding parameter values used in the VTT mould growth model, including growth coefficients (K_1), empirical constants (A , B , C), minimum relative humidity threshold ($RH_{min}^{\%}$), and material specific mould sensitivity factor (C_{mat}).

Sensitivity Class	Material	K_1 (M<1)	K_1 (M>1)	A	B	C	$RH_{min}^{\%}$	C_{mat}
Very sensible	Untreated wood; includes lots of nutrients for biological growth	1	2	1	7	2	80	1
Sensible	Planed wood; paper-coated products, wood-based boards	0.578	0.386	0.3	6	1	80	0.5
Medium resistant	Cement or plastic based materials, mineral fibers	0.072	0.097	0	5	1.5	85	0.25
Resistant	Glass and metal products, materials with efficient protective compound treatments	0.033	0.014	0	3	1.5	85	0.1

When surface conditions fall outside the favorable growth regime, mould activity decreases. For the reference wood material, the decline of the mould index is described by a piecewise function based on the duration of unfavorable exposure, as shown in Equation (9). For other materials, the rate of mould index decline is scaled using a material dependent coefficient, as demonstrated in Equation (10).

$$\frac{dM}{dt} = \begin{cases} -0.00133, & \Delta t \leq 6h \\ 0, & 6 < \Delta t \leq 24h \\ -0.000667, & \Delta t > 24h \end{cases} \quad (9)$$

$$\left(\frac{dM}{dt}\right)_{mat} = C_{mat} \left(\frac{dM}{dt}\right)_{wood} \quad (10)$$

The mould index $M(t)$ is obtained by numerically integrating the governing equations over time using the same temporal resolution as the hygrothermal input data.

Appendix B

Table B1. Summary of the primary aims and application focus of the studies evaluated in the systematic review.

Study	Aim of the Study	Data Type	Dataset Size	Feature Domain	AI Model	Target Output	Metrics Reported
1- Klosowski et al. (2026) [75]	Solve the ill-posed EIT inverse problem for moisture imaging in historic masonry with a multi-head network that suppresses over-smoothing and artifacts, verified in situ.	Synthetic FEM voltage data + real EIT measurements at a 14th-century church; dielectric and IRT reference	49,000 training and 1,000 validation cases; 448 voltages in, 3,050 FEM elements out; no separate test set	Ordered 448-element inter-electrode voltage sequence; output element-wise conductivity (dry 1, wet 10)	Multi-head fully connected NN (122 heads) for Image Reconstruction Regression; single-head dense baseline	Element-wise conductivity field rendered as 2D/3D moisture tomograms	Multi-head vs baseline: MSE 0.093 vs 0.205; PSNR 10.74 vs 7.34 dB; SSIM 0.850 vs 0.769; in-situ agreement qualitative only
2- Bektas Ekici et al. (2026) [90]	Introduce a six-class heritage surface deterioration dataset and a six-stage ablation framework quantifying the gains from common deep learning components, including biological growth.	RGB images of historic building surfaces (adobe, brick, stone, timber); released on IEEE DataPort	23,688 images across six classes, augmented to 39,340 then split ~58/31/11; augmentation before splitting leaks data	Visual texture, color and morphology learned end to end; ArcFace embeddings, MixUp/CutMix augmentation	Six-stage ablation on EfficientNet-B0 distilled into ConvNeXtV2-Tiny for Multi-Class Image Classification	Single class label per image: biological, chemical, cracks, human-induced, material loss or undamaged	A6 accuracy 0.9964, macro-F1 0.9966; biological F1 0.993; figures inflated by pre-split augmentation leakage
3- Bourgeois et al. (2026) [97]	Test how training dataset composition affects efflorescence segmentation on limestone masonry across eight CNNs, then deploy the chosen model to orthophoto mapping at the Monastery of Batalha.	RGB DSLR and drone imagery of limestone masonry + three public efflorescence datasets; SAM-assisted masks	100 own + 1,211 Kbell + 96 S2DS + 120 Efflore images at 512 x 512; four compositions (T1-T4)	Visual color, texture and morphology of salt deposits; geometric and photometric augmentation; ImageNet backbones	Eight CNN architectures (DeepLabV3+, HRNet, U-Net variants) for binary Semantic Segmentation	Pixel-level efflorescence masks aggregated into orthophoto-scale probability/area maps	Boundary IoU only: DeepLabV3+ up to 0.915; deployed HRNet-T4 (BIoU 0.762) chosen qualitatively, field output not quantified
4- Bayat Pour et al. (2026) [119]	Quantify how extruded mortar joints affect mould risk on timber studs behind brick veneer across four orientations, comparing ASHRAE and experimental water penetration criteria.	Delphin 1D hygrothermal simulations of a brick-veneer timber frame wall; SMHI Gothenburg climate 2018-2023	Eight case studies; 1,000-2,300 simulations each; 70/30 train/test; metamodel then predicts 10 ⁷ LHS scenarios	Extruded mortar depth, cavity air change rate, insulation conductivity, WDR and solar coefficients, material properties	Random Forests metamodel per case study for Tabular Regression, within Mould Reliability Analysis and the VTT model	Maximum mould index per scenario; 95th percentile per case study; ranked input sensitivity	Test R2 0.90-0.95 across all eight metamodels; 95th-percentile MMI 3.24-5.42; ES criterion lowers MMI by 7-15%
5- Arslan et al. (2026) [156]	Develop and validate an agentic RAG + LLM decision-support system (IEIS-DMP) integrating sensors, BIM and diagnostic reports for room-level damp and mould risk and remediation advice.	Indoor T/RH time series, BIM/IFC spatial model, expert diagnostic reports	Four sensor locations in one dwelling, five months at 1 s intervals; evaluated on 50 diagnostic questions	T/RH trends and threshold breaches; BIM geometry and zoning; expert observations of mould, ingress, odor	Gemma 3 (4B) LLM used zero-shot in an Agentic RAG architecture; standard single-pass RAG baseline	Natural-language damp/mould risk determinations with evidence and remediation recommendations	Completeness 95.2%, Accuracy 94.6% over 50 questions; standard RAG 82.4% / 80.1%

6- Barbosa et al. (2026) [142]	Validate a low-cost modular multisensor prototype detecting airborne fungal contamination indirectly through environmental signatures, using P. chrysogenum in a controlled chamber.	Chamber multisensor time series (PM, MOX gas, thermal array) at 2 s; SAS impaction CFU counts as ground truth	1,989 observations in four exposure classes, each recorded on a separate date; 80/20 splits over 50 seeds	15 variables: PM fractions, particle counts, gas resistance, NO ₂ , O ₃ , AQI, T, RH, pressure, thermal features	Extra Trees, SVM and SGD for Binary and Multi-Class Classification with ANOVA feature selection	Contamination class per 2 s observation (control vs cultures) as an indirect spore-presence indicator	Accuracy/AUC/F1 ~100% in nearly all tasks; pressure alone gave 100% in 5 of 6 pairs, suggesting session effects
7- Choi et al. (2026) [138]	Develop a MOF/MWCNT gas sensor array with ML for real-time detection of fungal MVOCs: single-VOC classification, mixture concentrations and fungal vs control gas.	Lab gas-chamber responses from a three-sensor array (QCM, resistive, EIS) to VOCs and fungal culture headspace	200 single-gas, 65 mixture and 177 fungal-gas samples; 70:30 or 50:50 splits with repeated 10-fold CV	Nine sensor features (normalized response, response and recovery times per channel); PCA before fungal classification	Random Forest for Multi-Class VOC Classification; CatBoost for mixture Regression and fungal/control Classification	VOC class; component concentrations (ppm) in mixtures; binary fungal vs control gas label	Single-gas RF accuracy 0.95; mixture R ² 0.59-0.93; fungal vs control accuracy 0.963 (54 test samples)
8- Hu et al. (2026) [76]	Combine hyperspectral imaging and XGBoost for interpretable detection and 3D mapping of bio-colonisation on heritage facades under hard visual interference, quantifying gains over RGB.	UAV RGB and VIS-NIR hyperspectral imagery of the Castle of Horst projected onto a 3D point cloud; open data and code	67,197 annotated pixels (27,068 bio-colonisation; 7,964 hard negatives); 80/20 split with 5-fold CV	3 RGB channels, 31 reflectance bands (455-835 nm) and their fusion; SHAP highlights the chlorophyll red edge	XGBoost binary classifier in six configurations (RGB, HSI, fusion; with/without hard negatives) for pixel Classification	Binary bio-colonisation label per pixel/point, deployed as a semantic 3D facade map	Best fusion with hard negatives: F1 99.4%, FPR 1.9%; RGB-clean FPR 73.4%; facade deployment qualitative only
9- Mansouri et al. [85] (2025)	Adapt YOLOv9 for automatic detection of visible and hidden mould regions on walls, addressing challenges of irregular shape and background camouflage.	RGB images from household disrepair claims	2,359 annotated images (2,063 train, 196 validation, 100 test)	Visual texture and spatial distribution of mould regions vs stains and surface defects	YOLOv9 (proposed adoption); YOLOv8 and YOLOv10 for Object Detection	Bounding boxes for mould regions	YOLOv9 avg classification 76%; mAP@0.5 0.537; balanced precision/recall ~50%
10- Aziz & Hardy [148] (2025)	Develop a predictive, explainable ML model to identify English homes at risk of developing damp, supporting a shift from reactive to proactive damp management.	Inspection records + Energy Performance Certificate (EPC) data	2,073 housing records; 1,630 damp and 443 non-damp	13 variables covering building characteristics and energy efficiency (cost, consumption, efficiency, EPC)	ANN, Decision Tree, XGBoost, Random Forest, SVM, Logistic Regression, KNN for Binary Classification	Has Damp: binary (1 = damp / 0 = non-damp)	Random Forest: accuracy 0.793; precision 0.533; recall 0.242; F 0.333; AUC 0.716
11- Chu et al. [81] (2025)	Develop an automated CNN-based system using infrared thermal imagery to detect moisture in lime-plastered brick walls of heritage buildings, replacing manual expert interpretation.	Paired visible and infrared thermal images (FLIR SC660) of heritage walls in Taiwan	4,000 thermal images; 80/20 train/test; +/-10 deg rotation augmentation	Raw pixel data from preprocessed thermal images (alignment, masking, segmentation)	AlexNet CNN (selected over GoogLeNet and InceptionV3) for Binary Image Classification	Binary classification: moisture vs no moisture per patch	Accuracy 97.26%; precision 97.2%; recall 97.0%; F1 97.2%

12- Baudrit et al. (2025) [152]	Predict mass loss of Guyanese wood-fibre insulation boards from fungal and termite attack using panel chemistry and physical properties, with SHAP interpretation.	Lab durability tests (EN 113-3 fungal, modified EN 118 termite) + panel chemical composition and density	10 panel types; 60 fungal and 30 per-termite-species observations; 5-fold CV, no independent test set	Holocellulose-to-lignin ratio, extractives content, panel density, moisture content (fungal target only)	Twelve regressors compared (tree ensembles, boosting, SVM, KNN, MLP) for Tabular Regression, with SHAP	Anhydrous mass loss (%) from white rot and two termite species (three regression targets)	Fungal R2 0.991 (CatBoost), MAE 0.344%; R. flavipes R2 0.885; C. dudleyi R2 0.740
13- Luo et al. [86] (2025)	Develop an automatic DL detection method for efflorescent areas on historical brick walls, applied to Wuhan Gao's Hospital to analyze moisture-driven causes of efflorescence.	RGB images of brick wall efflorescence with annotated bounding boxes	1,986 images: 1,311 training, 536 validation, 139 testing	Texture, color, brightness and shape of white crystalline efflorescence vs surrounding brick	YOLOv10n for single-class Object Detection of efflorescence on brick walls	Bounding boxes with confidence scores for efflorescence	Precision 90.7%; recall 86.5%; mAP@0.5 89.8%; F1 0.83 at 0.31 confidence
14- Chen et al. [82] (2025)	Develop a UAV-based DL framework for instance segmentation of pathologies on heritage gray brick masonry, including biological invasion and discoloration/deposits, at the Soochow Hospital site.	UAV-collected RGB facade images (DJI Mavic 2 Pro with Hasselblad camera)	895 annotated images; 721 train / 174 validation	Visual features (texture, color, shape, irregular boundaries); LabelMe instance masks; data augmentation	YOLOv8l-seg (plus YOLOv8s/m, v9e, v10l, v11l variants) for Object Detection and Segmentation	Segmentation masks for material loss, discoloration/deposits, cracks, spalling and biological invasion	YOLOv8l-seg mAP@50: segmentation 51.8%; detection 51.8%
15- Yang et al. [137] (2025)	Develop an e-nose with ML classification to identify two indoor mould species from their volatile emissions for objective, cost-effective IAQ monitoring.	Lab e-nose measurements of mould cultures on two substrates (agar + gypsum, agar + wheat)	36,595 labeled samples downsampled to 2,107 per class (7 classes); only 8 measurements per species; interpolate-then-shuffle split shares windows	Resistance signals from UV-activated SnO2 nanowire chemiresistive sensors responding to MVOCs	LDA in four configurations; best is an ensemble of 120 LDAs feeding a softmax regressor, with decision-boundary novelty detection	Mould species and substrate class, plus novelty detection for unknown samples	Average F1: LDA-0 83.7%, LDA-W 98.1%, LDA-SR 98.4%; novelty detection untested (no novel test samples)
16- Suchorab et al. [59] (2025)	Demonstrate the feasibility of e-nose coupled with AI models to estimate fungal MVOC concentrations in indoor air, supporting rapid detection of mycological contamination.	GC-MS concentrations + MOS sensor resistance signals	12 rooms tested; 8 chemicals per room; 17 sensors; 15-minute sampling	17 MOS sensors' relative resistance values; GC-MS concentrations (ug/cm3) as ground truth	Multilayer Perceptron (MLP) and Support Vector Machine (SVM) for Tabular Regression	Concentration of 8 fungal MVOC markers in indoor environments	SVM R2 0.975, RMSE 0.004, MAE 0.003 ug/cm3; MLP R2 0.758
17- Chen et al. (2025) [87]	Build an end-to-end shedthin tile workflow: instance segmentation of four pathologies including water stains, a severity index (STPI) and inpainting-based restoration previews.	RGB mobile-phone photos of tile soffits in ten Suzhou classical gardens; stitched roof panoramas at application	1,250 annotated photos with 26,162 labels; 905 train / 227 validation; 566-photo manual verification set	Visual color, tone, texture and morphology of four pathology classes; Ultralytics online augmentation	Improved YOLO11-seg (C3k2, C2PSA attention) for Instance Segmentation; YOLO and Mask R-CNN benchmarks	Instance masks for water stains, color aberration, surface scaling, excessive gaps; heat maps, area ratios, STPI	Validation mask mAP50 0.498 (water stains 0.436); manual verification 74.4% overall, water stains 79.3%

18- Meng et al. [118] (2025)	Build a model that uses a short sensor phase and outdoor weather to predict future in-wall temperature and RH, then assess mould risk.	In-wall sensor monitoring for Temp and RH; outdoor ERA5 climate data	Two years of sensor monitoring; 96 sensors on 48 wall spots (2 per wall)	Outdoor Temp, RH, air pressure, wind-driven rain	Hybrid 1D-CNN + LSTM for Time Series Regression	In-wall Temp and RH; post-processed VTT Mould Index	Temp R2 0.952, RMSE 2.134; RH R2 0.805, RMSE 3.419; VTT 97.9% no risk; max MI 1.06
19- Lee et al. [146] (2025)	Develop and compare seven ML models predicting indoor airborne fungal concentrations in public facilities from environmental variables, with SHAP identifying key drivers.	Indoor air sampling in 7 public facility types (South Korea); fungi quantified by qPCR	137 samples (Aug 2021 - Jul 2024); 80/20 train/test split	Facility type, floor, month, Temp, RH, coarse PM2.5-10, 2-day accumulated precipitation	Gradient Boosting, XGBoost, ENR, KNN, SVR, DT, RF for Tabular Regression	Indoor airborne fungal concentration, log(copy number/m3 + 1)	Best model GB: R2 0.78; RMSE 0.53; MAE 0.42 on test set
20- Santos et al. (2025) [88]	Compare deep learning segmentation architectures for mapping dark moisture stains on flat roofs from drone imagery, supporting automated inspection and preventive maintenance.	RGB drone images (DJI Mavic Pro) of waterproofed flat roofs at nine buildings, Brasilia	126 stained images with 2,501 polygon annotations; 10-fold CV at 70/20/10; resized 512 x 512	Color, tone and texture of dark persistent stains; HSV preprocessing; flip, brightness, flare, fog, rotation augmentation	FCN-ResNet50, DeepLabV3-ResNet101 and SegFormer MiT-B1 in 27 optimizer/LR configurations for binary Semantic Segmentation	Pixel-level dark moisture-stain masks across the flat roof surface	Best FCN with Adagrad 1e-2: precision 79.7%, recall 67.8%, F-score 73.1%, IoU 57.7%
21- Pakdehi et al. [149] (2025)	Develop a quantitative ML model predicting indoor mould spore counts in residential buildings after hurricane flooding, integrating flood, building and occupant characteristics.	Indoor air samples; building inspections and flood-depth hindcasting after Hurricanes Ida and Ian	60 homes (Louisiana 25, Northeast 8, South Florida 27); 70/30 train/test split	Maximum flood depth, air tightness, bathroom exhaust fan rate, roof age, window blind status	MLP neural network; compared against RF and XGBoost; SHAP for feature importance	Total indoor mould spore counts (spores/m3)	R2 0.38; RMSE 0.18 spores/m3; NRMSE 33%; SHAP: air tightness most important
22- Jiang et al. [117] (2025)	Introduce a framework combining a predictive model for indoor Temp and RH with NSGA-III to optimize air supply parameters for comfort, energy efficiency and mould prevention.	Sensor monitoring; outdoor data (ERA5-Land)	7,056 samples, hospital office (Apr-Jun 2024)	Temp, RH, CO2; supply/return air Temp, RH, velocity; outdoor Temp, RH	CNN + Transformer for Time Series Regression and Multi-Objective Optimization	Indoor Temp, RH; energy use, thermal comfort and VTT Mould Index in optimization	Temp, RH: R2 0.98-0.99; RMSE 0.045-0.144; VTT: 88% reduction
23- Dulin et al. [150] (2025)	Develop an ML approach predicting mould severity across building portfolios using existing facility management datasets, with SHAP revealing key predictors.	Inspection data + comments + building features + climate data	2,494 buildings in the U.S. Army Corps of Engineers BUILDER SMS dataset	21 variables in HVAC systems, building characteristics and inspection records	LightGBM for Tabular Regression	Mold/moisture severity index (0-1)	R2 0.883; RMSE 0.056; MSE 0.0032
24- Jiang et al. [113] (2025)	Develop AI models predicting spatially non-uniform temperature and RH at ceiling level in healthcare buildings for early detection of mould conditions.	Sensor data + ERA5-Land weather + CFD simulation data	7,056 samples (April-June 2024, 10-min intervals)	Outdoor Temp, RH; supply/return Temp, RH, air velocity	CNN + Transformer and GNN + Transformer for Time Series Regression	Temp and RH at six ceiling locations	CNN+Transformer R2 0.993; GNN+Transformer R2 0.941

25- Zhao et al. [127] (2025)	Develop a model predicting interior wall RH and temperature in light wood-framed walls across China's HSCW region, for regional mould risk assessment via VTT outputs.	Meteorological data + Delphin hygrothermal simulations + monitoring data	200 weather stations; hourly monitoring and simulation (Aug 2022 - Jan 2024)	Outdoor T/RH, indoor T/RH, WDR (ASHRAE method), solar radiation on wall (TMY)	1D-CNN + LSTM hybrid neural network for Time Series Regression	In-wall Temp and RH; VTT Mould Index	Simulated data: RH R2 0.976, Temp R2 0.985; monitoring data: RH R2 0.824, Temp R2 0.914
26- Kim et al. [143] (2025)	Use ML to isolate joint effects of indoor Temp and RH on airborne mould concentrations across 10 facility types, identifying T/RH combinations linked to Caution-Danger levels.	Indoor air samples; bacteria cultured on TSA, mould on PDA	4,048 samples (Aug 2021 - Jul 2023); 70/15/15 train/val/test split	Facility type (label-encoded), indoor temperature, RH	Random Forest regression of concentration thresholded to Safety vs Caution-Danger, inside a 50-seed outlier-removal procedure	Airborne mould concentration (CFU/m3) thresholded at 400 CFU/m3 into Safety vs Caution-Danger	Before refinement: Safety recall 91.9%, Caution-Danger recall 63.4%, balanced accuracy 77.6%, R2 0.43; refined R2 0.638 is circular
27- Fan et al. (2025) [83]	Combine UAV visible and thermal imaging with deep learning segmentation and parametric 3D visualization to detect facade cracks and water seepage on aging Shenzhen residential buildings.	Paired UAV visible and thermal infrared imagery + WHU and public crack datasets for training	900 WHU wall images (720/180); 1,892 crack images (1,514/378); 340 thermal seepage images	Visible texture and edge geometry for cracks; thermal temperature anomalies for seepage; standard augmentation	K-Net with UPerNet decoder for Semantic Segmentation of cracks and seepage; SegFormer for wall extraction	Masks for walls, cracks and seepage projected to per-defect 3D area and centroid in a Grasshopper tool	Crack mIoU 87.9%, Dice 93.2%; seepage mIoU 79.1%;
28- Staffa Junior et al. (2025) [104]	Develop and pilot a web platform integrating drone capture, cloud object detection and checklist reporting for eleven roof pathologies including algae, lichens and mosses.	RGB drone images of fiber-cement roofs on social housing, Salvador, Brazil, plus augmented images	Training 2: 3,404 images from 173 roofs; held-out test 104 images; pilot on 5 projects, 54 roofs	Visual size, color contrast and pattern regularity against fiber-cement tiles; rotation, zoom, flip augmentation	Microsoft Azure Custom Vision object detector (no-code transfer learning) in the Smart Inspects web platform	Bounding boxes for 11 roof pathology classes aggregated into checklist inspection reports	Test accuracy 75.0%, precision 77.1%, recall 71.2%, F1 74.0%; algae/lichens/mosses precision 70.8-73.5%
29- Alexakis et al. [77] (2024)	Develop a DL model capable of automatically detecting rising damp in historical masonries using infrared thermographic (IRT) imagery.	Field data (thermal and visible light images) + annotations	48 thermal images (42 with rising damp, 6 without); 70/20/10 split	Thermal gradients, emissivity, ambient temperature, humidity, material properties	PSP_R50 model for Semantic Segmentation	Binary segmentation masks for rising damp regions on IRT images	Accuracy 0.93; IoU 0.89; F-score 0.88; precision 0.87; recall 0.93
30- Tan et al. (2024) [95]	Integrate computer vision, AR and BIM for defect inspection, detecting and tracking five interior defect classes including mould and seepage in real-time HoloLens video.	RGB defect photos+ internet images; iPhone tracking video; live HoloLens 2 streams in deployment	12,032 detection images (60/20/20); 7,430 tracking frames (6:2:2); field trials with 20 inspectors	Visual texture, color and morphology of five defect classes; DeepSORT appearance and motion features	YOLOv5 for Object Detection + DeepSORT for Multi-Object Tracking, with a BIM-linked management platform	Boxes, classes and track IDs for crack, blister, peel, seepage, mould; defect dimensions in world space	AP@0.5 mould 0.939, seepage 0.931, mAP 0.950; tracking IDF1 0.53-0.67; inspection efficiency gain 78.6%
31- Fu & Angkawisitpan [98] (2024)	Develop a DL framework combining YOLOv5 and Swin Transformer to classify, segment and detect six surface defect categories on heritage masonry, including moss, alkalization and staining.	RGB digital camera images of exterior heritage walls at Gulang Island, China	2,400 sub-images (400 per class, 6 classes); 80/20 split; 550 additional validation sub-images	Visual features (texture, color, edges, defect patterns) learned end to end	Swin Transformer for Semantic Segmentation; benchmarked against FCN, PSPNet, DANet, DeepLabv3+, SegFormer	Segmentation masks for moss, alkalization and staining	Overall mIoU 90.96%, mAcc 95.78%; alkalization 91.0%, moss 94.9%, staining 92.3%

32- Choi et al. [134] (2024)	Develop a reliable approach detecting indoor fungal contamination by identifying MVOCs via mass spectrometry and ML, finding a small set of reliable VOC indicators.	SPME-GC-MS headspace gas from 3 fungi x 3 substrates and 8 indoor VOC sources	261 samples with 848 VOC variables; 70/30 train/test split	GC-MS peak areas, 1,087 VOCs grouped into 19 classes; simplified model uses 5 key VOCs	Random Forest for Binary Classification	Classification label: MVOC or non-MVOC	RF accuracy 100% (full model); simplified RF accuracy 96.2%
33- Bayat Pour & Kahangi Shahreza (2024) [120]	Assess how extruded mortar joints behind clay brick veneer affect mould risk on timber studs under high and low WDR Swedish climates, comparing ASHRAE and experimental criteria.	Delphin 1D hygrothermal simulations (verified against 2D); SMHI climate 2018-2023 for Gothenburg and Rensjon	Eight case studies; 1,200-3,000 simulations each; 70/30 train/test; five-year simulation period	Extruded mortar depth, cavity air change rate, insulation conductivity, WDR/solar coefficients, brick properties; 4-8 selected per case	Random Forests metamodel per case for Tabular Regression, with LHS, Spearman and feature-importance sensitivity; VTT model	Maximum mould index per scenario; 95th percentile per case study; ranked input importance	Test R2 0.85-0.96, MAE 0.026-0.044; 95th-percentile MMI 5.52-5.89 (Gothenburg) vs 0.94-1.25 (Rensjon)
34- Zou & Deng (2024) [89]	Combine an improved YOLOv8-seg model with refined 3D reconstruction to extract, visualize and quantify material deterioration on a masonry tower, including mildew and bio-disease.	RGB deterioration images for training; laser scanning and photogrammetry orthophotos of Huiguang Tower for application	1,468 annotated images (70/20/10) with augmentation; applied to six elevations of a nine-story tower	Visual texture, color, morphology and scale; RFACnv spatial attention and multi-head self-attention	YOLOv8x-seg improved with RFACnv, MHSA and SlideLoss for Instance Segmentation; YOLO benchmarks	Masks for cracking, defect, mildew and bio-disease; per-floor counts and areas on orthographic elevations	F1 0.638, mAP50 0.619 (no per-class results); field mask overlap 95.8%; 66.4% time saving
35- Aggarwal et al. [121] (2024)	Develop a PLS regression metamodel replacing hygrothermal simulations to predict mould index and rank moisture severity of brick-veneer wood-frame walls across Global Warming scenarios and runs.	Simulations + ensemble climate projections for Ottawa; historical (1991-2021) + 7 GW periods	120 run-wise simulations (15 runs x 8 GW periods, 31 years hourly); 45 runs train / 75 test	5 run-averaged climate variables: temperature, RH, wind speed, wind-driven rain, normal solar radiation	PLS for Tabular Regression; run-wise variant (31-year averages) and yearly variant	Maximum mould index, average mould index, dMI (cumulative hours with MI > 3 over 31 years)	Run-wise PLS: avg MI R2 0.70, RMSE 0.23; max MI R2 0.41; dMI R2 0.65; yearly PLS avg MI R2 0.54
36- Kulisz et al. (2024) [107]	Extend EIT wall-moisture reconstruction from a single measurement vector to five noise-perturbed vectors fed to a multi-branch bidirectional LSTM, testing accuracy and robustness gains.	EIT voltages at two basement stands of an 1869 Katowice tenement; FEM training data; dielectric, gravimetric and thermal validation	32,000 simulated cases (31,000 train / 1,000 test); 448 voltages per vector, five vectors per input; 11,458 voxels	Ordered 448-element voltage sequences; output voxel-wise conductivity mapped to moisture	Multi-branch bidirectional LSTM (21 layers) for Image Reconstruction Regression vs single-vector input	Voxel-wise conductivity forming 2D/3D images of internal wall moisture	Five sequences vs single: MSE 2.86 vs 3.66; SSIM 0.868 vs 0.835; ICC 0.886 vs 0.806; field agreement qualitative
37- Bayat Pour & Abdul	Develop an inutility decision analysis method combining probabilistic hygrothermal analysis (RF metamodel),	Delphin 1D simulations of three timber	1,000-2,300 simulations per case study; 70/30	Seven stochastic boundary conditions + six material	Random Forests metamodel per case for Tabular	95th-percentile mould index over 10-year	R2 0.832-0.965; P(MI3) 32.7% / 2.3% / 0.0%; CS2 optimal

Hamid (2024) [122]	life cycle costing and LCA to select the optimal moisture-safe wall assembly.	wall assemblies; Meteoronorm typical-year Gothenburg climate	split; metamodel predicts 10^7 LHS scenarios per case	properties per layer, reduced by parametric feature selection	Regression, with LHS and the VTT mould model	inspections, converted to failure probability P(MI3) and an inutility index	(inutility index 0.293)
38- Łagód et al. [136] (2023)	Assess the potential of electronic nose and AI data analysis for detecting and evaluating mould risk in buildings by analyzing VOCs in indoor air.	Sensor-based air sampling data collected from indoor air	11 sources (9 rooms + clean air + decayed timber); 1,672 observations	Relative resistance values of eight MOS gas sensors, Temp and RH	MLP and Random Forest for Multi-Class Classification	6-class severity: Air, None, Fine Stains, Visible, Significant, Totally stricken	Accuracy: MLP 100%; Random Forest 100% (train + test)
39- Salonvaara et al. [123] (2023)	Develop an ML metamodel predicting mould index and maximum moisture content of residential wood-frame and masonry wall assemblies, replacing hygrothermal simulations.	WUFI Pro 1D simulations; 22 US locations across all 8 climate zones	48,855 simulated wall cases (87% wood-frame); 80/10/10 split; 5-year simulations per case	Assembly inputs (R-value, moisture storage of layers) + 10 annual-average climate variables	Google AutoML Tables (Vertex AI) for Tabular Regression	Max mould index (5-yr max, interior/exterior sides of cavity) and maximum moisture content	Mould index R2: new materials 0.974; new climates 0.901; new materials + climates 0.916
40- Lee et al. [147] (2023)	Develop AI models using common IAQ sensor data to predict real-time concentrations of airborne biological particles (bacteria-, fungi- and pollen-like) indoors.	IAQ sensor data + fluorescence-based bioaerosol counts as ground truth	15,310 data points (commercial office) and 15,203 (shopping mall), 1-min intervals	Temp, RH, CO ₂ , TVOCs, PM _{2.5} , PM ₁₀	Linear/Lasso Regression, RF, XGBoost, MLP, RNN and LSTM for Time Series Regression	Time series concentrations of bacteria-, fungi- and pollen-like particles, plus PM _{2.5} /PM ₁₀	LSTM: 60-80% accuracy for bioaerosols, ~90% for PM prediction
41- Aggarwal et al. (2023) [124]	Transfer a brick-veneer PLS mould-index metamodel to stucco walls via a scaling correction, and quantify mould risk across a 15-member climate ensemble, historical and future.	DELPHIN 1D simulations of wood-frame walls; CanRCM4 ensemble climate for Ottawa, Vancouver and St John's	15 runs x 31 years x 2 periods x 3 cities x 2 claddings; screening covers 7,440 points per city and period	Yearly averages of five climate variables: T, RH, wind speed, wind-driven rain, normal solar radiation	PLS regression metamodel for Tabular Regression; stucco via linear correction of the brick model	Maximum mould index (VTT) at OSB sheathing; year rankings and Moisture Reference Year selection	R2 0.51-0.93 by city and period; rankings largely reproduced (Salonvaara ratios 90-97%); P(MI>3) rises under future climate
42- Bruno et al. (2023) [99]	Develop a Mask R-CNN approach for automatic detection and instance segmentation of moist areas and biological colonization on historic buildings from a small heterogeneous dataset.	On-site RGB photos of historic buildings in south Italy,	Moist area 248 images (173/49/26); biological colonization 142 images (100/27/15)	Visual texture, color and morphology; COCO polygon annotations	Two binary Mask R-CNN models (ResNet101 + FPN, COCO pretrained) for Instance Segmentation	Boxes, labels and masks for moist area and biological colonization	Moist area testing mAP 0.74-0.80; biological colonization 0.18-0.26; non-standard mAP definition
43- Marín-García et al. [96] (2023)	Develop a YOLOv5 model detecting efflorescence on exposed brick facades and classifying the required intervention (clean vs repair), supporting automated intervention maps.	RGB images of ceramic brick facades of buildings in Andalusia, Spain	765 facade images, 392 selected; 4,704 labeled brick instances, 2	Visual features (whitish surface salts, texture, deterioration patterns) learned end to end	YOLOv5 (small and large variants) for two-class Object Detection	Bounding boxes for efflorescent bricks with binary severity: clean vs repair	YOLOv5l best mAP@0.5 0.894; YOLOv5s 0.851; confusion matrix accuracy: clean 85%, repair 82%

			classes; 80/20 split				
44- Ruiz-Jimenez et al. [135] (2023)	Evaluate VOC emissions from fungal cultures, insulation materials and indoor air with SPME-Arrow TD-GC-MS, applying ML to distinguish fungal strains, materials and contaminated vs control houses.	VOC measurements from lab fungal cultures, insulation materials and indoor air	140 fungal, 3 insulation material and 52 indoor air samples from 24 houses	VOC concentration profiles from TD-GC-MS; features grouped by functional class	SVM for Classification; SIMCA for screening contaminated vs control houses	Fungal species from VOCs; houses with vs without mold	SVM indoor air classification >91% accuracy; SIMCA screening 84%
45- D'Orazio et al. [153] (2023)	Develop a neural network predicting microalgae growth on porous building facades, including wind-driven rain effects across EU climates and orientations.	Lab experimental growth data; TMY hourly climate data for case-study application	135 experimental growth curves (5 bricks x 9 conditions x 3 replicates), hourly over 1 year; 80/20 split	Coverage ratio, WDR-corrected RH, porosity, air temperature, roughness (MRMR-ranked); RH forced to 100% when WDR > 0	Feed-forward Neural Network (3 hidden layers) for Tabular Regression, applied iteratively as a time series	Hourly growth rate (dC/dT) of microalgae coverage, integrated into the coverage curve	R2 0.82; RMSE 0.00127; MAE 0.0003; Pearson correlation 0.853
46- Dafico et al. (2023) [114]	Develop NN models predicting material moisture content from the IRT-measurable surface thermal gradient, moving from qualitative to quantitative moisture diagnosis.	WUFI Pro 1D simulations of drying of initially saturated wall materials; no experimental or in-situ data	120 simulation scenarios (4 materials x 30 climates); 4,800 sampled sets; 80/20 split with cross-validation	Material type, air T and RH; plus surface thermal gradient for Models II-III; normalized inputs	Feedforward MLPs for Tabular Regression; Linear Regression and Decision Trees rejected as weaker	Model I: maximum surface thermal gradient; Models II-III: moisture content (kg/m3)	Model I R2 0.9999; Model II R2 0.957 (MAPE 21%); Model III R2 0.992 (MAPE 4.1%); simulation-only
47- Tijssens et al. [128] (2023)	Apply the pre-trained CNN metamodel at large scale to derive a climate-specific criteria matrix for safe timber frame wall designs across 96 compositions and 54 European climates.	Climate for Culture projection data; plus Delphin simulations for validation	135,168 CNN runs (96 walls x 1,408 samples) + 5,120 Delphin 2D runs	Climate time series (T, RH, WDR, solar); scalar inputs (orientation, WDR exposure); material moisture retention and vapor resistance	Pre-trained CNN for Time Series Regression, used for inference only; no new training	Hygrothermal response aggregated into climate-severity criteria for wall composition	Reproduced from the source model: Mould Index WB RMSE 0.169, MAE 0.077; VB RMSE 0.015, MAE 0.003
48- Yong et al. (2023) [102]	Identify prompt features that maximize zero-shot and few-shot CLIP performance for building defect classification and detection, including mildew.	RGB images of five residential defect types from web crawling and open datasets; prompts from construction and general dictionaries	No training data; 500-image balanced classification set; 1,650-image detection set (32:1); 5,100-image crack benchmark	Cosine similarity in CLIP's 512-D embedding space between images and textual, visual and ensemble prompts	CLIP off the shelf for zero-shot and few-shot Multi-Class Classification and detection	Defect class per image (crack, mildew, nail popping, peeling, wrinkling); defect vs non-defect detection	DK-ensemble zero-shot F1 0.835 (mildew up to 0.842); best multimodal accuracy 0.933; beats supervised crack baseline
49- Bayat Pour et al. [125] (2023)	Develop a probabilistic mould reliability and sensitivity analysis framework using a Random Forests metamodel trained on hygrothermal simulations to	Delphin hygrothermal simulations +	1,000-1,900 simulations per case study (3 cases); expanded	Solar absorption, indoor/outdoor surface heat transfer, long-wave	Random Forests for Tabular Regression as a metamodel; combined with Latin	Maximum mould index at wall surface; probability of	R2: CS1 0.855, CS2 0.958, CS3 0.942; MAE 0.007-0.011; failure probability:

	predict maximum mould index in external walls.	climate data (Lund, Sweden)	to 10 ⁷ scenarios via metamodel; 70/30 split	emissivity, vapor diffusion, WDR, material properties	Hypercube Sampling and the VTT mould model	failure P(MI1) and reliability R(MI1)	CS1 10.08%, CS2 5.88%, CS3 0.05%
50- Kłosowski et al. (2023) [108]	Improve EIT moisture imaging by adding a second LSTM stage refining the first-stage reconstruction, verified on a historic building where sampling is not permitted.	Synthetic FEM voltage data + real EIT measurements at the Golden Gate, Gdansk; dielectric and thermal reference	31,000 first-stage + 30,000 second-stage simulated cases; 448 voltages in, 11,458 voxels out; no independent test set	448-element voltage vector as single-step sequence; output voxel-wise conductivity (dry 1, wet 10)	Two-stage cascade of bidirectional LSTM networks for Image Reconstruction Regression	Voxel-wise conductivity forming 2D/3D images of internal wall moisture	Stage 1 to Stage 2 (5 cases): MSE 5.27 to 2.04; SSIM 0.37 to 0.53; ICC 0.64 to 0.83; field check qualitative
51- Mostafa et al. (2023) [91]	Automate condition assessment of built-up roofs with a two-stage CNN detecting and classifying moisture-related defects, coupled to budget-constrained repair optimization.	RGB video frames of roof surfaces from 21 campus buildings, phone camera at fixed height and area per frame	5,088 images after augmentation (2,480 defective); 75/25 split; 11 buildings held out; 490-image case study	Visual texture and color of ponding and vegetation vs sound membrane; defect area from frame counts	Two sequential nine-layer CNNs (defect/no defect, then defect type) + Excel Solver linear program	Defect vs no defect; ponding vs vegetation; damage area and repair/clean fund allocation	CNN1 ~95% accuracy, CNN2 97%; case study 90% correct, no missed defects; 21 buildings in under 3 h
52- Kłosowski et al. (2023) [109]	Develop a hybrid CNN + LSTM model improving EIT reconstructions of wall moisture, plus a classifier selecting the best model per measurement case.	EIT voltages from a portable tomograph at Pelplin abbey cloisters; FEM training data; dielectric and thermal validation	40,000 simulated cases (30,000 train / 10,000 test) with 5% Gaussian noise; 11,297-element mesh	448 voltage differences per case as ordered sequence; output pixel-wise conductivity (dry 1, wet 10)	1D CNN, LSTM and CNN+LSTM hybrid for Image Reconstruction Regression; decision-tree model selector	Pixel-wise conductivity forming 2D/3D images of internal wall moisture	Test MSE: hybrid 0.609 vs CNN 7.53 and LSTM 6.97; ICC 0.913; selector accuracy 99.1%; field check visual
53- Kavkler et al. [155] (2022)	Identify factors driving mould on the Celje Ceiling and build a predictive model relating sample attributes and position to mould presence.	Tabular data with binary material features, ordinal orientation variables, binary mould label	535 labeled samples; 133 attributes per sample	Binary material composition features plus ordinal spatial orientation features	Decision trees (J48), JRip, bagging, random forest in WEKA for Binary Classification; InfoGain and Relief ranking	Class label per sample: mould present or absent; ranked associated factors	Cross-validated accuracy: baseline 51%; best models 62% (Random Forest, orientation)
54- Kłosowski & Rymarczyk [110] (2022)	Develop a non-invasive method for identifying and visualizing moisture distribution in building walls by transforming EIT measurements into images.	Numerical simulation data + continuous voltage vectors + tomographic images	40,000 simulated samples (EIDORS generated)	96 voltage measurements from 16 electrodes on the surface of a damp brick wall	CNN and LSTM for Image Reconstruction Regression	Continuous pixel-level moisture/conductivity field (dry = 1, damp = 10)	CNN RMSE 1.32-1.78, ICC 0.72-0.94; LSTM RMSE 1.03-1.26, ICC 0.85-0.97
55- Kim et al. (2022) [92]	Classify wall quality into five defect types including efflorescence and damp	RGB wall images from	5,000 images (1,000 per class);	Visual texture, color and morphology;	Conv2D + four pretrained backbones	Wall defect class per image: crack,	Best ResNet-exponential:

	patches with a Conv2D ResNet transfer model, comparing six activation layers.	public repositories (Kaggle, SDNET2018); regions of interest extracted	80/20 split; 10 epochs per configuration	Gaussian derivative filtering; ImageNet-pretrained backbones	(Xception, VGG19, DenseNet, ResNet) with six activations for Multi-Class Classification	hole, efflorescence, damp patch or spall	precision 0.921, recall 0.913, accuracy 0.915; reported F-score 0.998 inconsistent; no per-class results
56- Cui et al. (2022) [103]	Develop a few-shot facade defect classifier (weight imprinting, contrastive learning, hard negative mining) so rare classes such as efflorescence are recognized from 1-10 samples.	RGB images of plastered and painted HDB facades in Singapore, cropped 224 x 224 around defects	10,352 images; five base classes plus two novel classes (efflorescence 30, exposed-rebar spalling 30); K = 1-10 shots	Texture embeddings from ImageNet-pretrained ResNet-50; cosine similarity to class proxies	Extensible classifier by weight imprinting with auxiliary contrastive loss and hard negative mining for few-shot Classification	Class label across seven facade classes including efflorescence	1-shot novel-class accuracy 63.5% (baseline 35.8%); 10-shot 83.0%; efflorescence 1-shot F1 0.547
57- Aggarwal et al. [126] (2022)	Develop an AI model predicting mould index and hygrothermal performance of wood frame wall assemblies across Canadian climate zones (Ottawa, Vancouver, St. John's).	DELPHIN simulated data from hourly hygrothermal analyses of brick-clad wood frame walls	192 training samples per city (16 years x 6 orientations x 2 climate periods); validation on 31 years per city	Climate variables: temperature, RH, wind speed, wind-driven rain, solar radiation	Partial Least Squares Regression (PLSR) for Tabular Regression	Maximum annual mould index	R2/RMSE: Ottawa 0.81/0.50; Vancouver 0.90/0.34; St. John's 0.89/0.31
58- Bozigar et al. [151] (2022)	Predict in-home exposures including mould for children with asthma from EHR and geospatial data, and quantify built-environment effects.	Clinical records of children with asthma + home exposure reports + geospatial housing and neighborhood attributes	Mould n = 583 (190 positive); 5-fold CV, no independent test set; 86 features screened by elastic net	86 multilevel individual, housing and neighborhood geospatial features	SuperLearner stacked ensemble (RF, BART, XGBoost, GLMs, GAMs) for Binary Classification; G-computation for effects	Presence/absence of cockroaches, rodents, mould and bedroom carpeting/rugs	Mould AUC 0.54 (chance level, excluded from effect estimation); other outcomes AUC 0.64-0.65
59- Fan & Chung [179] (2022)	Train and compare three supervised ML classifiers for pixel-level detection of efflorescence on concrete surfaces, assessing the effect of light source conditions.	RGB images of efflorescence and normal concrete building surfaces	500 pixel checkpoints; ~3:2 efflorescence-to-normal ratio; manually labeled training regions	Pixel-level RGB spectral values and texture; feature scaling/normalization	SVM (selected), Maximum Likelihood and Random Forest for Binary Image Classification	Binary pixel classification: efflorescence vs normal concrete surface	SVM accuracy 90.2%, F1 0.88, AUC 0.902; ML 89.8%; RF 87.0%
60- Kung et al. (2021) [105]	Combine a CNN with UAV imagery for detection and localization of four wall-tile defects (efflorescence, spalling, cracking, defacement) for real-time inspection.	RGB wall-tile images from phones, cameras and drones, resized 224 x 224; datasets released publicly	5,680 training images (efflorescence 1,382); 10% validation; 200-image held-out test set	Visual texture, color and morphology learned end to end; CAM heatmaps for localization	Fine-tuned ImageNet-pretrained VGG-16 for Multi-Class Classification with class activation mapping	Defect class per image plus CAM localization heatmap	Validation accuracy 86%; efflorescence precision 0.91, recall 0.80; no undamaged class modeled

61- Bansal et al. [129] (2021)	Apply SVR to forecast long-term (26-year) hygrothermal responses (T, RH) and mould index of wood frame and massive timber walls, training only on the first 5 years.	DELPHIN simulations + climate projections (1986-2016 and 2062-2092); Ottawa, Calgary, Vancouver	31-year hourly series per case; 5 years train / 26 years test; 2 wall types x 3 cities x multiple runs	4 LASSO-selected climate variables (WDR, RH, T, direct radiation); sliding windows of 48 h (T) and 168 h (RH)	Support Vector Regression (SVR) for Time Series Regression	Hourly T and RH at outer OSB (wood frame) and CLT (massive timber); post-processed Mould Index	Temperature RMSEP 0.04-0.12 C, R2 0.99; RH (WF) R2 0.46-0.75; RH (MT) R2 0.27-0.89
62- Rymarczyk et al. (2021) [111]	Compare ML against Gauss-Newton for the EIT inverse problem imaging moisture inside a historic building, using many single-output models per pixel.	EIT voltages from a purpose-built tomograph on a brick pillar; FEM training data; dielectric, microwave and gravimetric validation	44,000 simulated cases (70/15/15); ~7,756-element output mesh; 40 validation point measurements	448 voltage readings per vector; output pixel-wise conductivity calibrated to indicate moisture	Per-pixel Linear Regression (SVM and least squares learners) and MLPs (448-12-1) for Tabular Regression; Gauss-Newton baseline	Pixel-wise conductivity forming 2D/3D moisture images of the pillar	ANN best: RMSE 0.743, ICC 0.984; Gauss-Newton RMSE 4.38, ICC 0.554; field comparison visual
63- Perez & Tah [84] (2021)	Develop a DL object detection model identifying four building defect types including mould in real time using a smartphone camera.	RGB images of building surfaces collected from mobile phones, cameras and web sources	876 total images with 219 mould class (175 training, 44 testing)	Visual features of building surfaces: texture, color, edges, defect patterns	Single Shot Detector (SSD) with MobileNet for Object Detection and Classification	Bounding boxes, class labels and confidence for mould, stain, cracks and paint deterioration	Mould class: accuracy 0.81; recall 0.85; precision 0.66
64- Guo et al. (2021) [100]	Propose rule-based deep learning producing ISO 4628-conformant evaluation areas for six facade defect categories including biological growth.	RGB images of plastered and painted HDB facades in Singapore at ~1 m distance; LabelMe/COCO annotations	2,055 images (70/15/15); 2,820-8,715 instances depending on merge threshold (biological growth 162-255)	Visual texture, color and morphology via ResNet-50 + FPN; rule-derived distance and mask-ratio quantities	Mask R-CNN for Instance Segmentation, extended with localization, proposal-weighting and rectification rules	Instance masks and ISO 4628 evaluation areas for six defect categories including biological growth	Mean localization mAP50 0.509 (no rules) to 0.651 (all rules); segmentation 0.464 to 0.584; no per-class AP
65- Rymarczyk et al. (2021) [78]	Compare ML against classical Total Variation for the EIT inverse problem detecting moisture inside the walls of historic buildings.	EIT voltages (16 or 32 electrodes) at the Partisan Hill colonnade, Wroclaw; FEM training data with added noise	35,000 simulated cases (70/15/15); four reconstruction variants of 5,102-13,004 elements	96 or 448 voltage measurements per vector; output pixel-wise conductivity (dry 1, wet 10)	LARS, Elastic Net and per-pixel shallow ANNs for Tabular Regression; Total Variation baseline	Pixel-wise conductivity forming 2D/3D moisture images inside the wall	Elastic Net best: mean MSE 3.03, ICC 0.939; Total Variation MSE 17.06, ICC 0.736; checked against dielectric, microwave, gravimetric
66- Tijssens et al. [130] (2021)	Develop and optimize a CNN metamodel to replace time-consuming hygrothermal simulations of timber frame walls, enabling probabilistic	Delphin 6.0 simulations + Climate for Culture	5,760 samples (3,456 train / 1,152 validation / 1,152 test); 4-year	Climate time series (T, RH, WDR, solar) plus scalar wall and material inputs	CNN for Time Series Regression; hyperparameters tuned via Grey Wolf Optimizer; 5-fold CV	T/RH time series at weather and vapor barriers to compute max	Mould Index WB RMSE 0.169, MAE 0.077; VB RMSE 0.015, MAE 0.003

	assessment of mould and condensation risks.	projection data (2020-2050)	simulation per sample			Mould Index and condensation	
67- Hatır et al. (2021) [101]	Detect and map nine deterioration types on the Gumusler rock-cut monastery, including biological colonisation, efflorescence and deposits, replacing labor-intensive visual survey.	RGB drone (DJI Phantom 3) and camera images of the monastery; orthophotos from PhotoScan	1,740 images, 80/20 split; 13,385 labels across 9 classes (biological colonisation 2,184; efflorescence 936)	Visual texture, color and morphology; varied distances and lighting for invariance	Mask R-CNN with ResNet-101 backbone and RPN for Instance Segmentation	Instance masks for nine deterioration classes aggregated into deterioration maps	mAP 98.2%; AP 100% for biological colonisation, deposits and efflorescence; likely optimistic
68- May Tzuc et al. (2021) [115]	Estimate T and RH inside a concrete wall behind a double-skin green facade under emulated Nordic climate, and rank influential variables by global sensitivity analysis.	Sensor measurements from an Accelerated Weathering Laboratory emulating a Tampere typical year; 5-min intervals	8,064 samples; 70/30 split plus external validation; two walls, two foliage separations, three seasons	Five inputs: microclimate RH and T, global RH, ambient T, facade separation distance; normalized	ANN (5-42-2, Levenberg-Marquardt) for multi-output Tabular Regression; MLR/MPR/SRM benchmarks; PAWN sensitivity	RH and temperature inside the concrete wall; ranked input influence	Test R2 0.9917 (RH) and 0.9995 (T); ambient temperature most influential (40.7%)
69- Kłosowski et al. [112] (2021)	Develop a non-destructive moisture detection method for brick walls combining EIT with LSTM to improve reconstruction accuracy and speed.	Real EIT voltage measurements, synthetic EIT simulation data, moisture meter validation	30,000 simulated EIT samples for training + real EIT from a historic basilica wall for testing	448 real-valued EIT voltage measurements from electrode pairs on the wall	LSTM and ANN for Image Reconstruction Regression	Spatial image of moisture distribution inside the brick wall	LSTM RMSE 0.55-1.45, ICC 0.92-0.97; ANN RMSE 0.69-3.33, ICC 0.81-0.97
70- Perez and Tah [180] (2020)	Improve CNN classification accuracy by identifying and removing outlier images from training datasets using t-SNE dimensionality reduction and statistical outlier detection.	RGB images collected from mobile phones, cameras and web sources	2,523 total images with 631 mould class (448 training, 183 testing)	Visual appearance (texture, color, defect patterns) learned by a CNN	CNN based on pretrained VGG-16 with transfer learning for Image Classification	Defect label per image: normal, mould, stain or paint deterioration	Mould class: precision 0.89; recall 0.90; F1 0.89
71- Liu et al. [144] (2019)	Predict indoor culturable fungi concentrations using ML and identify the most effective minimal set of environmental input variables.	Sensor monitoring + lab-based microbiological analysis	85 residential units; 70% train and 30% test	PM10, PM2.5, RH, temperature, CO2 from indoor and outdoor sensors	Kernel-based Extreme Learning Machine (K-ELM) for Tabular Regression	Indoor culturable fungi concentration (CFU/m3)	RMSE 168.3 CFU/m3; R2 0.762; MAPE 0.439
72- Wang et al. (2019) [106]	Develop an automatic DL system detecting and locating efflorescence and spalling on historic brick masonry, deployed for real-time on-site use via IP webcam and smartphone.	RGB orthophotos of the Palace Museum Wall; on-site smartphone/webcam video	500 annotated images with 1,484 objects (471 efflorescence); 400 train / 100 test; 33 configurations trialed	Visual texture and color distinguishing salt deposits and spalling from sound black brick	Faster R-CNN (ResNet101 + RPN) for Object Detection; deployed to Android via TensorFlow	Bounding boxes with class and confidence for efflorescence and spalling	Best mAP 0.950 (efflorescence AP 0.999, spalling 0.900); 0.132 s per 500 x 500 image

73- Perez et al. [181] (2019)	Develop and evaluate a DL approach automatically detecting and localizing damp-related building defects (mould, stains, paint deterioration) from images.	RGB images collected from mobile phones, cameras and web sources	2,622 total images with 717 mould class (534 training, 183 testing)	Image color, texture and pattern cues of dampness defects	Pretrained VGG-16, ResNet-50 and Inception for Image Classification	Class label (mould, stain, deterioration, normal) + CAM heatmap	Mould class: precision 0.90; recall 0.91; F1 0.91
74- Tijskens et al. [116] (2019)	Compare MLP, RNN (LSTM/GRU) and CNN architectures as metamodels for predicting hygrothermal time series (T, RH, moisture content) in a masonry wall.	Delphin 5.8 simulations + Climate for Culture data (4 Belgian cities)	80 Sobol-sampled simulations, 4 years hourly; 64 train / 16 test; no validation set (early stopping on test)	Climate time series only (exterior T, RH, wind, solar; derived WDR; interior T/RH per EN 15026)	MLP, LSTM, GRU and CNN for Time Series Regression	Seven hygrothermal time series (T, RH, MC) at three wall positions; no damage index computed	Standardized-output RMSE: CNN 0.12, LSTM 0.18, GRU 0.19, MLP 0.50; R2 CNN 0.99; CNN ~10x faster to train
75- Tijskens et al. [131] (2019)	Optimize CNN hyperparameters via Grey Wolf Optimizer to predict hygrothermal time series of a masonry wall, deriving design guidelines for CNN metamodels.	Delphin 5.8 simulations + Climate for Culture data (4 Belgian cities); 3 brick types	960 samples (768 train / 192 validation / 256 test); 4-year hourly simulation per sample	6 climate time series (T, RH, WDR, solar) + 10 scalar wall and brick inputs	CNN for Time Series Regression	Time series of T, RH and saturation degree at three wall positions; post-processed into max Mould Index	Hygrothermal R2 0.985-0.999; max Mould Index RMSE 0.027, MAE 0.005, R2 0.702
76- Rymarczyk et al. (2018) [79]	Develop a non-destructive electrical tomography system with custom electrodes, comparing ML reconstruction algorithms for imaging moisture inside walls, especially historic masonry.	EIT voltages from a hybrid tomograph on concrete-block and brick wall models plus a real object; FEM training data	6,140 FEM-generated cases (70/15/15); electrode arrays 2x8 to 4x8; 8,099-pixel output lattice	192 voltage readings per measurement cycle; output pixel-wise conductance mapped to moisture	8,099 per-pixel MLPs for Tabular Regression; LARS and ElasticNet; Gauss-Newton baseline	Pixel-wise conductance forming 2D/3D moisture images inside the wall	ANN testing MSE 2.0e-5; R close to 1 across subsets; reconstructions compared across methods
77- Liu et al. [145] (2018)	Estimate indoor airborne culturable fungi concentration from IAQ indicators, avoiding slow lab incubation.	Sensor monitoring + lab-based microbiological analysis	249 data groups from 85 residential flats in Baoding, China	Indoor PM10, PM2.5, RH, temperature, CO2; outdoor PM10, PM2.5	ANN and SVM for Static Supervised Regression	Indoor culturable fungi concentration (CFU/m3)	Prediction accuracy: ANN 83.3%; SVM 64.0%
78- Tran & Hoang (2017) [154]	Develop an ML model predicting colonization time of the green alga K. flaccidum on mortar surfaces from material bioreceptivity properties, informing facade maintenance planning.	Laboratory experiments on mortar sprinkled with algal suspension	539 lab experiments from 49 mortar mixtures; 10-fold CV plus a 90/10 split (485/54)	Surface roughness, porosity, surface pH, permeability, cement type (OPC/CAC), colonization phase	LS-SVR with RBF kernel tuned by the Flower Pollination Algorithm for Tabular Regression; ANN and MLR benchmarks	Algal colonization time (days) to reach a given surface coverage	Test RMSE 4.55 days, R2 0.94; LM-ANN 5.45/0.92; MLR 13.34/0.53; 86% within +/-5 days
79- Freire et al. [132] (2017)	Predict the hygrothermal behavior of building roofs and mould growth risk on concrete tiles using ML, reducing the computational cost of full hygrothermal simulations.	Simulated data from a 2D heat-air-moisture model + VTT mould growth +	2,444 sample sets from 1.5 years of simulation; 50% train, 50% test	External temperature, external RH, direct solar radiation, diffuse solar radiation	Least Squares Support Vector Machines (LS-SVM) for Time Series Regression	Vapor flux, sensible heat flux, latent heat flux, mould growth risk index	R2: vapor flux 0.971; sensible heat flux 0.984; mould growth risk 0.9998

external TRY
weather data

References

1. Klepeis, N.E., et al., *The National Human Activity Pattern Survey (NHAPS): a resource for assessing exposure to environmental pollutants*. Journal of Exposure Science & Environmental Epidemiology, 2001. **11**(3): p. 231-252.
2. Sharpe, R.A., et al., *Making the Case for “Whole System” Approaches: Integrating Public Health and Housing*. International Journal of Environmental Research and Public Health, 2018. **15**(11): p. 2345.
3. Gabel, C., et al., *Seasonal indoor air quality, self-reported health and comfort amongst tenants living at Danish multi-family social housing sites*. Indoor and Built Environment, 2025. **34**(1): p. 135-148.
4. ElSharkawy, M.F., *Indoor air quality in Saudi residential homes*. Indoor and Built Environment, 2024. **33**(3): p. 583-600.
5. Jia, L., et al., *Synergistic Impact on Indoor Air Quality: The Combined Use of Air Conditioners, Air Purifiers, and Fresh Air Systems*. Buildings, 2024. **14**(6): p. 1562.
6. Du, C., B. Li, and W. Yu, *Indoor mould exposure: Characteristics, influences and corresponding associations with built environment—A review*. Journal of Building Engineering, 2021. **35**: p. 101983.
7. International Energy, A., *Global Status Report for Buildings and Construction 2019*. 2019, International Energy Agency: Paris.
8. Government, H.M., *The Climate Change Act 2008 (2050 Target Amendment) Order 2019*. 2019, HM Government: London.
9. Arriagada-Bustos, R., et al., *Study of interior hygrometry profiles for the hygrothermal assessment of social housing envelopes in humid temperate climates*. Journal of Building Engineering, 2025. **103**: p. 112031.
10. Hashemi, A. and M. Dungrani, *Indoor Environmental Quality and Health Implications of Building Retrofit and Occupant Behaviour in Social Housing*. Sustainability, 2025. **17**(1): p. 264.
11. Department for Levelling Up, H. and Communities, *English Housing Survey 2021 to 2022: headline report*. 2022, Department for Levelling Up, Housing and Communities: London.
12. Department for Business Energy and Industrial Strategy, *Energy Follow-Up Survey: thermal comfort, damp and ventilation*. 2021, Department for Business, Energy and Industrial Strategy: London, UK.
13. Scottish, G., *Scottish House Condition Survey: 2023 key findings*. 2025, Gov.scot.
14. Zhang, B., et al., *Dampness and mould in Chinese homes and sick building syndrome (SBS) symptoms – Associations with climate, family size, cleaning and ventilation*. Building and Environment, 2023. **245**: p. 110878.
15. Asthma Australia, *Homes, Health and Asthma in Australia: Understanding who is at risk of asthma or allergies in their home, what actions people take to protect themselves, and the barriers to action*. 2022, Asthma Australia: Canberra, Australia.
16. Lawrence Berkeley National Laboratory. *Prevalence of Building Dampness*. 2022; Available from: <https://iaqscience.lbl.gov/prevalence-building-dampness>.
17. Heseltine, E. and J. Rosen, *WHO guidelines for indoor air quality: dampness and mould*. 2009.

18. Fisk, W.J., Q. Lei-Gomez, and M.J. Mendell, *Meta-analyses of the associations of respiratory health effects with dampness and mold in homes*. Indoor air, 2005. **17**(LBNL--59363).
19. Sharpe, R., C. Thornton, and N. Osborne, *Modifiable factors governing indoor fungal diversity and risk of asthma*. Clinical & Experimental Allergy, 2014. **44**(5): p. 631-641.
20. Benton, E., *Damp and mould—the big picture. How do we tackle the damp and mould crisis in social housing: lessons from the UK*. Frontiers in Environmental Health, 2024. **Volume 3 - 2024**.
21. Ministry of Housing, C.a.L.G., *Awaab's Law: Guidance for social landlords – Timeframes for repairs in the social rented sector*. 2025.
22. Cromarty, H., *Social Housing (Regulation) Bill [HL] 2022-23: Progress of the Bill*, in *House of Commons Library Research Briefing*. 2023, UK Parliament: London.
23. Ministry of Housing Communities and Local Government, Department of Health and Social Care, and UK Health Security Agency, *Understanding and addressing the health risks of damp and mould in the home*. 2024, UK Government: London.
24. Hodgkin, R. and T. Sasse, *Tackling the UK's energy efficiency problem: What the Truss government should learn from other countries*. 2022, Institute for Government.
25. Boardman, B., *Home Truths: A Low-carbon Strategy to Reduce UK Housing Emissions by 80% by 2050*. 2007: University of Oxford's Environmental Change Institute.
26. Haghighat, M., E. MohammadiSavadkoobi, and N. Shafiabady, *Applications of Explainable Artificial Intelligence (XAI) and interpretable Artificial Intelligence (AI) in smart buildings and energy savings in buildings: A systematic review*. Journal of Building Engineering, 2025. **107**: p. 112542.
27. Akhavan Shams, S., H. Ge, and L. Wang, *Hygrothermal modeling in mass timber constructions: Recent advances and machine learning prospects*. Journal of Building Engineering, 2024. **96**: p. 110500.
28. Song, W. and J. Calautit, *Impact of occupancy behavior on building energy efficiency: What's next in detection and monitoring technologies?* Next Energy, 2025. **8**: p. 100350.
29. Rosca, C.-M. and A. Stancu *Machine Learning Applications with Sensors for Indoor Air Quality Research*. Sensors, 2026. **26**, 2909 DOI: 10.3390/s26092909.
30. Vereecken, E. and S. Roels, *Review of mould prediction models and their influence on mould risk evaluation*. Building and Environment, 2012. **51**: p. 296-310.
31. Battersby, S. and V. Ezratty, *Dampness in Dwellings: Causes, Effects and Remedies*. 2025.
32. Yang, C.S. and P.A. Heinsohn, *Sampling and Analysis of Indoor Microorganisms*. 2006. 1-273.
33. Adan, O. and R. Samson, *Fundamentals of mold growth in indoor environments and strategies for healthy living*. 2011.
34. Adan, O., *On the fungal defacement of interior finishes*. 1994.
35. Mendell, M.J., J.M. Macher, and K. Kumagai, *Measured moisture in buildings and adverse health effects: A review*. Indoor Air, 2018. **28**(4): p. 488-499.
36. Ayerst, G., *The effects of moisture and temperature on growth and spore germination in some fungi*. Journal of Stored Products Research, 1969. **5**(2): p. 127-141.

37. Flannigan, B., R. Samson, and J. Miller, *Microorganisms in home and indoor work environments: Diversity, health impacts, investigation and control: Second edition*. 2016. 1-517.
38. Viitanen, H., et al., *Moisture and Bio-deterioration Risk of Building Materials and Structures*. Journal of Building Physics, 2010. **33**(3): p. 201-224.
39. Samson, R., et al., *Food and Indoor Fungi*. 2010.
40. Mensah-Attipoe, J. and O.O. Toyinbo, *Fungal Growth and Aerosolization from Various Conditions and Materials*, in *Fungal Infection*, É. Silva de Loreto and J. Simoni Moraes Tondolo, Editors. 2019, IntechOpen: London.
41. Hung, L.-L., S. Caulfield, and J. Miller, *Recognition, Evaluation, and Control of Indoor Mold, second edition Edited by Ling-Ling Hung, Steven Caufield and J. David Miller; American Industrial Hygiene Association*. 255 pages Ami. 2020.
42. Miller, J.D., *Fungal bioaerosols as an occupational hazard*. Current Opinion in Allergy and Clinical Immunology, 2023. **23**(2).
43. Mensah-Attipoe, J., et al., *Susceptibility of green and conventional building materials to microbial growth*. Indoor Air, 2015. **25**(3): p. 273-284.
44. Zhao, D., et al., *Chemical composition of material extractives influences microbial growth and dynamics on wetted wood materials*. Scientific Reports, 2020. **10**(1): p. 14500.
45. Vanpachtenbeke, M., et al., *Performance of wood and wood-based materials regarding fungal decay*. E3S Web Conf., 2020. **172**: p. 20010.
46. Hyvärinen, A., et al., *Fungi and actinobacteria in moisture-damaged building materials — concentrations and diversity*. International Biodeterioration & Biodegradation, 2002. **49**(1): p. 27-37.
47. Johansson, P., et al., *Laboratory study to determine the critical moisture level for mould growth on building materials*. International Biodeterioration & Biodegradation, 2012. **73**: p. 23-32.
48. Hoang, C.P., et al., *Resistance of green building materials to fungal growth*. International Biodeterioration & Biodegradation, 2010. **64**(2): p. 104-113.
49. Singh, J., *Dry Rot and Other Wood-Destroying Fungi: Their Occurrence, Biology, Pathology and Control*. Indoor and Built Environment, 1999. **8**(1): p. 3-20.
50. Piecková, E., et al., *Building materials vs. fungal colonization - Model experiments*. Vol. 11. 2007. 71-78.
51. Laks, P.E., D.L. Richter, and G. Larkin, *Fungal susceptibility of interior commercial building panels*. Forest Products Journal, 2002. **52**: p. 41-44.
52. Sharpe, R.A., et al., *Fuel poverty increases risk of mould contamination, regardless of adult risk perception & ventilation in social housing properties*. Environment International, 2015. **79**: p. 115-129.
53. Sharpe, R.A., et al., *Cross-sectional study of housing tenures, risk of indoor mold growth & adult asthma in the UK*. Archives of Environmental & Occupational Health, 2025. **80**(5-6): p. 125-137.
54. British Standards, I., *BS 5250:2021 Management of moisture in buildings: Code of practice*. 2021.
55. Chen, X., et al. *Monitoring, Human Health Risk Assessment and Optimized Management for Typical Pollutants in Indoor Air from Random Families of University Staff, Wuhan City, China*. Sustainability, 2017. **9**, 1115 DOI: 10.3390/su9071115.

56. Kim, K.-H., E. Kabir, and S.A. Jahan, *Airborne bioaerosols and their impact on human health*. Journal of Environmental Sciences, 2018. **67**: p. 23-35.
57. Schenkel, D., et al., *A meta-analysis approach for assessing the diversity and specificity of belowground root and microbial volatiles*. Frontiers in Plant Science, 2015. **Volume 6 - 2015**.
58. Eggleston, P.A. and R.K. Bush, *Environmental allergen avoidance: An overview*. Journal of Allergy and Clinical Immunology, 2001. **107**(3): p. S403-S405.
59. Suchorab, Z., et al., *Estimation of Microbial VOC Concentration of Fungal Origin Using Gas Sensor Array and Artificial Intelligence*. Ecological Chemistry and Engineering S, 2025. **32**(2): p. 171-188.
60. Hukka, A. and H.A. Viitanen, *A mathematical model of mould growth on wooden material*. Wood Science and Technology, 1999. **33**(6): p. 475-485.
61. Ojanen, T., et al. *Classification of material sensitivity-New approach for mould growth modeling*. in *Proceedings 9th Nordic Symposium on Building Physics NSB 2011, Tampere, Finland, 29 May-2 June 2011*. 2011. Tampere University of Technology.
62. Liberati, A., et al., *The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration*. Journal of Clinical Epidemiology, 2009. **62**(10): p. e1-e34.
63. Falagas, M.E., et al., *Comparison of PubMed, Scopus, Web of Science, and Google Scholar: strengths and weaknesses*. The FASEB Journal, 2008. **22**(2): p. 338-342.
64. Meho, L.I. and Y. Rogers, *Citation counting, citation ranking, and h-index of human-computer interaction researchers: A comparison of Scopus and Web of Science*. Journal of the American Society for Information Science and Technology, 2008. **59**(11): p. 1711-1726.
65. Hosseini, M.R., et al., *Critical evaluation of off-site construction research: A Scientometric analysis*. Automation in Construction, 2018. **87**: p. 235-247.
66. Falana, J., R. Osei-Kyei, and V.W.Y. Tam, *Towards achieving a net zero carbon building: A review of key stakeholders and their roles in net zero carbon building whole life cycle*. Journal of Building Engineering, 2024. **82**: p. 108223.
67. Mousavi, S., et al., *Data-driven prediction and optimization toward net-zero and positive-energy buildings: A systematic review*. Building and Environment, 2023. **242**: p. 110578.
68. Hunde, J.M., et al., *Data-driven and physics-based modeling approaches and their integration in building digital twins: A systematic review*. Journal of Building Engineering, 2025. **114**: p. 114214.
69. Ohene, E., A.P.C. Chan, and A. Darko, *Prioritizing barriers and developing mitigation strategies toward net-zero carbon building sector*. Building and Environment, 2022. **223**: p. 109437.
70. Haddaway, N.R., et al., *PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis*. Campbell Systematic Reviews, 2022. **18**(2): p. e1230.
71. Hagen, M.H., et al., *Digital psychosocial follow-up for survivors of childhood critical illness: protocol for a systematic literature review*. Systematic Reviews, 2025. **15**(1): p. 30.

72. Hirt, J., et al., *Guidance on terminology, application, and reporting of citation searching: the TARCiS statement*. BMJ, 2024. **385**: p. e078384.
73. Sepasgozar, S., et al. *A Systematic Content Review of Artificial Intelligence and the Internet of Things Applications in Smart Home*. Applied Sciences, 2020. **10**, 3074 DOI: 10.3390/app10093074.
74. Saleem, R., et al., *Explaining deep neural networks: A survey on the global interpretation methods*. Neurocomputing, 2022. **513**: p. 165-180.
75. Kłosowski, G., et al., *A multi-head neural network architecture for EIT inverse problem resolution in wall moisture imaging*. Measurement, 2026. **283**: p. 122107.
76. Hu, X., et al., *Physically interpretable machine learning for robust façade bio-colonisation detection: Comparative insights from hyperspectral and RGB imaging under hard visual interference*. Building and Environment, 2026. **300**: p. 114712.
77. Alexakis, E., et al., *A novel application of deep learning approach over IRT images for the automated detection of rising damp on historical masonries*. Case Studies in Construction Materials, 2024. **20**: p. e02889.
78. Rymarczyk, T., et al. *Historical Buildings Dampness Analysis Using Electrical Tomography and Machine Learning Algorithms*. Energies, 2021. **14**, 1307 DOI: 10.3390/en14051307.
79. Rymarczyk, T., G. Kłosowski, and E. Kozłowski *A Non-Destructive System Based on Electrical Tomography and Machine Learning to Analyze the Moisture of Buildings*. Sensors, 2018. **18**, 2285 DOI: 10.3390/s18072285.
80. Wang, J., et al., *Computer vision to advance the sensing and control of built environment towards occupant-centric sustainable development: A critical review*. Renewable and Sustainable Energy Reviews, 2024. **192**: p. 114165.
81. Chu, Y.-C., et al. *Automated Wall Moisture Detection in Heritage Sites Based on Convolutional Neural Network (CNN) for Infrared Imagery*. Applied Sciences, 2025. **15**, 6495 DOI: 10.3390/app15126495.
82. Chen, X., J. He, and S. Wang, *Deep learning-driven pathology detection and analysis in historic masonry buildings of Suzhou*. npj Heritage Science, 2025. **13**(1): p. 197.
83. Fan, Y., et al., *UAV and Deep Learning for Automated Detection and Visualization of Façade Defects in Existing Residential Buildings*. Sensors, 2025. **25**(23): p. 7118.
84. Perez, H. and J.H.M. Tah, *Deep learning smartphone application for real-time detection of defects in buildings*. Structural Control and Health Monitoring, 2021. **28**(7): p. e2751.
85. Mansouri, T., et al., *A Newly Adopted YOLOv9 Model for Detecting Mould Regions Inside of Buildings*. International Journal of Pattern Recognition and Artificial Intelligence, 2024. **39**(01): p. 2450025.
86. Luo, X., et al., *Brick wall efflorescence detection technology using target detection illustrated by Wuhan Gao's Hospital*. npj Heritage Science, 2025. **13**(1): p. 144.
87. Chen, X., et al., *Intelligent Detection and Virtual Restoration of Shedthin Tiles in Suzhou's Classical Gardens*. International Journal of Architectural Heritage, 2025: p. 1-21.
88. Santos, L.M.A.d., et al., *Mapping stains on flat roofs using semantic segmentation based on deep learning*. Case Studies in Construction Materials, 2025. **22**: p. e04106.

89. Zou, J. and Y. Deng, *Intelligent assessment system of material deterioration in masonry tower based on improved image segmentation model*. Heritage Science, 2024. **12**(1): p. 252.
90. Ekici, B.B., N.B. Avci, and S. Ekici, *A Six-Stage Ablation-Driven Benchmarking Framework for Deep Learning-Based Deterioration Classification in Heritage Structures*. IEEE Access, 2026. **14**: p. 12422-12434.
91. Mostafa, K., et al., *Using image analysis to quantify defects and prioritize repairs in built-up roofs*. Facilities, 2023. **41**(7-8): p. 498-509.
92. Kim, B., et al. *Deep Learning Activation Layer-Based Wall Quality Recognition Using Conv2D ResNet Exponential Transfer Learning Model*. Mathematics, 2022. **10**, 4602 DOI: 10.3390/math10234602.
93. Perez, H. and J.H.M. Tah, *Improving the Accuracy of Convolutional Neural Networks by Identifying and Removing Outlier Images in Datasets Using t-SNE*. Mathematics, 2020. **8**(5): p. 662.
94. Perez, H., J.H.M. Tah, and A. Mosavi, *Deep Learning for Detecting Building Defects Using Convolutional Neural Networks*. Sensors, 2019. **19**(16): p. 3556.
95. Tan, Y., et al., *Building defect inspection and data management using computer vision, augmented reality, and BIM technology*. Automation in Construction, 2024. **160**: p. 105318.
96. Marín-García, D., et al., *Deep learning model for automated detection of efflorescence and its possible treatment in images of brick facades*. Automation in Construction, 2023. **145**: p. 104658.
97. Bourgeois, I., et al., *Image-based automatic detection of efflorescence on limestone heritage constructions: Application to the Monastery of Batalha*. Journal of Building Engineering, 2026. **129**: p. 116924.
98. Fu, X. and N. Angkawisittpan, *Detecting surface defects of heritage buildings based on deep learning*. 2024. **33**(1).
99. Bruno, S., R.A. Galantucci, and A. Musicco, *Decay detection in historic buildings through image-based deep learning*. VITRUVIO-International Journal of Architectural Technology and Sustainability, 2023. **8**: p. 6-17.
100. Guo, J., Q. Wang, and Y. Li, *Evaluation-oriented façade defects detection using rule-based deep learning method*. Automation in Construction, 2021. **131**: p. 103910.
101. Hatır, M.E., İ. Ince, and M. Korkanç, *Intelligent detection of deterioration in cultural stone heritage*. Journal of Building Engineering, 2021. **44**: p. 102690.
102. Yong, G., et al., *Prompt engineering for zero-shot and few-shot defect detection and classification using a visual-language pretrained model*. Computer-Aided Civil and Infrastructure Engineering, 2023. **38**(11): p. 1536-1554.
103. Cui, Z., et al., *Few-shot classification of façade defects based on extensible classifier and contrastive learning*. Automation in Construction, 2022. **141**: p. 104381.
104. Staffa Junior, L.d.B., et al., *Web platform for building roof maintenance inspection using UAS and artificial intelligence*. International Journal of Building Pathology and Adaptation, 2023. **43**(1): p. 4-28.
105. Kung, R.-Y., et al., *Application of Deep Learning and Unmanned Aerial Vehicle on Building Maintenance*. Advances in Civil Engineering, 2021. **2021**(1): p. 5598690.
106. Wang, N., et al., *Automatic damage detection of historic masonry buildings based on mobile deep learning*. Automation in Construction, 2019. **103**: p. 53-66.

107. Kulisz, M., et al., *The use of the multi-sequential LSTM in electrical tomography for masonry wall moisture detection*. Measurement, 2024. **234**: p. 114860.
108. Kłosowski, G., et al., *Use of the double-stage LSTM network in electrical tomography for 3D wall moisture imaging*. Measurement, 2023. **213**: p. 112741.
109. Kłosowski, G., et al. *Using Machine Learning in Electrical Tomography for Building Energy Efficiency through Moisture Detection*. Energies, 2023. **16**, 1818 DOI: 10.3390/en16041818.
110. Kłosowski, G. and T. Rymarczyk, *APPLICATION OF CONVOLUTIONAL NEURAL NETWORKS IN WALL MOISTURE IDENTIFICATION BY EIT METHOD*. Informatyka, Automatyka, Pomiary w Gospodarce i Ochronie Środowiska, 2022. **12**(1): p. 20-23.
111. Rymarczyk, T., et al. *Comparison of Machine Learning Methods in Electrical Tomography for Detecting Moisture in Building Walls*. Energies, 2021. **14**, 2777 DOI: 10.3390/en14102777.
112. Kłosowski, G., et al. *The Concept of Using LSTM to Detect Moisture in Brick Walls by Means of Electrical Impedance Tomography*. Energies, 2021. **14**, 7617 DOI: 10.3390/en14227617.
113. Jiang, K., et al., *Predictive modelling and multi-scenario validation of indoor micro-climate thermal-humidity conditions*. Journal of Building Engineering, 2025. **111**: p. 113474.
114. Dafico, L.C.M., et al., *Machine learning models applied to moisture assessment in building materials*. Construction and Building Materials, 2023. **405**: p. 133330.
115. May Tzuc, O., et al., *Modeling of hygrothermal behavior for green facade's concrete wall exposed to nordic climate using artificial intelligence and global sensitivity analysis*. Journal of Building Engineering, 2021. **33**: p. 101625.
116. Tijskens, A., S. Roels, and H. Janssen, *Neural networks for metamodelling the hygrothermal behaviour of building components*. Building and Environment, 2019. **162**: p. 106282.
117. Jiang, K., et al., *Optimization of air supply parameters for predicting indoor temperature and humidity under indoor mold index constraints*. Energy and Buildings, 2025. **345**: p. 116050.
118. Meng, X., et al., *Long-term hygrothermal performance assessment of on-site wood-framed walls based on sensor monitoring and machine learning*. European Journal of Wood and Wood Products, 2025. **83**(2): p. 64.
119. Bayat Pour, M., S. Kahangi Shahreza, and A. Abdul Hamid, *Influence of extruded mortar joints on mould growth in timber frame wall with brick veneer: Probabilistic and machine learning modelling*. Journal of Building Physics, 2026: p. 17442591251414130.
120. Bayat Pour, M. and S. Kahangi Shahreza, *Impact of extruded mortar joints on the hygrothermal performance of brick veneer walls: A probabilistic study*. Journal of Building Engineering, 2024. **94**: p. 109936.
121. Aggarwal, C., et al., *Long-term hygrothermal performance assessment of wood-frame walls considering climate uncertainties using partial least squares (PLS) regression*. Energy and Buildings, 2024. **307**: p. 113953.
122. Bayat Pour, M. and A. Abdul Hamid, *Towards a sustainable decision analysis approach for moisture-safe building envelope design*. Journal of Building Engineering, 2024. **97**: p. 110955.

123. Salonvaara, M., et al. *Application of Machine Learning to Assist a Moisture Durability Tool*. Energies, 2023. **16**, 2033 DOI: 10.3390/en16042033.
124. Aggarwal, C., H. Ge, and M. Defo, *Assessing mould growth risk of wood-frame walls using partial least squares (PLS) regression considering climate model uncertainties*. Building and Environment, 2023. **238**: p. 110374.
125. Bayat Pour, M., et al., *Robust probabilistic modelling of mould growth in building envelopes using random forests machine learning algorithm*. Building and Environment, 2023. **243**: p. 110703.
126. Aggarwal, C., et al., *Hygrothermal performance assessment of wood frame walls under historical and future climates using partial least squares regression*. Building and Environment, 2022. **223**: p. 109501.
127. Zhao, Y., et al., *Regional assessment of mold growth risk in light wood-framed wall envelope based on meteorological data-driven and neural network model*. European Journal of Wood and Wood Products, 2025. **83**(2): p. 86.
128. Tijskens, A., H. Janssen, and S. Roels, *Probabilistic hygrothermal assessment of various timber frame wall compositions*. Building and Environment, 2023. **243**: p. 110619.
129. Bansal, N., M. Defo, and M.A. Lacasse, *Application of Support Vector Regression to the Prediction of the Long-Term Impacts of Climate Change on the Moisture Performance of Wood Frame and Massive Timber Walls*. Buildings, 2021. **11**(5): p. 188.
130. Tijskens, A., S. Roels, and H. Janssen, *Hygrothermal assessment of timber frame walls using a convolutional neural network*. Building and Environment, 2021. **193**: p. 107652.
131. Tijskens, A., H. Janssen, and S. Roels *Optimising Convolutional Neural Networks to Predict the Hygrothermal Performance of Building Components*. Energies, 2019. **12**, 3966 DOI: 10.3390/en12203966.
132. Freire, R.Z., G.H.d. Santos, and L.d.S. Coelho *Hygrothermal Dynamic and Mould Growth Risk Predictions for Concrete Tiles by Using Least Squares Support Vector Machines*. Energies, 2017. **10**, 1093 DOI: 10.3390/en10081093.
133. Mehraban, M.H., et al. *AI-Driven Prediction of Building Energy Performance and Thermal Resilience During Power Outages: A BIM-Simulation Machine Learning Workflow*. Buildings, 2025. **15**, 3950 DOI: 10.3390/buildings15213950.
134. Choi, Y., B. Kang, and D. Kim, *Effective detection of indoor fungal contamination through the identification of volatile organic compounds using mass spectrometry and machine learning*. Environmental Pollution, 2024. **363**: p. 125195.
135. Ruiz-Jimenez, J., et al., *Evaluation of VOCs from fungal strains, building insulation materials and indoor air by solid phase microextraction arrow, thermal desorption–gas chromatography-mass spectrometry and machine learning approaches*. Environmental Research, 2023. **224**: p. 115494.
136. Łagód, G., et al. *Application of Dimensionality Reduction and Machine Learning Methods for the Interpretation of Gas Sensor Array Readouts from Mold-Threatened Buildings*. Applied Sciences, 2023. **13**, 8588 DOI: 10.3390/app13158588.
137. Yang, H., et al., *Electronic Nose for Indoor Mold Detection and Identification*. Advanced Sensor Research, 2026. **5**(2): p. e00124.

138. Choi, Y., S. Kim, and D. Kim, *Machine learning-integrated metal-organic frameworks/multi-walled carbon nanotubes sensor array for real-time detection of fungal VOCs*. Journal of Hazardous Materials, 2026. **505**: p. 141493.
139. Georgakopoulos, D.G., et al., *Microbiology and atmospheric processes: biological, physical and chemical characterization of aerosol particles*. Biogeosciences, 2009. **6**(4): p. 721-737.
140. Chen, Q. and L.M. Hildemann, *The Effects of Human Activities on Exposure to Particulate Matter and Bioaerosols in Residential Homes*. Environmental Science & Technology, 2009. **43**(13): p. 4641-4646.
141. Haas, D., et al., *The concentrations of culturable microorganisms in relation to particulate matter in urban air*. Atmospheric Environment, 2013. **65**: p. 215-222.
142. Barbosa, M.I., et al., *Integrated PM–MOX–Thermal Sensing for Monitoring Bioaerosol Dynamics in Controlled Indoor Environments*. Sensors, 2026. **26**(11): p. 3521.
143. Kim, D., et al., *Revisiting the joint effect of temperature and relative humidity on airborne mold and bacteria concentration in indoor environment: A machine learning approach*. Building and Environment, 2025. **270**: p. 112548.
144. Liu, Z., et al., *Predicting the concentration of indoor culturable fungi using a kernel-based extreme learning machine (K-ELM)*. International Journal of Environmental Health Research, 2020. **30**(3): p. 344-356.
145. Liu, Z., et al., *Exploring the potential relationship between indoor air quality and the concentration of airborne culturable fungi: a combined experimental and neural network modeling study*. Environmental Science and Pollution Research, 2018. **25**(4): p. 3510-3517.
146. Lee, B.G., et al., *Machine learning models for predicting indoor airborne fungal concentrations in public facilities utilizing environmental variables*. Environmental Pollution, 2025. **368**: p. 125684.
147. Lee, J.Y., et al., *Artificial intelligence-based prediction of indoor bioaerosol concentrations from indoor air quality sensor data*. Environment International, 2023. **174**: p. 107900.
148. Aziz, G. and A. Hardy, *A predictive model for damp risk in english housing with explainable AI*. Scientific Reports, 2025. **15**(1): p. 12658.
149. Pakdehi, M., et al., *Modeling the latent impacts of extreme floods on indoor mold spores in residential buildings: Application of machine learning algorithms*. Environment International, 2025. **196**: p. 109319.
150. Dulin, S., et al., *Predicting mold severity in buildings using interpretable machine learning*. Journal of Building Engineering, 2025. **112**: p. 113901.
151. Bozigar, M., et al., *In-home environmental exposures predicted from geospatial characteristics of the built environment and electronic health records of children with asthma*. Annals of Epidemiology, 2022. **73**: p. 38-47.
152. Baudrit, C., et al., *Biological durability of Guyanese fibre insulation boards in tropical context – part 2: predicting fungal and termite resistance using explainable machine learning*. Wood Material Science & Engineering, 2025: p. 1-13.
153. D'Orazio, M., E. Quagliarini, and A. Gianangeli, *Influence of the rain on algae growth on building facades. A predictive model based on neural networks*. Building and Environment, 2023. **246**: p. 110990.

154. Tran, T.-H. and N.-D. Hoang, *Estimation of algal colonization growth on mortar surface using a hybridization of machine learning and metaheuristic optimization*. Sādhanā, 2017. **42**(6): p. 929-939.
155. Kavkler, K., et al., *A multidisciplinary study of biodeteriorated Celje Ceiling, a tempera painting on canvas*. International Biodeterioration & Biodegradation, 2022. **170**: p. 105389.
156. Arslan, M., et al., *Integrated environmental information system for damp and mould prevention*. Developments in the Built Environment, 2026. **26**: p. 100893.
157. Qiao, J., et al., *Experimental and simulation investigation of mould growth risk on typical residential indoor materials: A comparative study*. Journal of Building Engineering, 2026. **119**: p. 115381.
158. Johansson, P., L. Lång, and C.-M. Capener, *How well do mould models predict mould growth in buildings, considering the end-user perspective?* Journal of Building Engineering, 2021. **40**: p. 102301.
159. Jiang, Z., et al., *Physics-informed machine learning for building performance simulation-A review of a nascent field*. Advances in Applied Energy, 2025. **18**: p. 100223.
160. Jiang, Z., X. Wang, and B. Dong, *Physics-Informed Modularized Neural Networks for Building Dynamic Modeling: A Smart Buildings Hackathon Case Study*, in *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems*. 2025, Association for Computing Machinery. p. 924–927.
161. Wang, X., et al., *Physics-consistent input convex neural network-driven reinforcement learning control for multi-zone radiant ceiling heating and cooling systems: An experimental study*. Energy and Buildings, 2025. **327**: p. 115105.
162. Kashinath, K., et al., *Physics-informed machine learning: case studies for weather and climate modelling*. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 2021. **379**(2194).
163. Hu, X. and R.H. Assaad, *A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies*. Journal of Building Engineering, 2024. **86**: p. 108901.
164. Mehraban, M.H., A.A. Alnaser, and S.M.E. Sepasgozar, *Building Information Modeling and AI Algorithms for Optimizing Energy Performance in Hot Climates: A Comparative Study of Riyadh and Dubai*. Buildings, 2024. **14**(9): p. 2748.
165. Mehraban, M.H., et al., *AI-enhanced automation of building energy optimization using a hybrid stacked model and genetic algorithms: Experiments with seven machine learning techniques and a deep neural network*. Results in Engineering, 2025. **26**: p. 104994.
166. de las Morenas, J., L.M. Belmonte, and R. Morales, *Designing an AI-driven digital twin architecture for building energy prediction*. Journal of Building Engineering, 2025. **113**: p. 113966.
167. Khosh-Amadi, N., et al., *Development of a spatially contextualised and AI-enabled digital twin for asthma-specific indoor air quality management*. Journal of Building Engineering, 2026. **118**: p. 115027.

168. Kim, C.H. and D.W. Jung, *AI-based dynamic predictive control and energy optimization for semiconductor FAB HVAC systems through digital twin technology*. Journal of Building Engineering, 2026. **117**: p. 114824.
169. Iqbal, F. and S. Mirzabeigi, *Digital Twin-Enabled Building Information Modeling–Internet of Things (BIM-IoT) Framework for Optimizing Indoor Thermal Comfort Using Machine Learning*. Buildings, 2025. **15**(10): p. 1584.
170. Ghansah, F.A., et al., *An enabling framework towards digital twins implementation model for sustainable smart facility management*. Smart and Sustainable Built Environment, 2026: p. 1-30.
171. Ghansah, F.A., *Digital twins for smart building at the facility management stage: a systematic review of enablers, applications and challenges*. Smart and Sustainable Built Environment, 2024. **14**(4): p. 1194-1229.
172. Ghansah, F.A., et al., *Generative digital twins for facility management in the built environment: an expert-informed systematic literature review approach*. Facilities, 2026: p. 1-28.
173. Mehraban, M.H., et al., *Automated Image-to-BIM Using Neural Radiance Fields and Vision-Language Semantic Modeling*. Buildings, 2025. **15**(24): p. 4549.
174. Holme, J., et al., *Culturable mold in indoor air and its association with moisture-related problems and asthma and allergy among Swedish children*. Indoor Air, 2010. **20**(4): p. 329-340.
175. Chen, Y., et al., *Correlating surface mold contamination with airborne pollution under mild indoor air disturbance: A case study of Aspergillus niger*. Building and Environment, 2024. **266**.
176. Lundberg, S.M. and S.-I. Lee, *A unified approach to interpreting model predictions*. Advances in neural information processing systems, 2017. **30**.
177. Ghansah, F.A., et al., *Digital twins implementation for sustainable smart facility management in the built environment: underlying critical challenges via embedded mixed method design*. Expert Systems with Applications, 2026. **328**: p. 132940.
178. Mehraban, M.H., et al., *Transfer learning for improving building energy performance and occupant thermal comfort prediction across climates and building types to address the data scarcity challenge*. Energy and Built Environment, 2026.
179. Fan, C.-L. and Y.-J. Chung *Supervised Machine Learning–Based Detection of Concrete Efflorescence*. Symmetry, 2022. **14**, 2384 DOI: 10.3390/sym14112384.
180. Perez, H. and J.H.M. Tah *Improving the Accuracy of Convolutional Neural Networks by Identifying and Removing Outlier Images in Datasets Using t-SNE*. Mathematics, 2020. **8**, 662 DOI: 10.3390/math8050662.
181. Perez, H., J.H.M. Tah, and A. Mosavi *Deep Learning for Detecting Building Defects Using Convolutional Neural Networks*. Sensors, 2019. **19**, 3556 DOI: 10.3390/s19163556.

